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| TITLE                           | MODEL  |
| Bounce RF Troubleshooting Guide | Bounce |

**Introduction:**

This document covers the Board Test RF sections for all variants of Bounce. It includes the schematic and layout of each block to help troubleshoot factory failures and analysis.

Need to update pg numbers:

**Sequence:**

|                                   |                     |
|-----------------------------------|---------------------|
| PCB PLACEMENT & CONNECTOR MAPPING | <a href="#">2</a>   |
| WTR                               | <a href="#">3</a>   |
| MAIN ASM LB&HB:                   | <a href="#">7</a>   |
| DIVERSITY ASM LB HB               | <a href="#">9</a>   |
| PA LB&MB                          | <a href="#">11</a>  |
| PA HB                             | <a href="#">13</a>  |
| B1 LTE&WCDMA TRX                  | <a href="#">15</a>  |
| B2&B25 LTE,WCDMA B2, CDMA BC1 TRX | <a href="#">25</a>  |
| B3 LTE&WCDMA TRX                  | <a href="#">34</a>  |
| B4 LTE&WCDMA TRX                  | <a href="#">49</a>  |
| B5 LTE, WCDMA B5                  | <a href="#">62</a>  |
| B7 LTE TRX                        | <a href="#">74</a>  |
| B8 LTE & WCDMA TRX                | <a href="#">122</a> |
| B12&B17 LTE TRX                   | <a href="#">131</a> |
| B13 LTE TRX                       | <a href="#">146</a> |
| B20 LTE TRX                       | <a href="#">155</a> |
| B28 LTE TRX                       | <a href="#">167</a> |
| B34&B39 LTE&TDSCDMA TRX           | <a href="#">175</a> |
| B40 LTE TRX                       | <a href="#">185</a> |
| B38&B41 LTE TRX                   | <a href="#">196</a> |
| RF MIPI DEVICES                   | <a href="#">199</a> |

|                               |           |
|-------------------------------|-----------|
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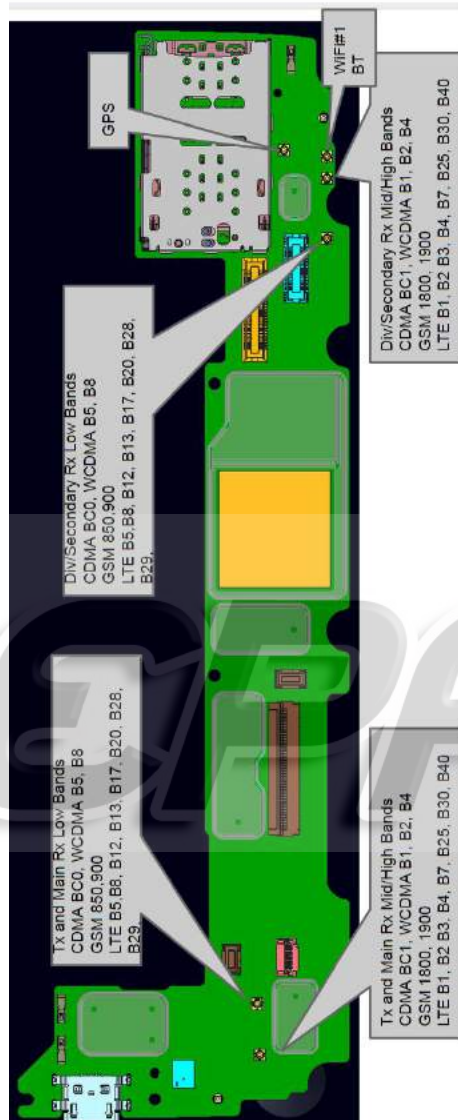
TITLE

Bounce RF Troubleshooting Guide

MODEL

Bounce

## PCB Component Placement and Connector Mapping



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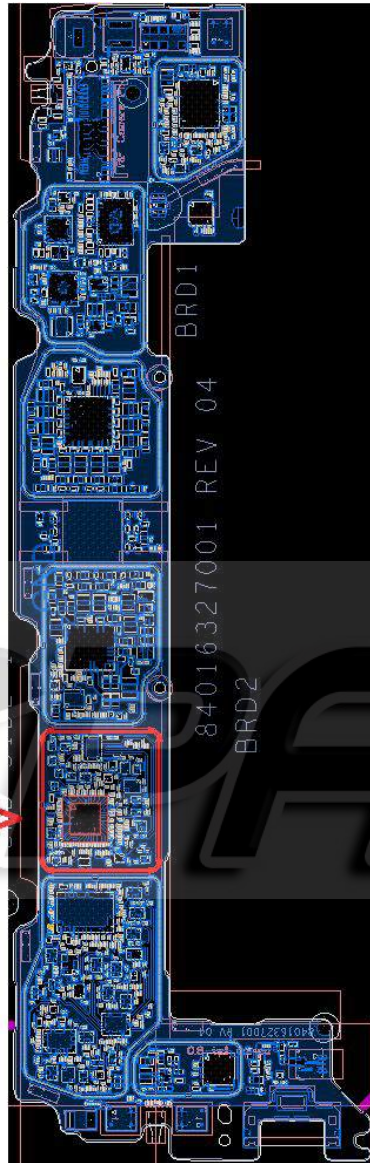
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Bounce RF Troubleshooting Guide

MODEL

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WTR- Layout:



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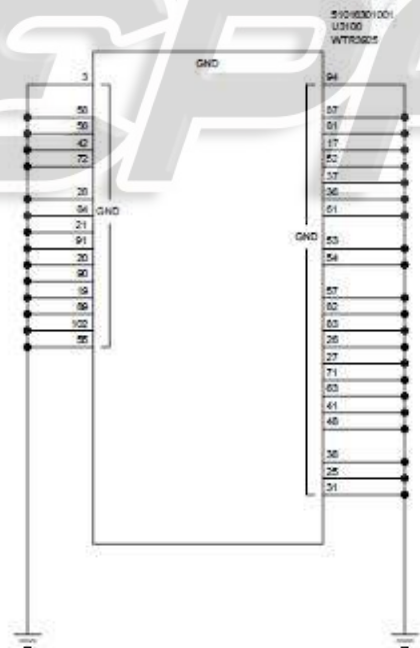
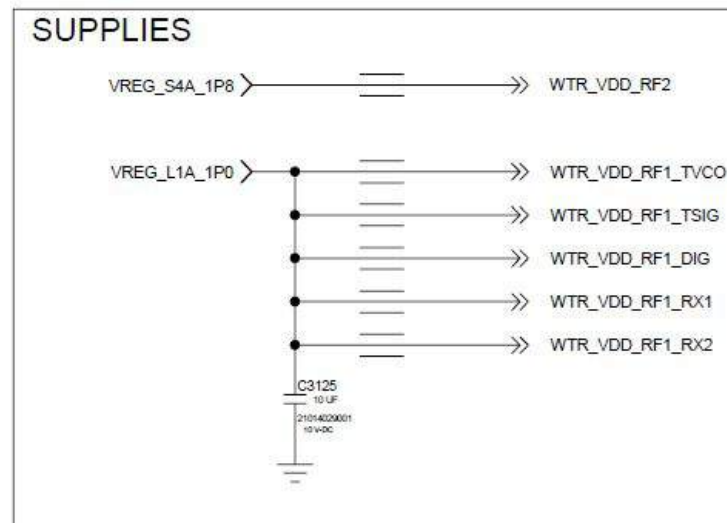
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Bounce RF Troubleshooting Guide

MODEL

Bounce

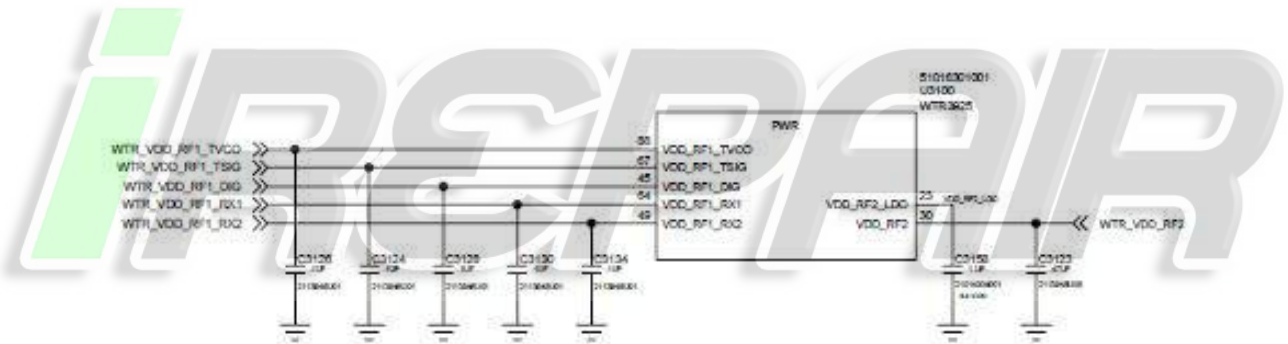
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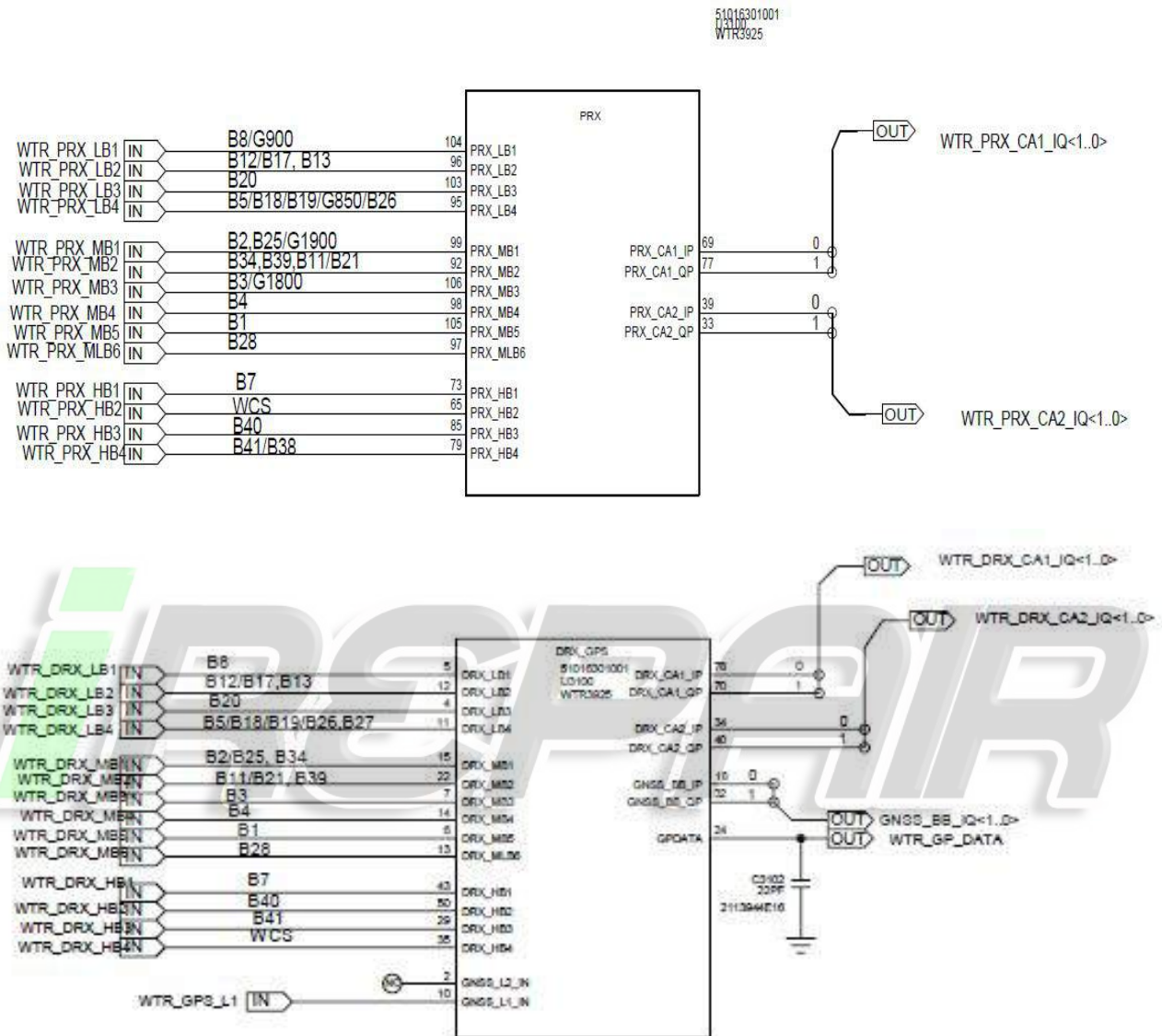
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MODEL

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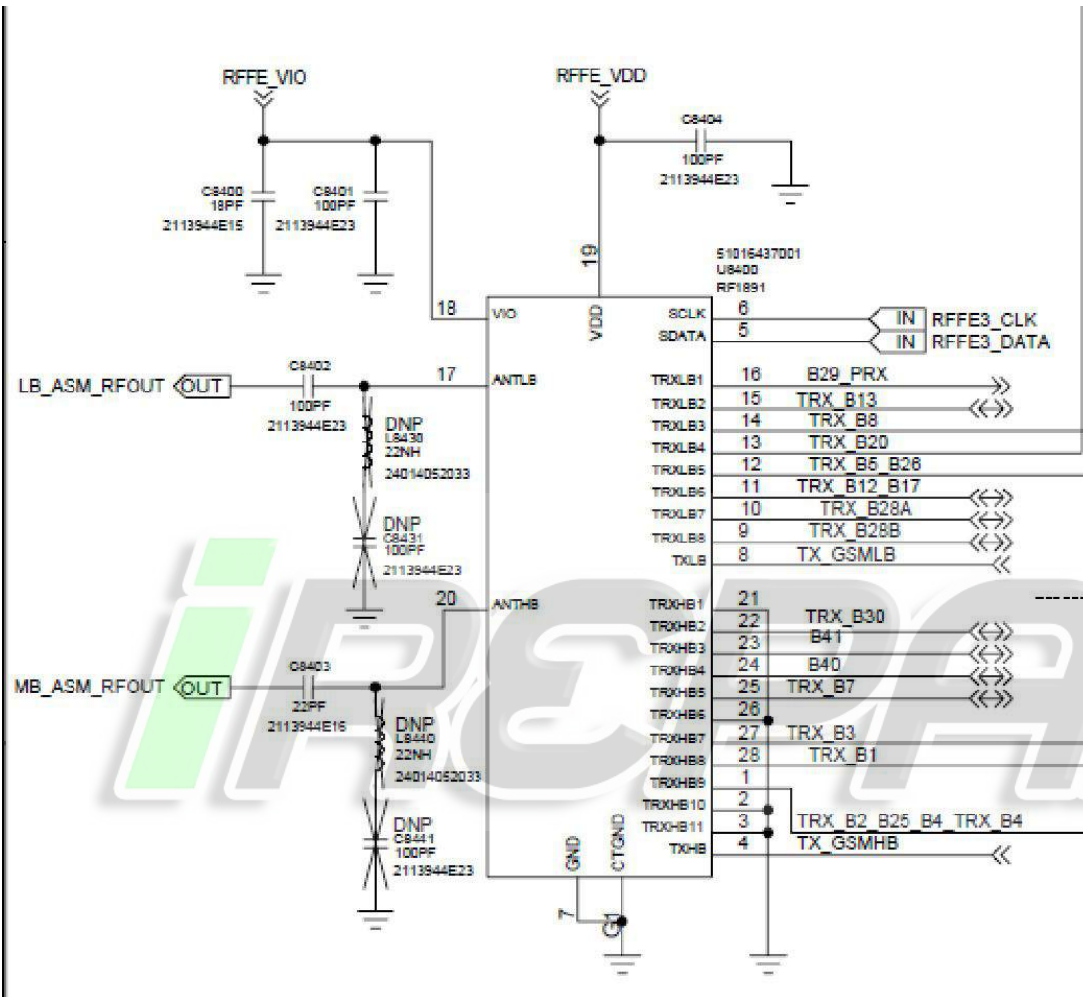
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MODEL

Bounce

# MAIN ASM (Antenna Switch Module) LB & HB – Schematic



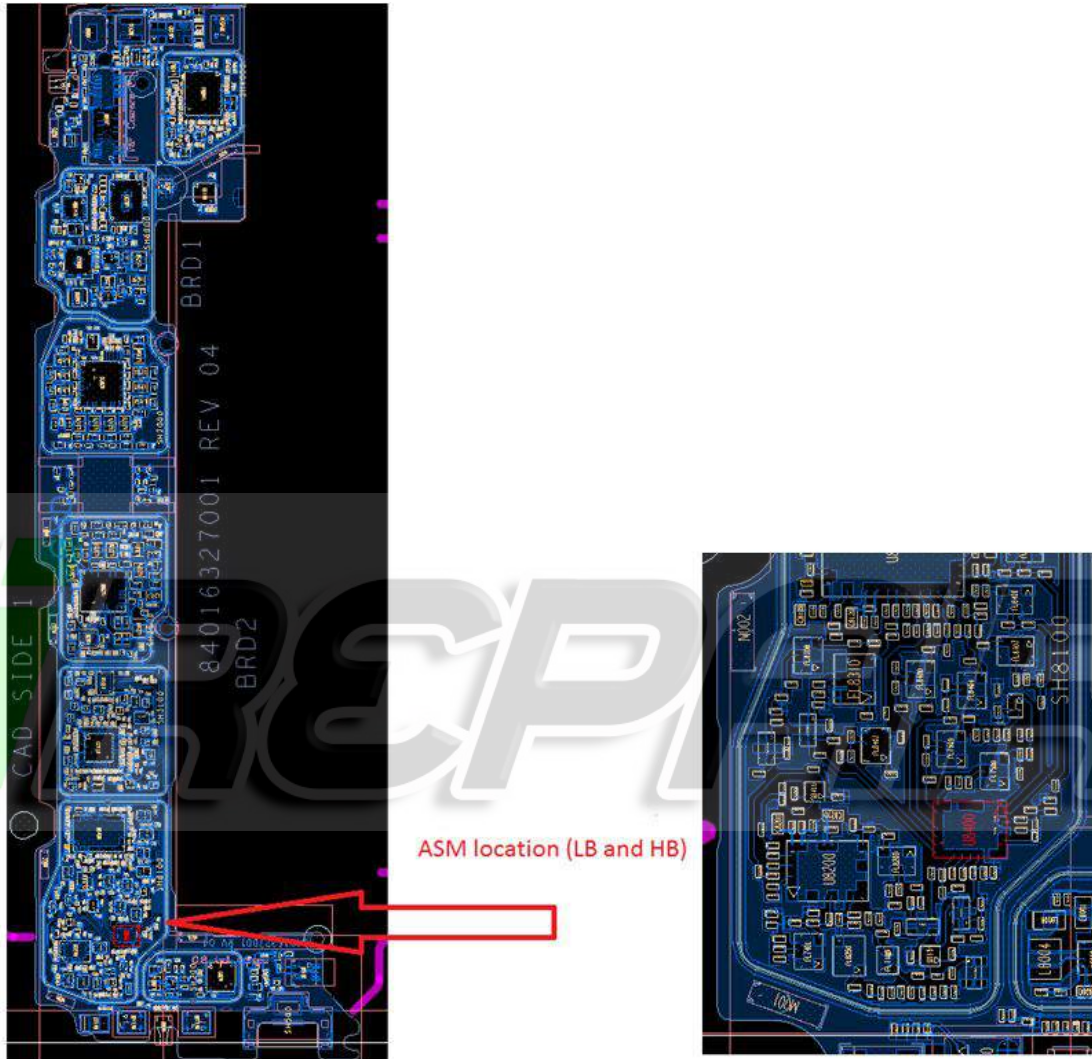
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|-----------------------------------------------------|----------------------------|
| <p>TITLE</p> <p>Bounce RF Troubleshooting Guide</p> | <p>MODEL</p> <p>Bounce</p> |
|-----------------------------------------------------|----------------------------|

## MAIN ASM (Antenna Switch Module)

### LB & HB - Layout



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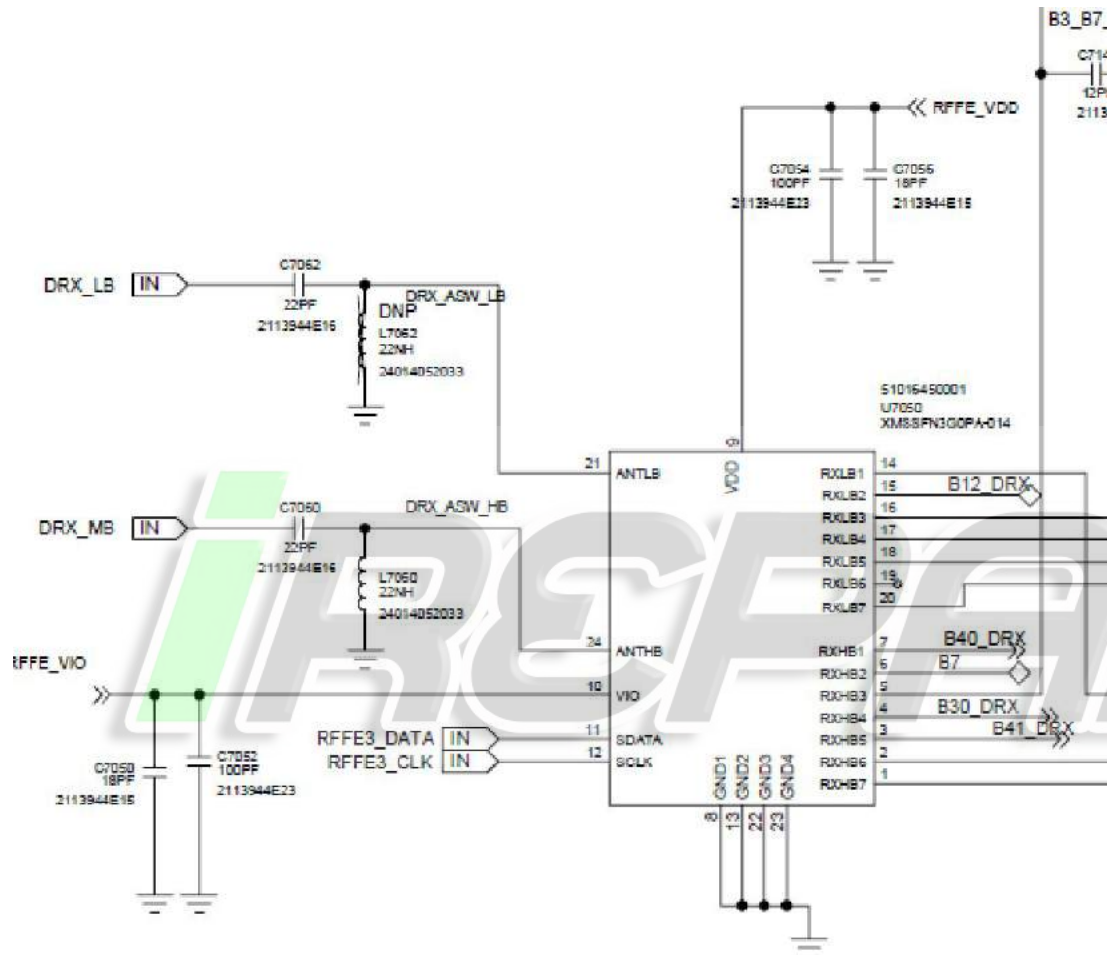
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MODEL

Bounce

Diversity ASM (Antenna Switch Module):

LB & HB – Schematic



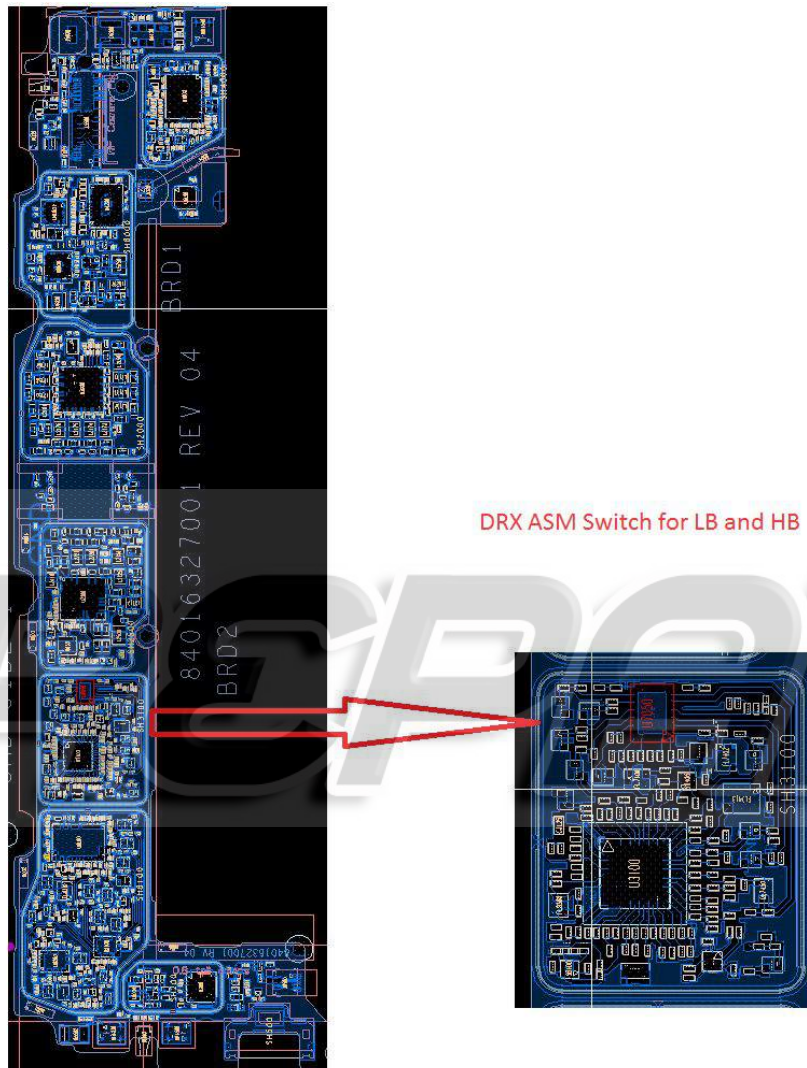
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|------------------------------------------------------------|-----------------------------------|
| <p>TITLE</p> <p><b>Bounce RF Troubleshooting Guide</b></p> | <p>MODEL</p> <p><b>Bounce</b></p> |
|------------------------------------------------------------|-----------------------------------|

Diversity ASM (Antenna Switch Module):

LB & MB – Layout



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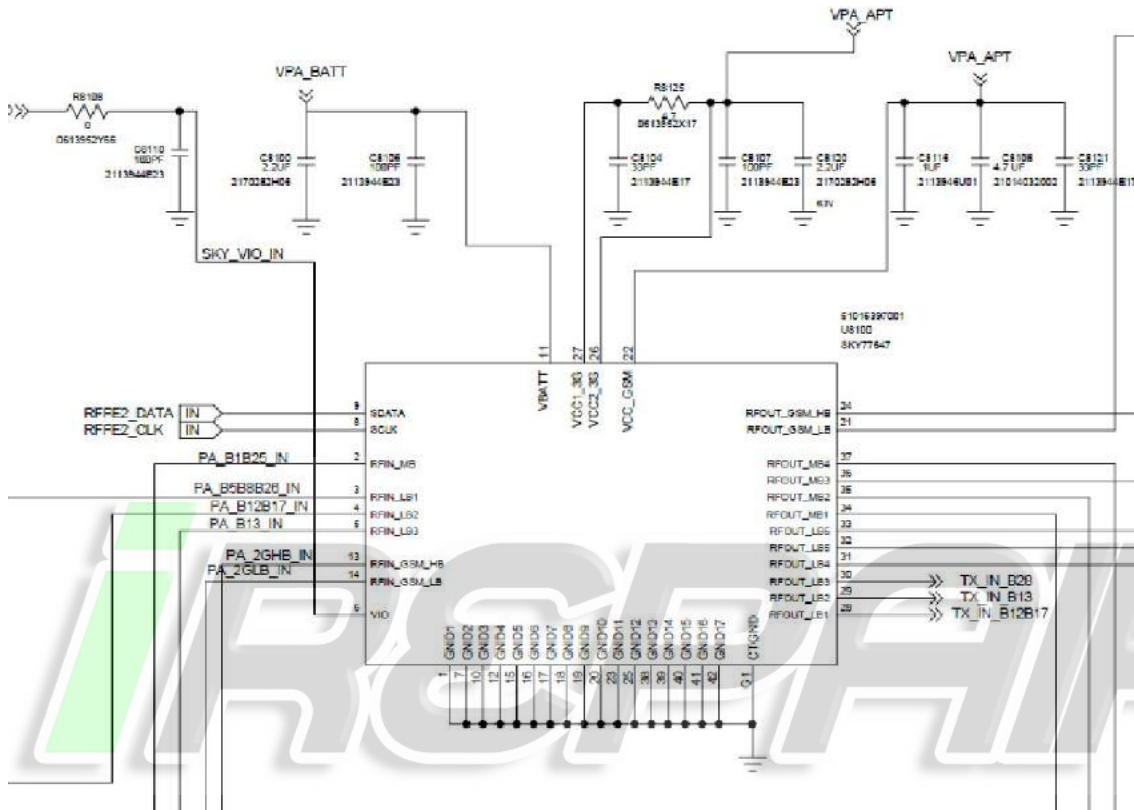
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MODEL

Bounce

## PA LB &amp; MB – Schematic

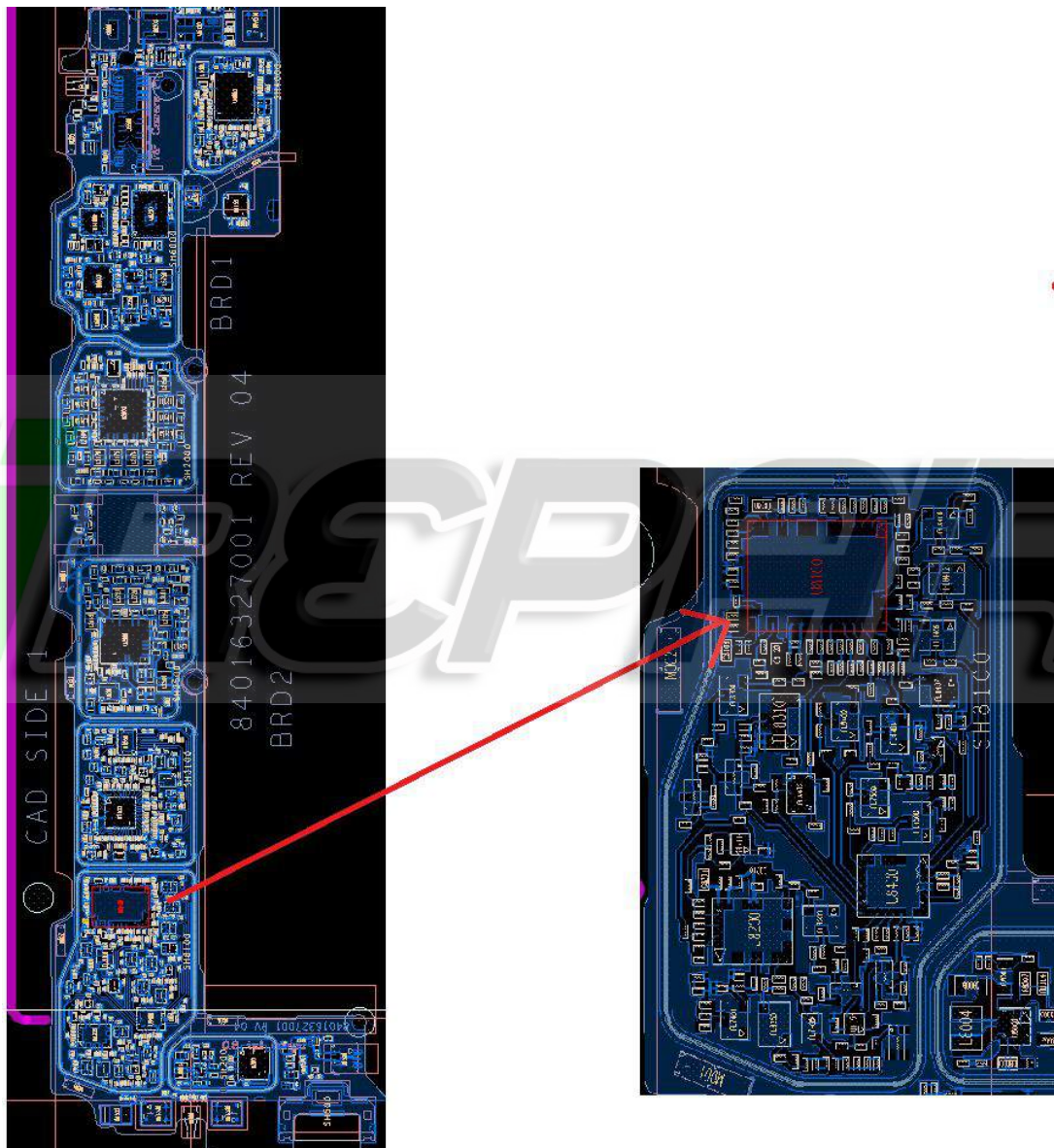


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|------------------------------------------------------------|-----------------------------------|
| <p>TITLE</p> <p><b>Bounce RF Troubleshooting Guide</b></p> | <p>MODEL</p> <p><b>Bounce</b></p> |
|------------------------------------------------------------|-----------------------------------|

## PA LB & MB – Layout



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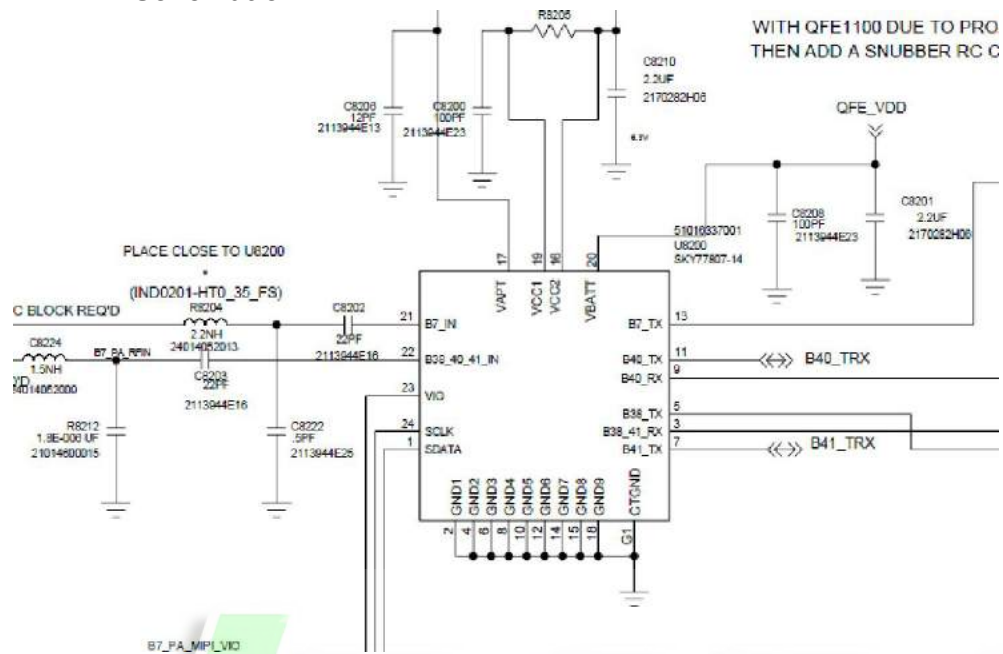
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MODEL

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## PA HB – Schematic



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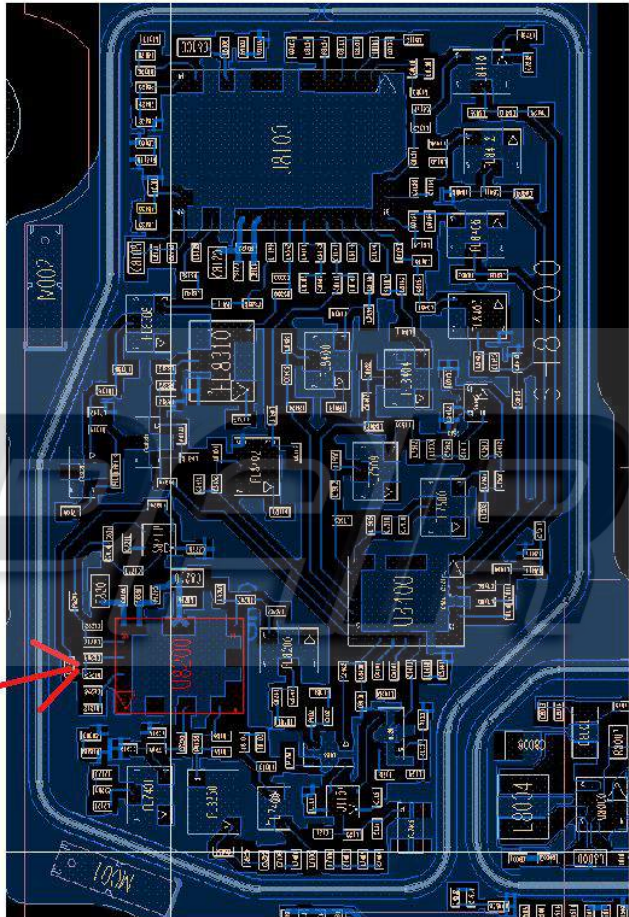
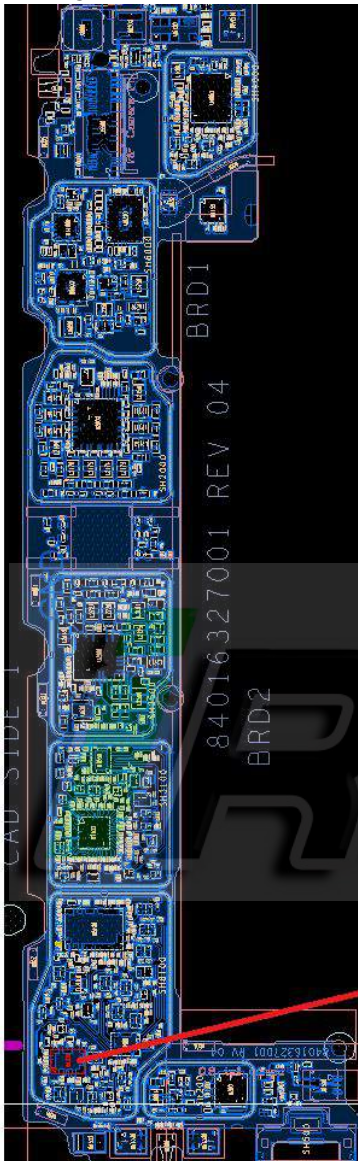
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MODEL

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## PA HB – Layout



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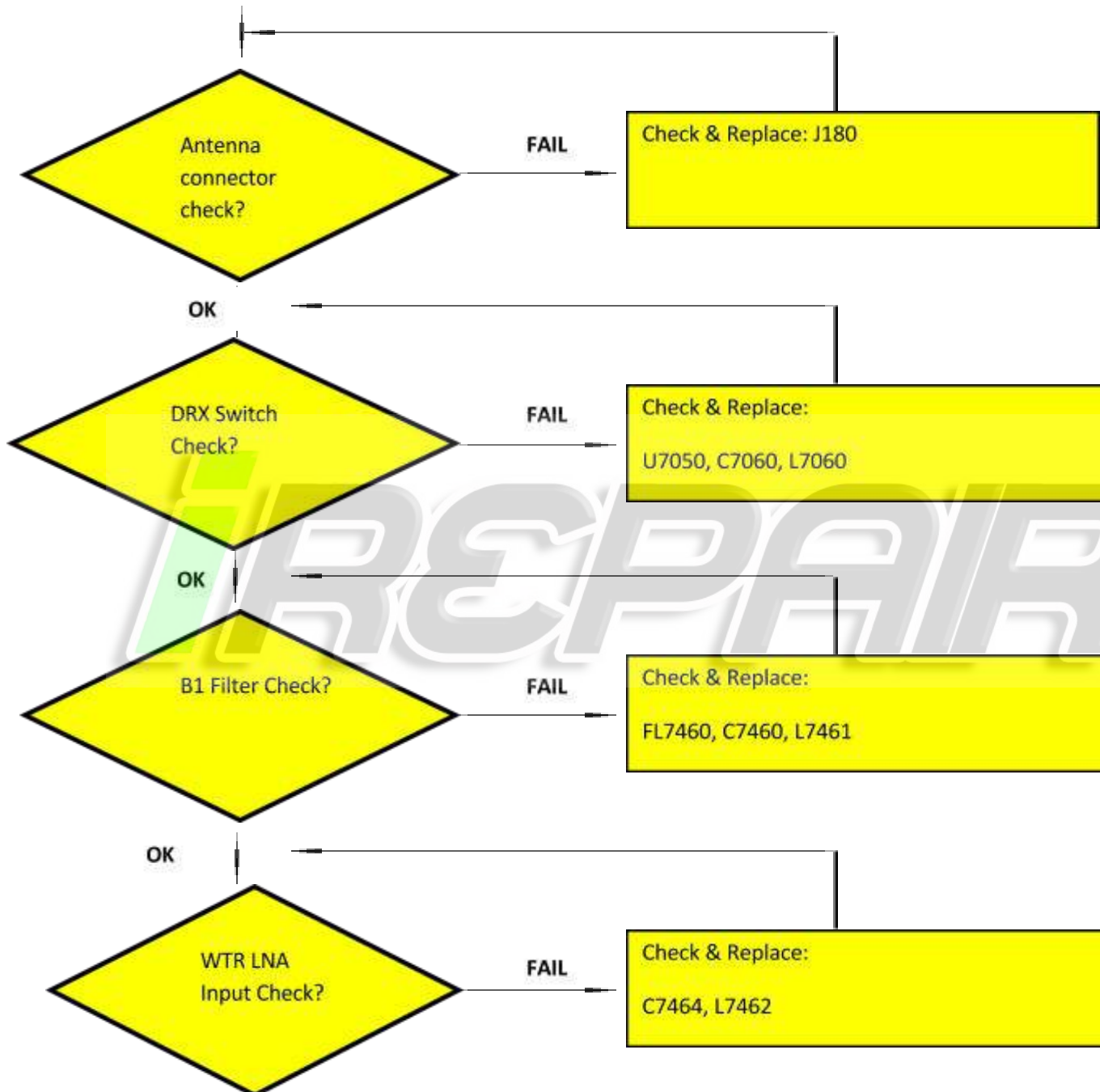
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Bounce RF Troubleshooting Guide

MODEL

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## B1 LTE &amp; WCDMA DRX



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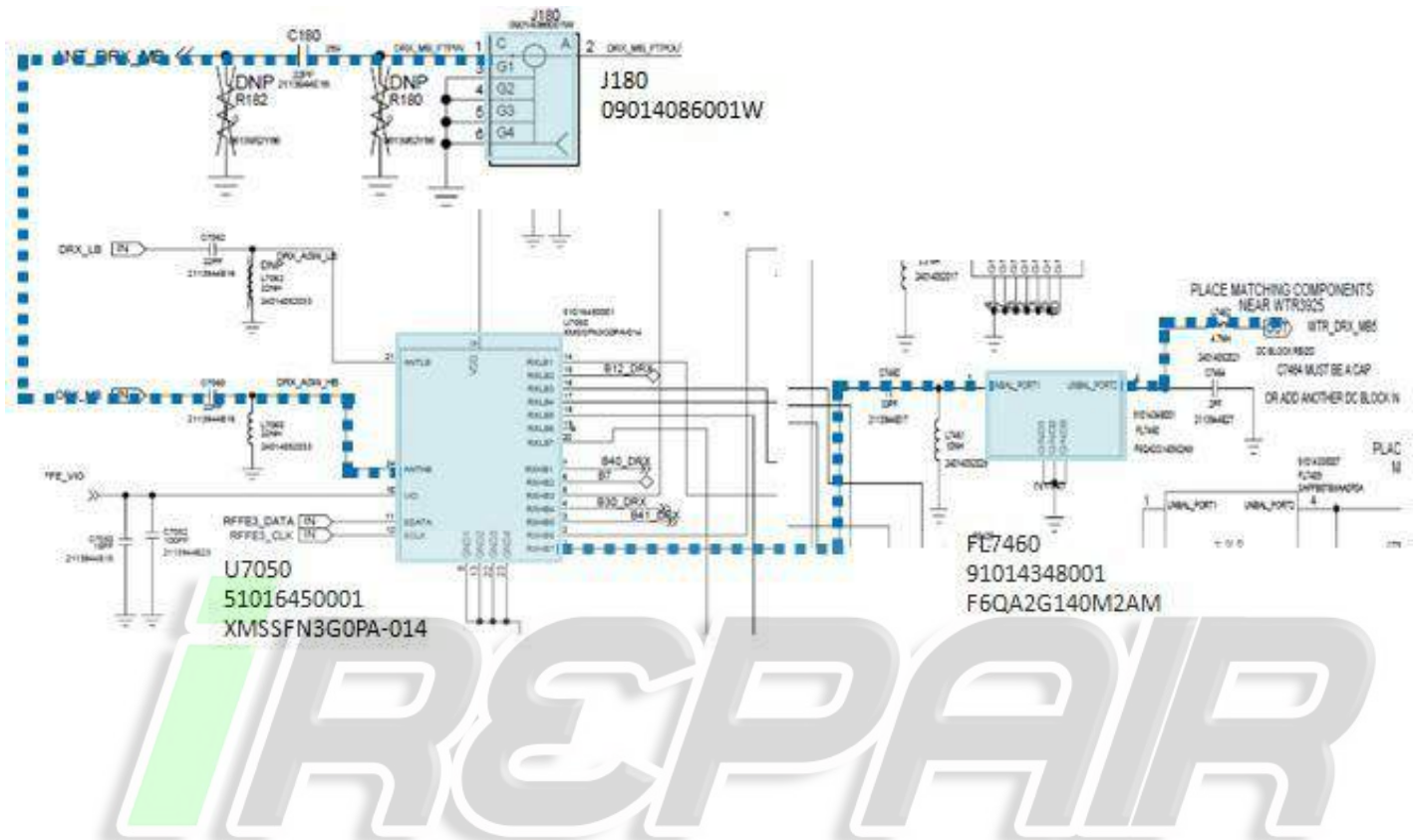
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Bounce RF Troubleshooting Guide

MODEL

Bounce



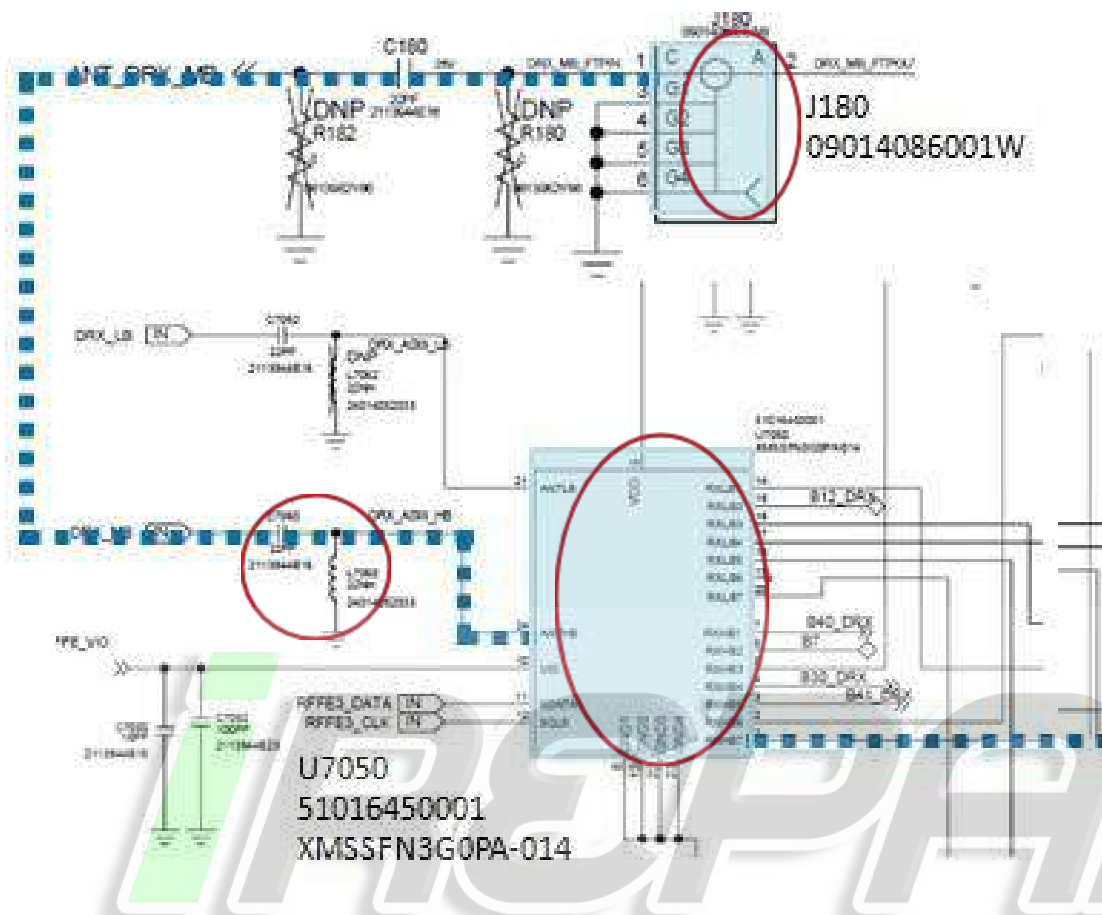
## PAGE 17 OF 204 PAGES

TITLE

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## MODEL

## Bounce



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## PAGE 18 OF 204 PAGES

## Bounce RF Troubleshooting Guide

## Bounce



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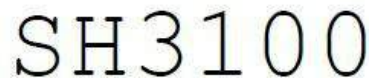
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## PAGE 19 OF 204 PAGES

## Bounce



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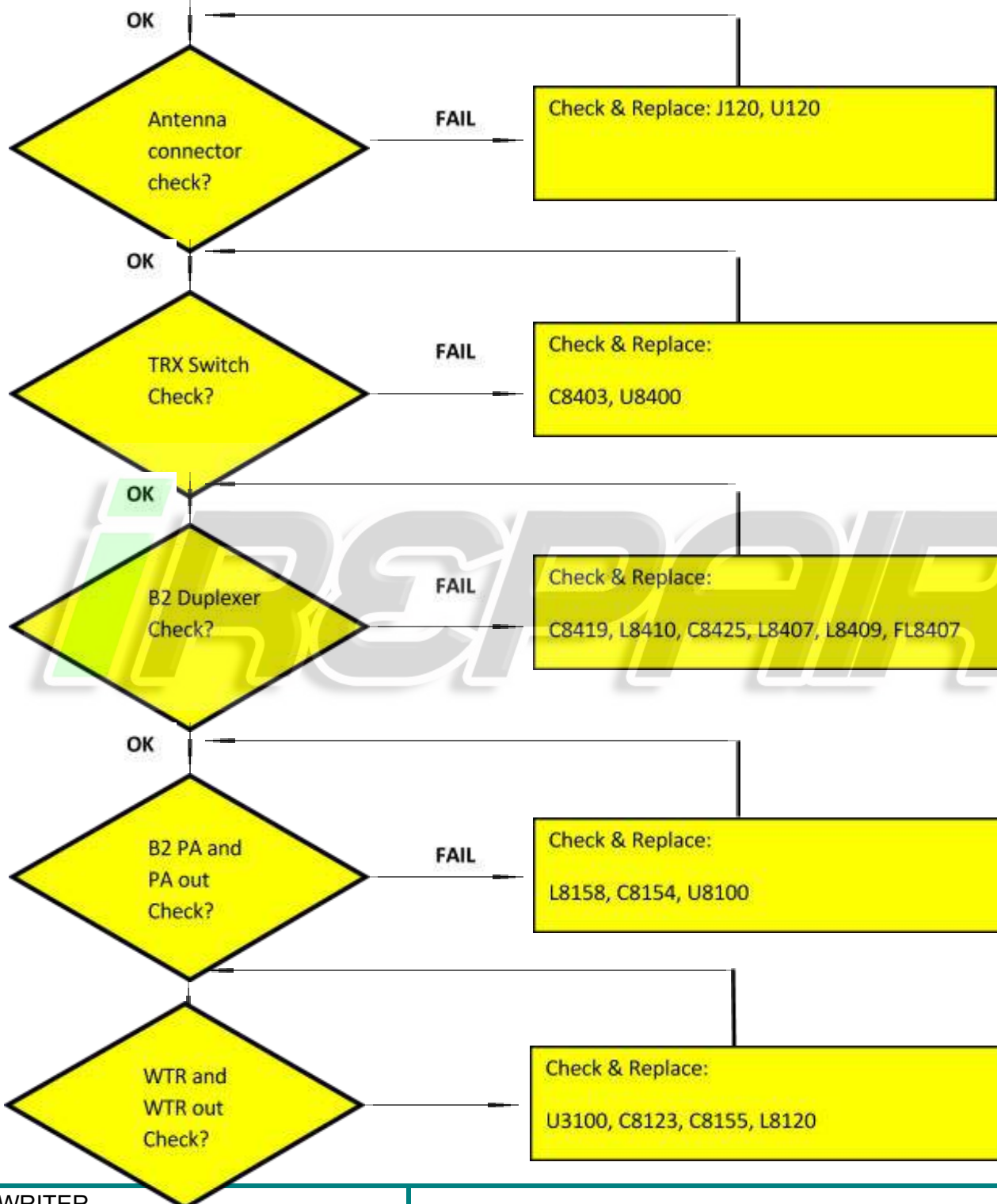
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MODEL

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## B2 WCDMA, LTE B2/25, CDMA 1900 – TX



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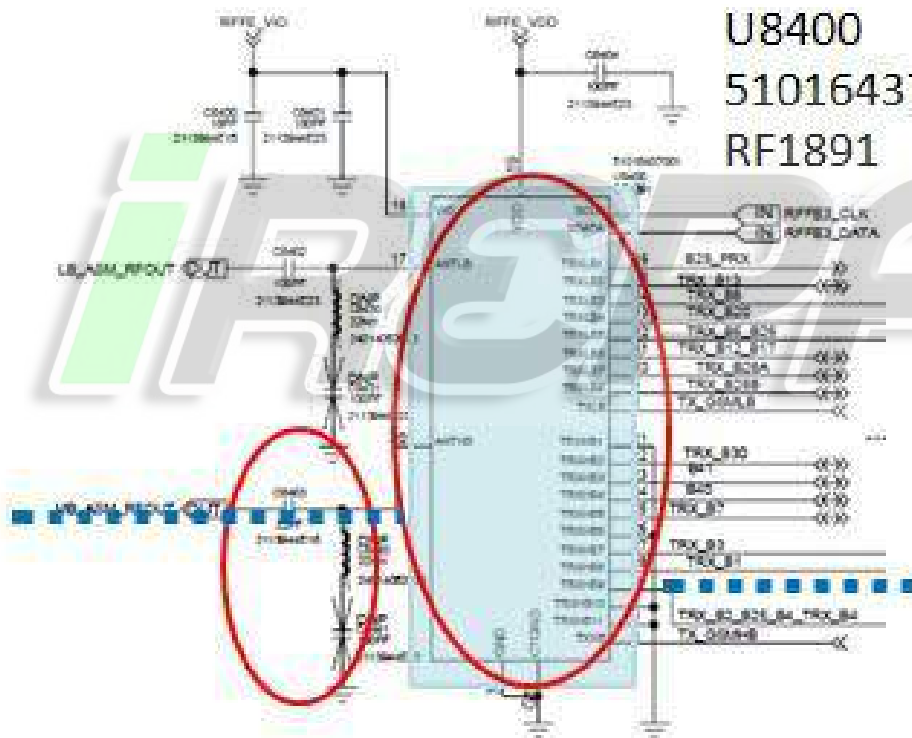
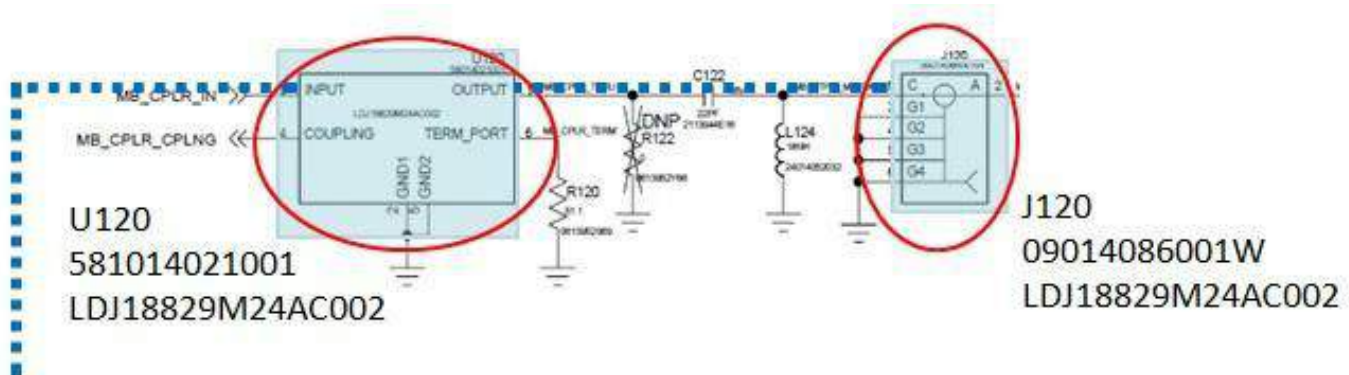
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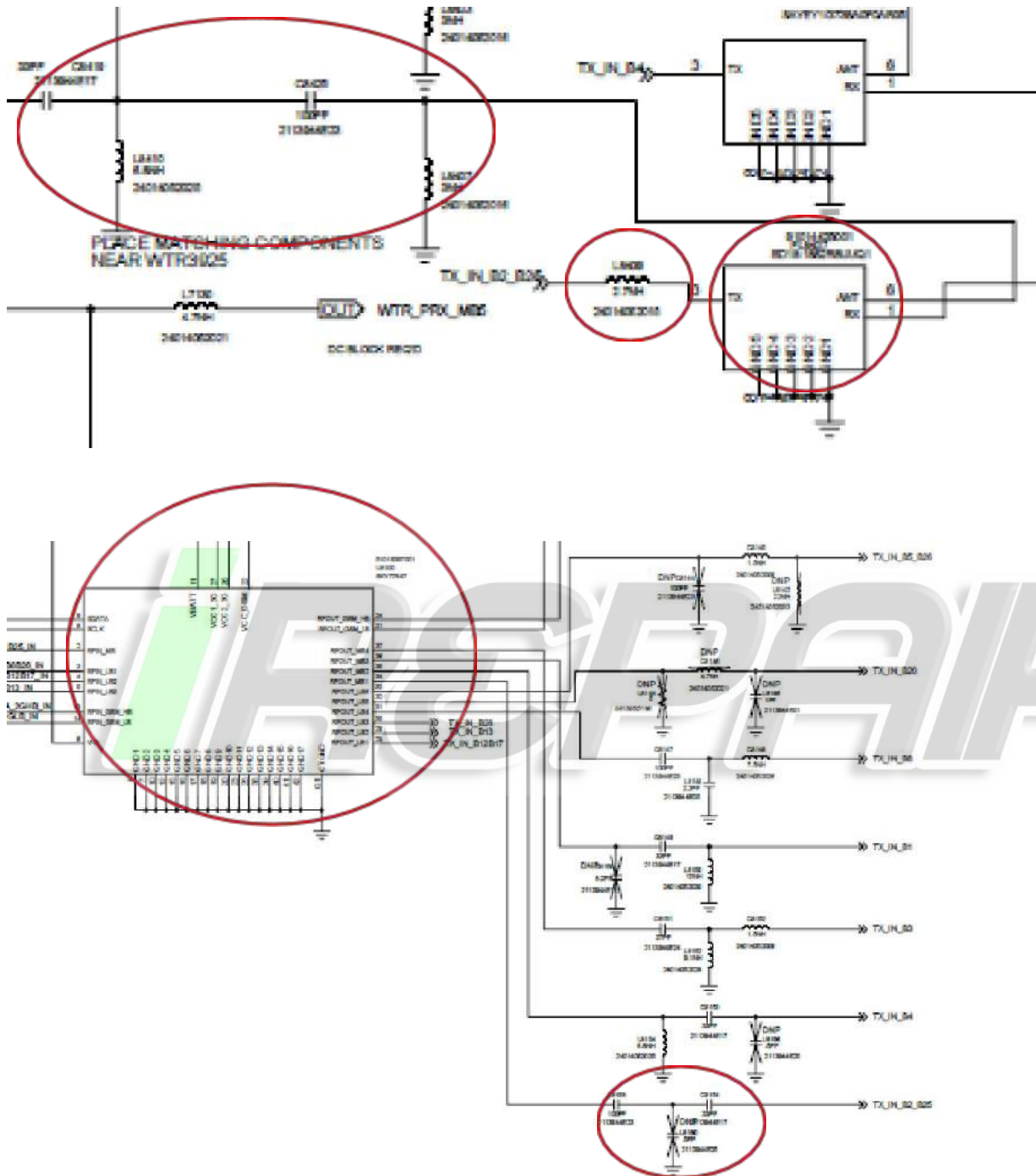
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TITLE

Bounce RF Troubleshooting Guide

MODEL

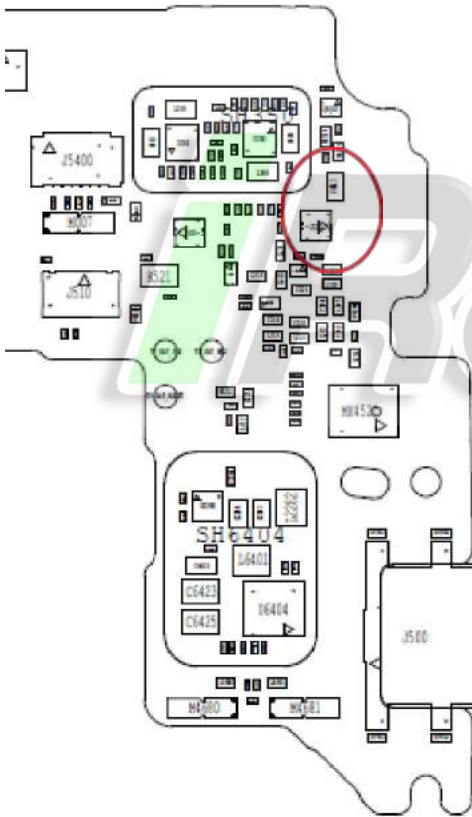
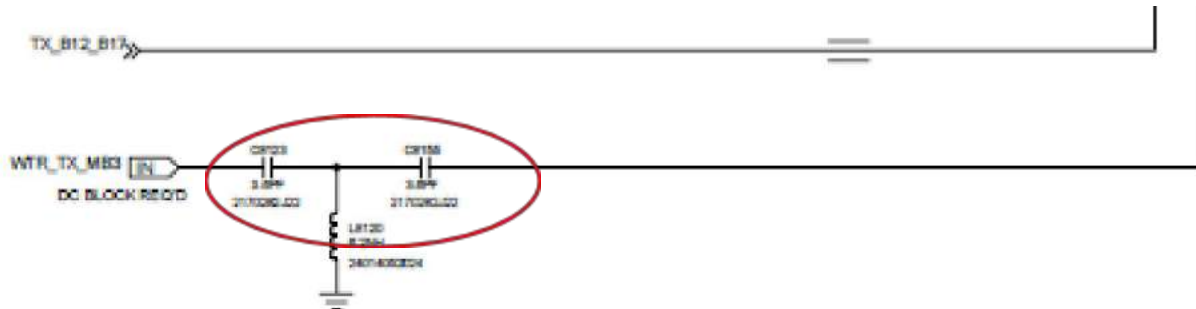
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| TITLE<br><b>Bounce RF Troubleshooting Guide</b> | MODEL<br><b>Bounce</b> |
|-------------------------------------------------|------------------------|



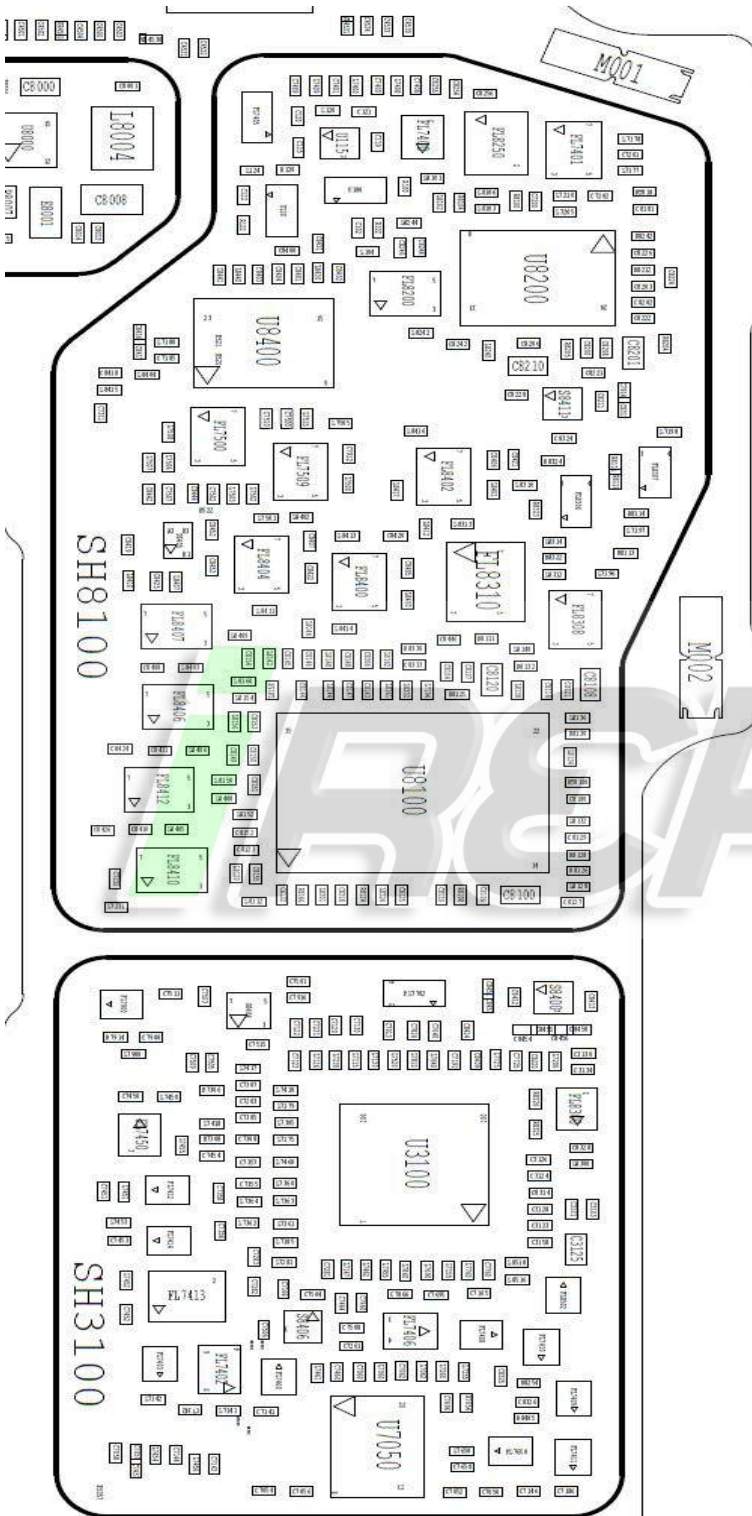
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TITLE

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MODEL

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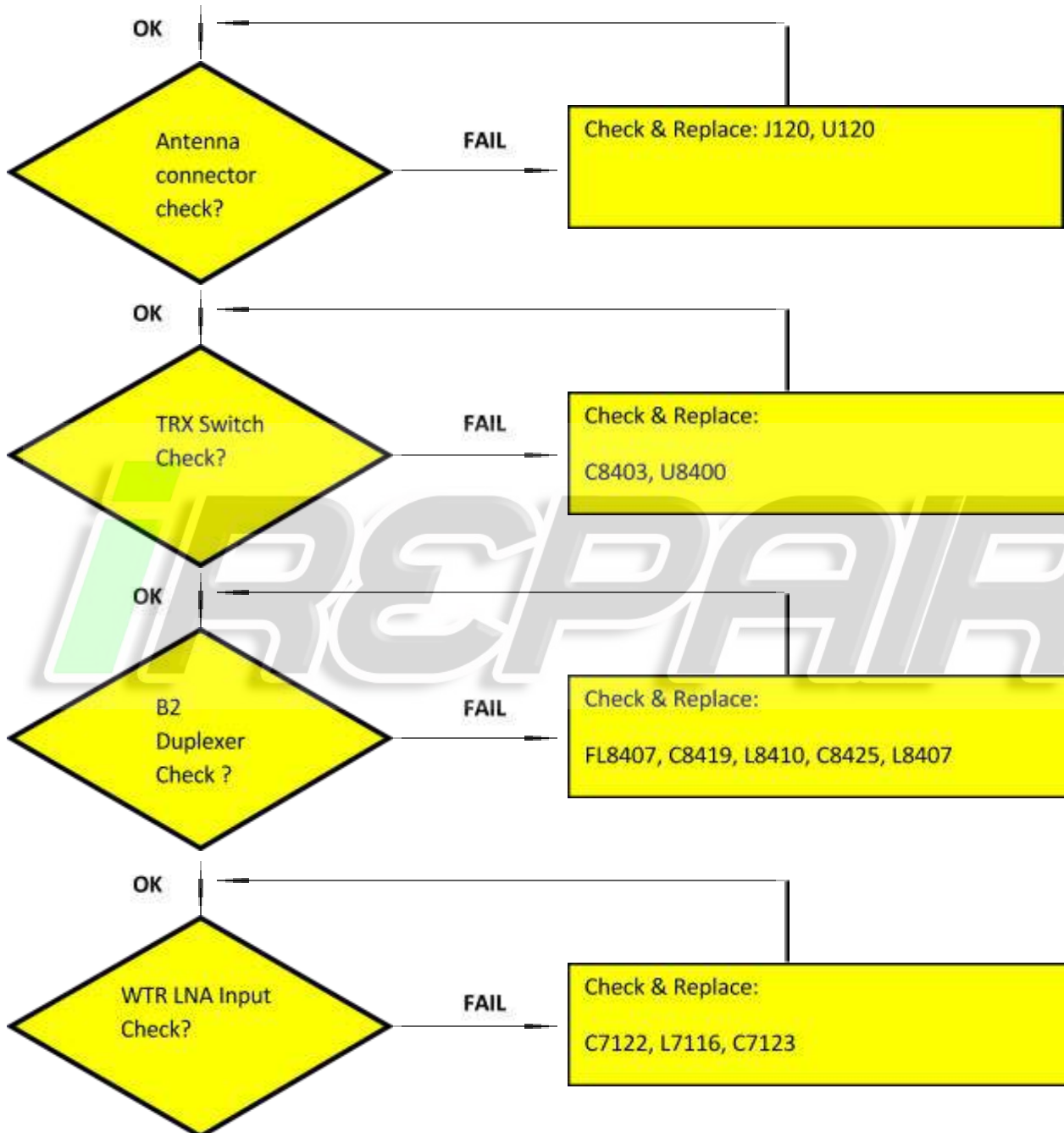
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Bounce RF Troubleshooting Guide

MODEL

Bounce

B2 WCDMA, LTE B2/25, CDMA 1900 - PRX

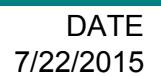


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## PAGE 26 OF 204 PAGES

## Bounce



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**PAGE 27 OF 204 PAGES**

## Bounce

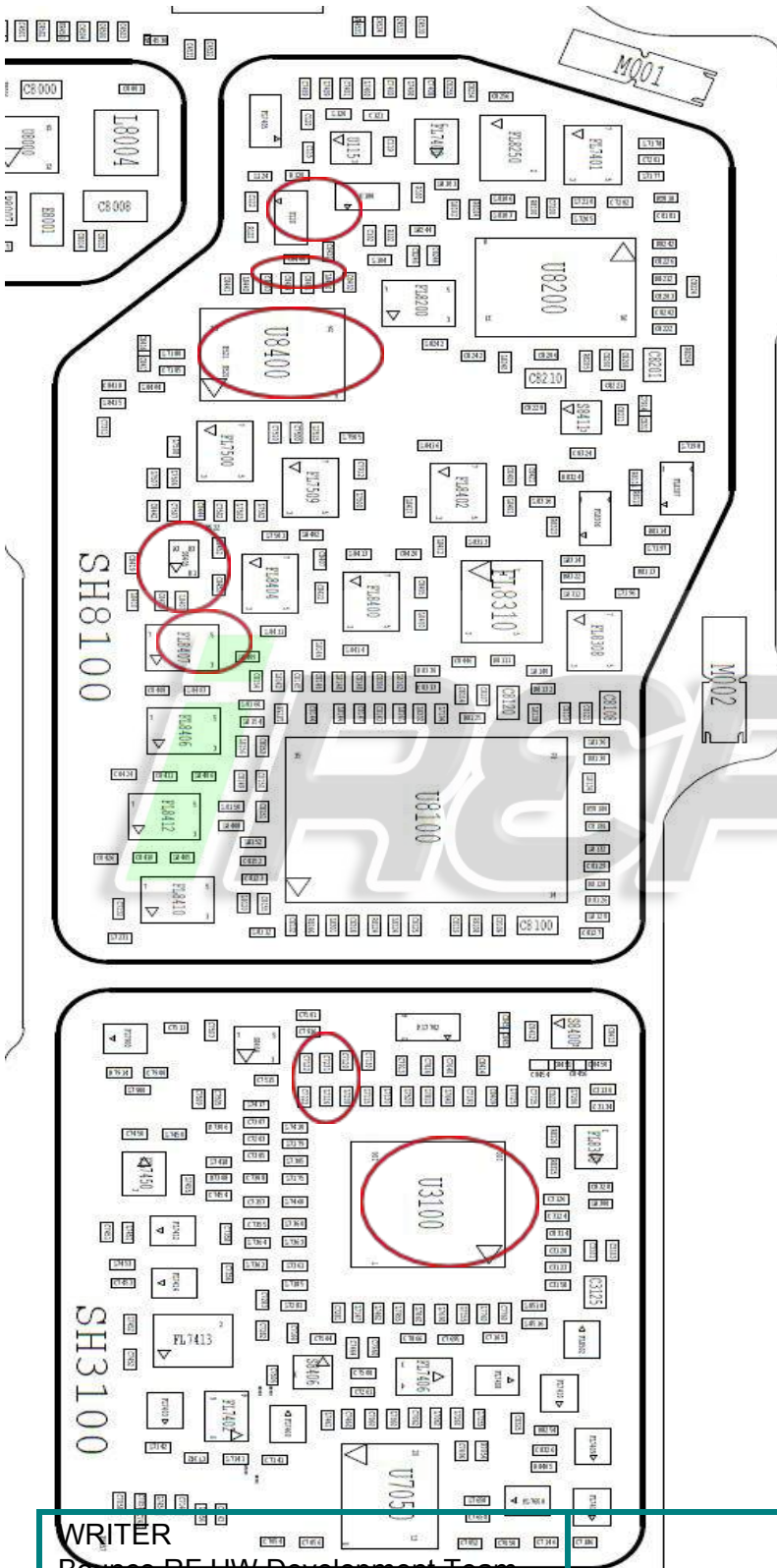


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MODEL

Bounce



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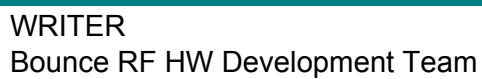
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## B2 WCDMA, LTE B2/25, CDMA 1900 – DRX



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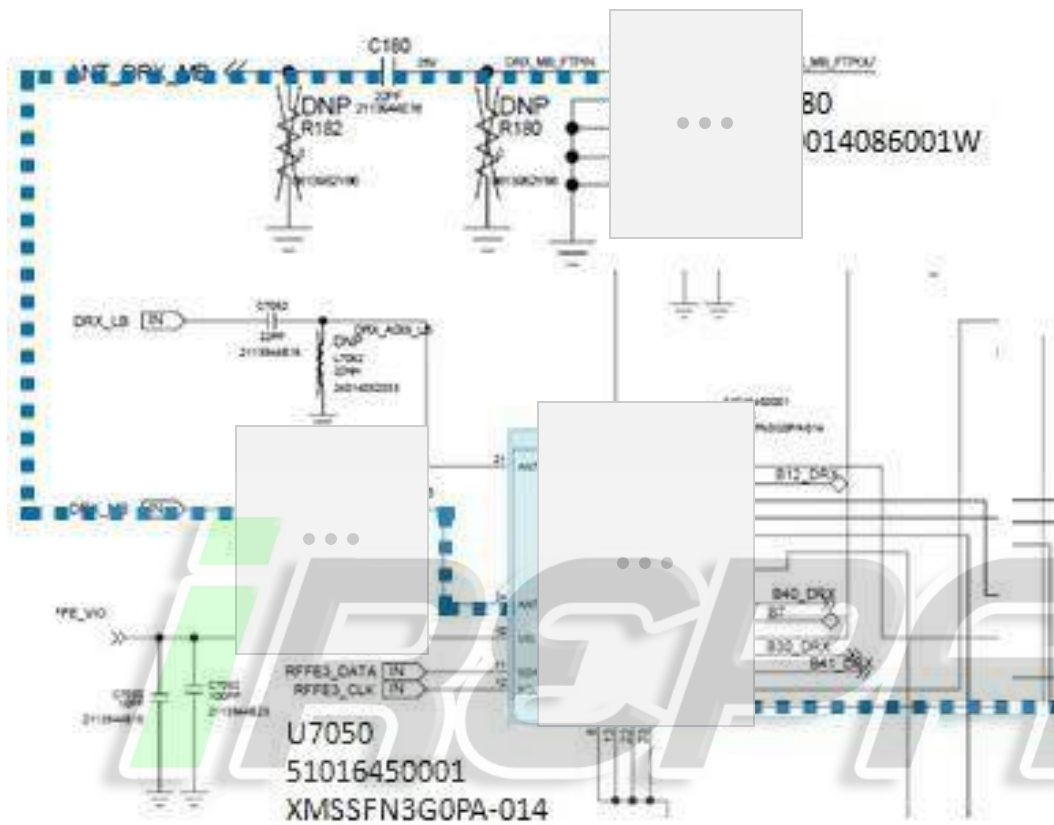
**PAGE 31 OF 204 PAGES**

## TITLE

## Bounce RF Troubleshooting Guide

## MODEL

## Bounce



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DATE  
7/22/2015

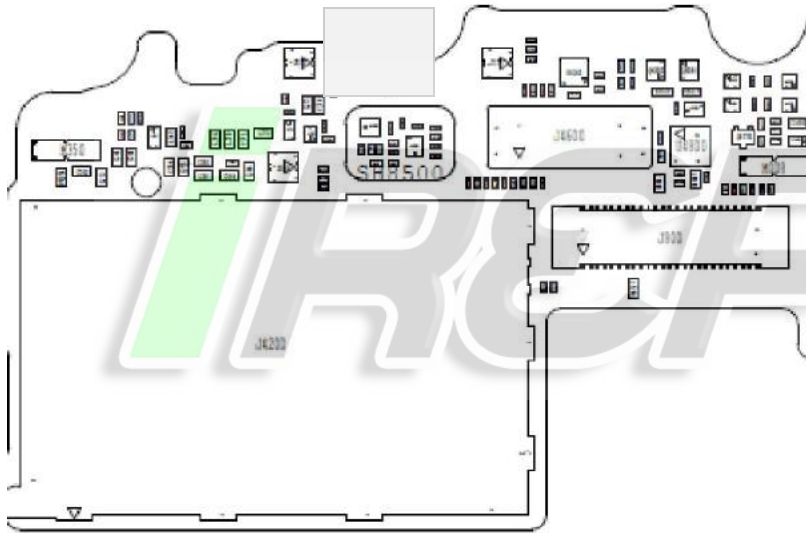
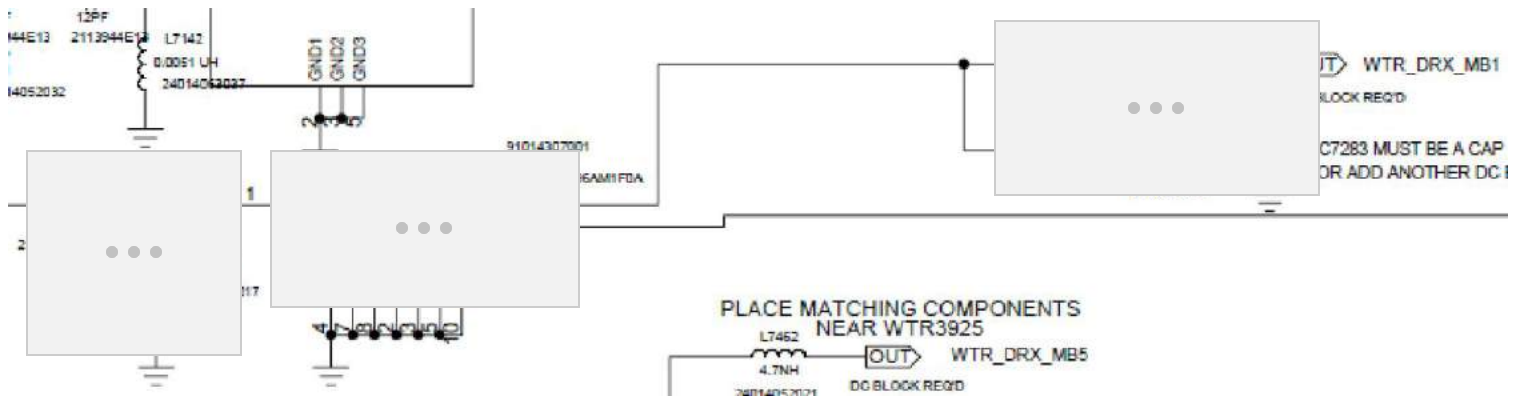
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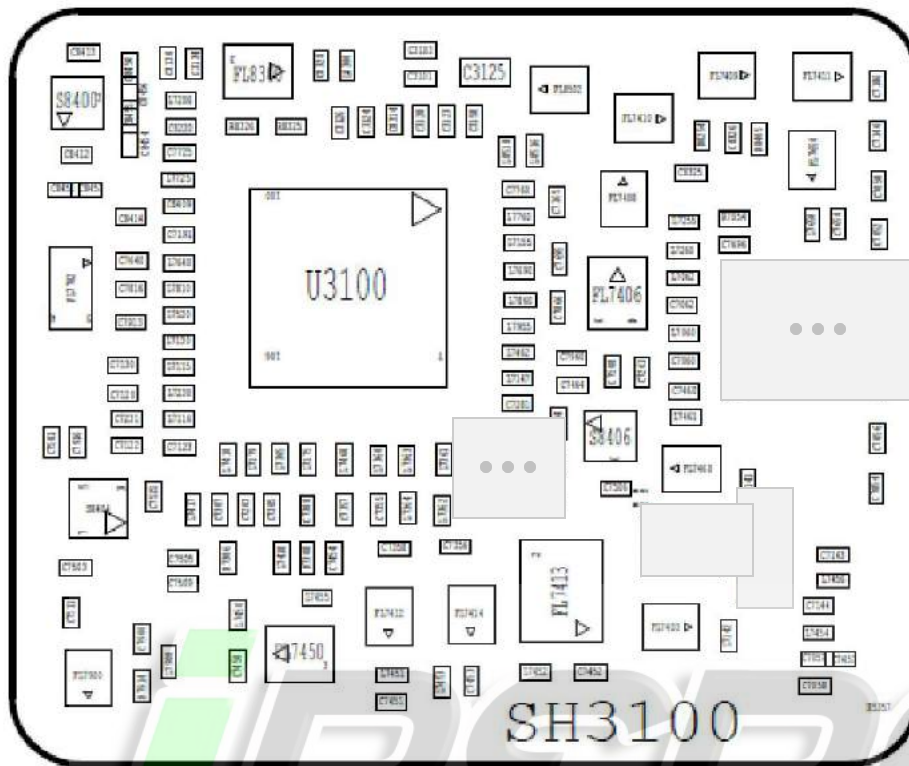
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TITLE

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MODEL

Bounce



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|                                 |        |
|---------------------------------|--------|
| TITLE                           | MODEL  |
| Bounce RF Troubleshooting Guide | Bounce |

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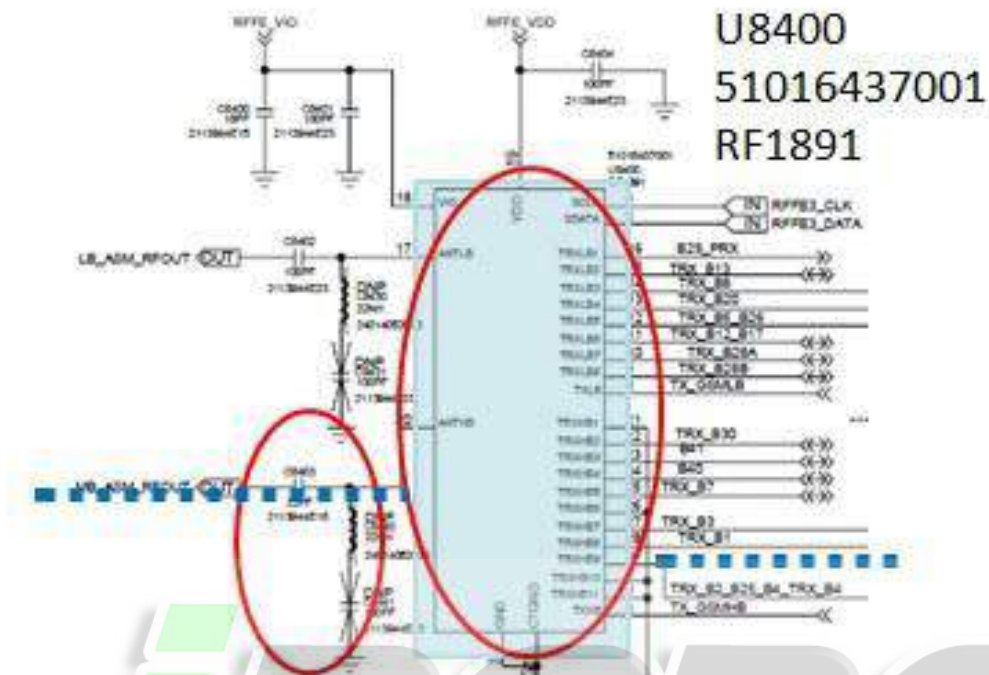


## PAGE 35 OF 204 PAGES

## Bounce

## Bounce RF Troubleshooting Guide

## Bounce



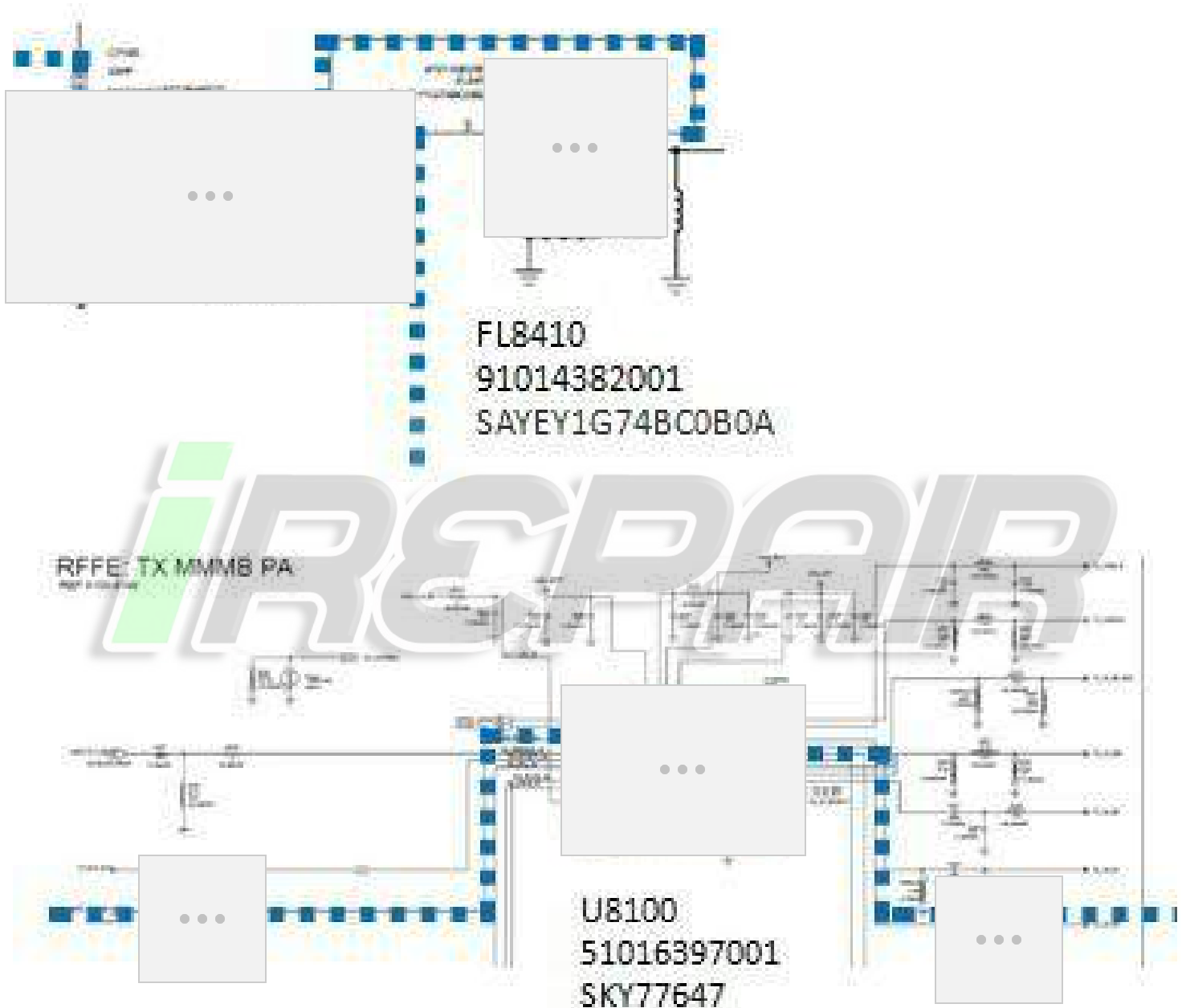
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MODEL

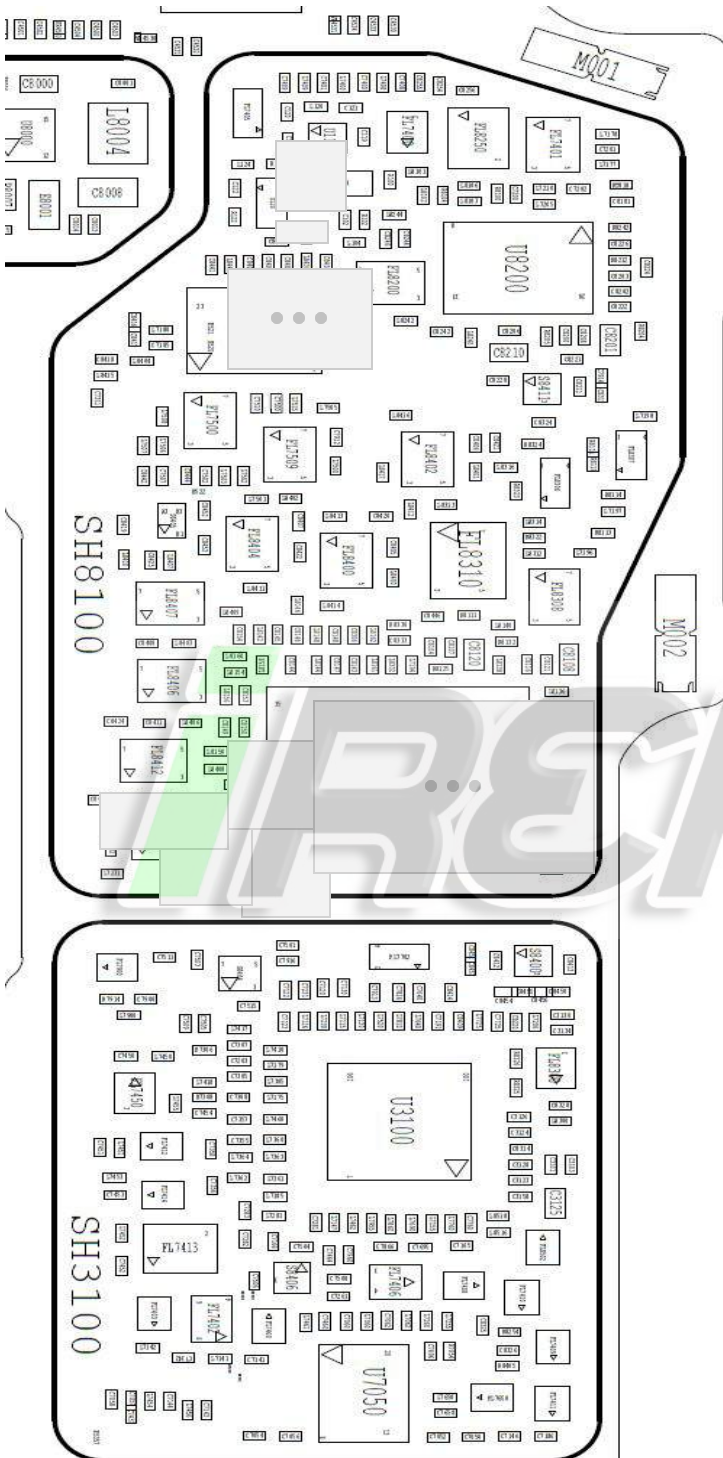
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|                                                            |                                   |
|------------------------------------------------------------|-----------------------------------|
| <p>TITLE</p> <p><b>Bounce RF Troubleshooting Guide</b></p> | <p>MODEL</p> <p><b>Bounce</b></p> |
|------------------------------------------------------------|-----------------------------------|



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TITLE

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MODEL

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LTE B3 PRX



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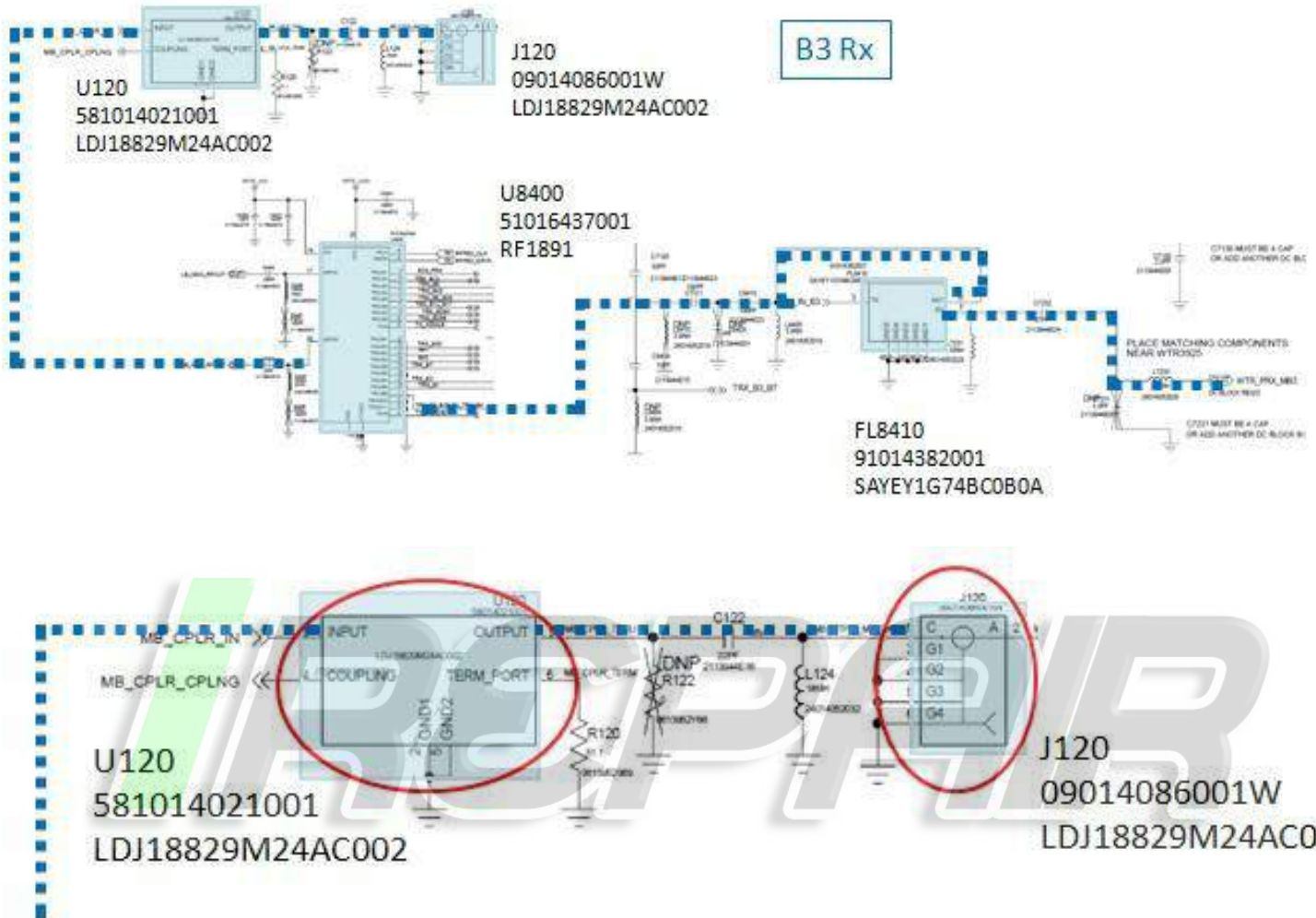
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MODEL

Bounce



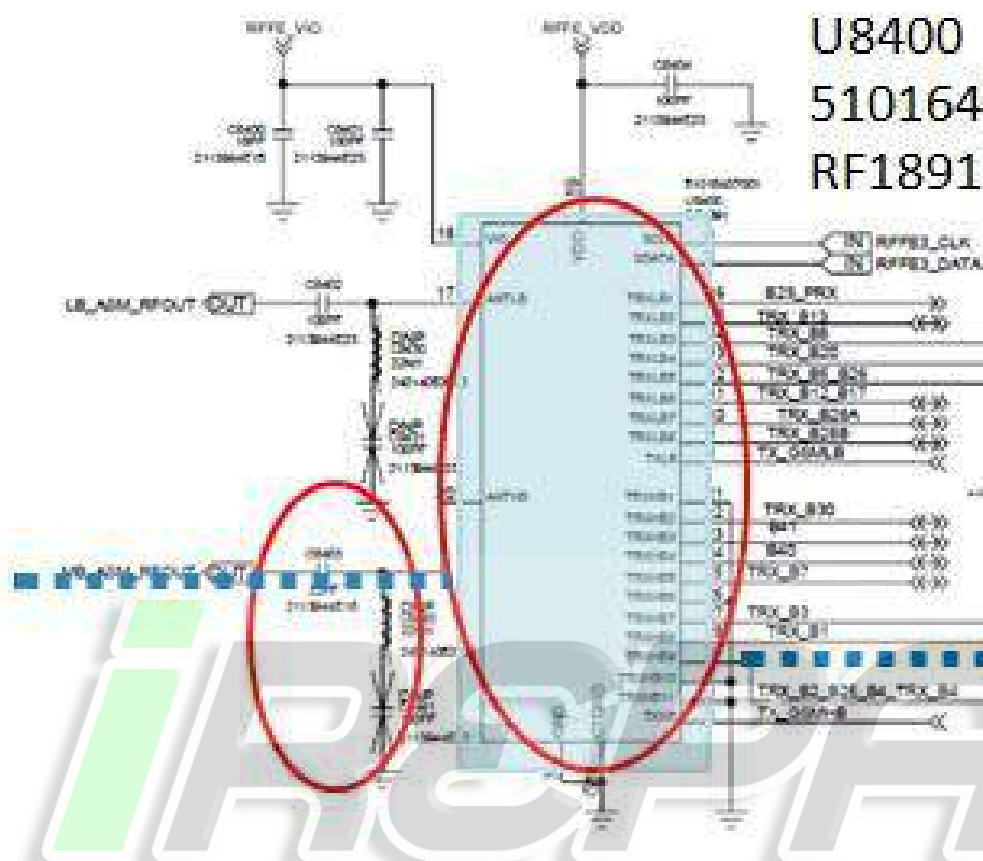
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## Bounce



DATE  
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| TITLE                           | MODEL  |
|---------------------------------|--------|
| Bounce RF Troubleshooting Guide | Bounce |



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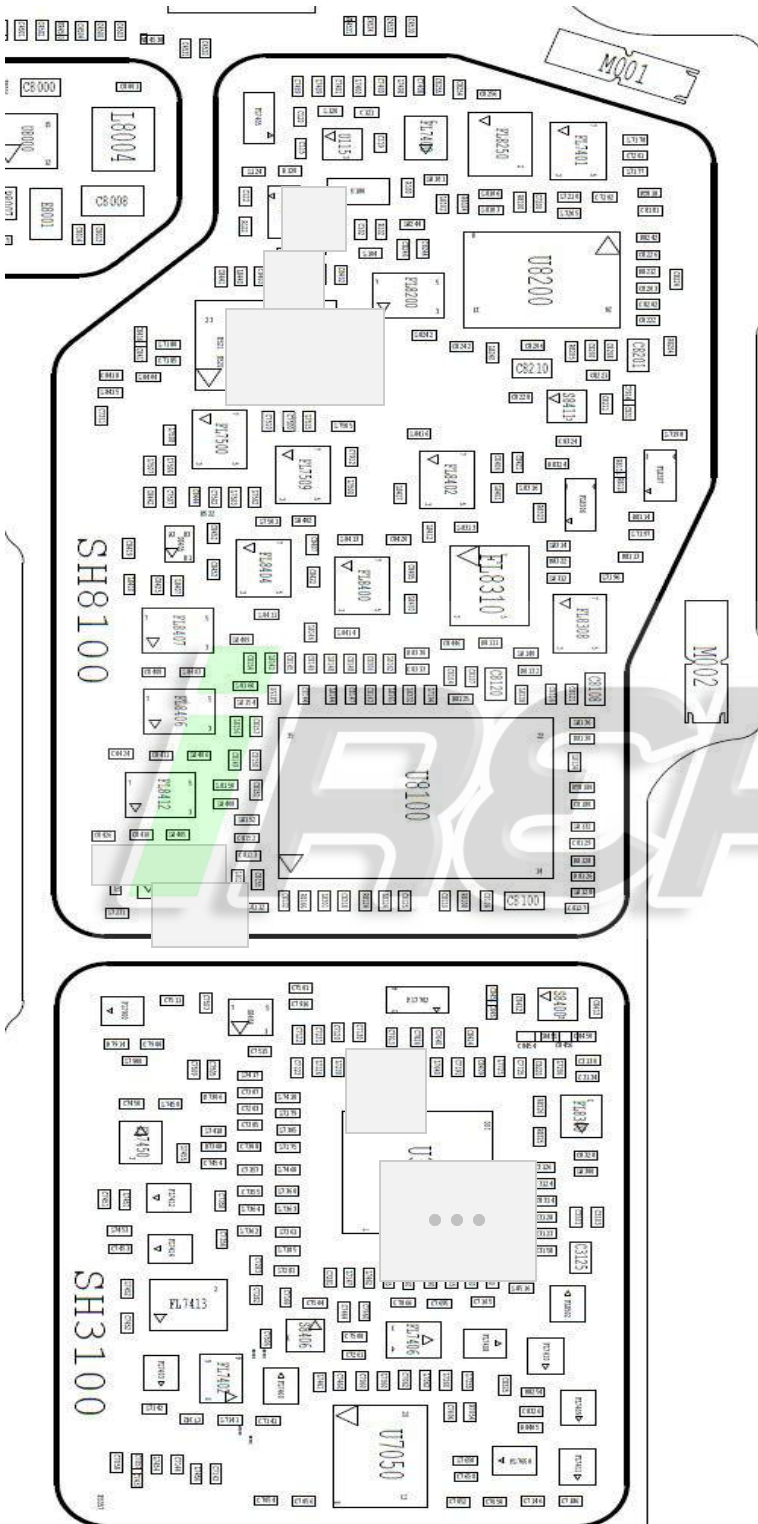
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TITLE

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MODEL

Bounce



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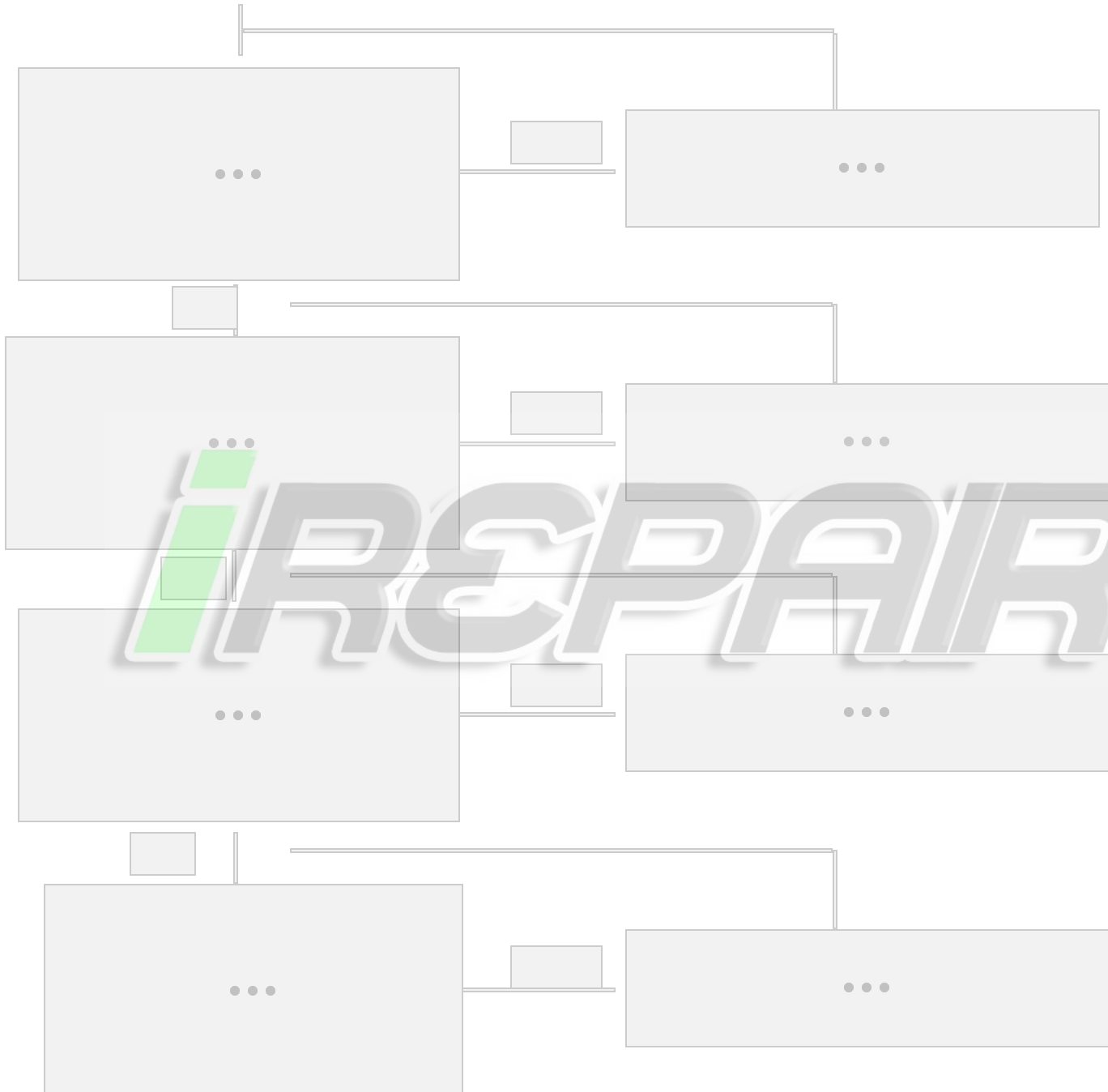
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MODEL

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## LTE B3 DRX



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TITLE

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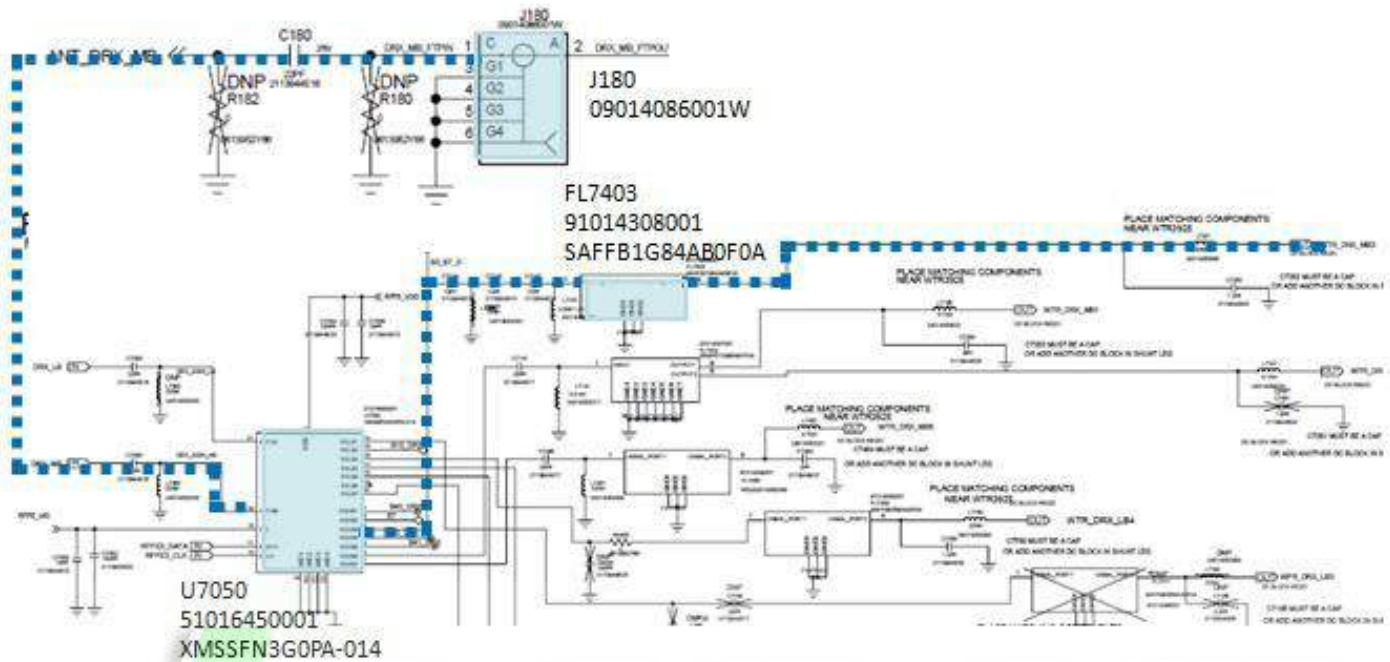
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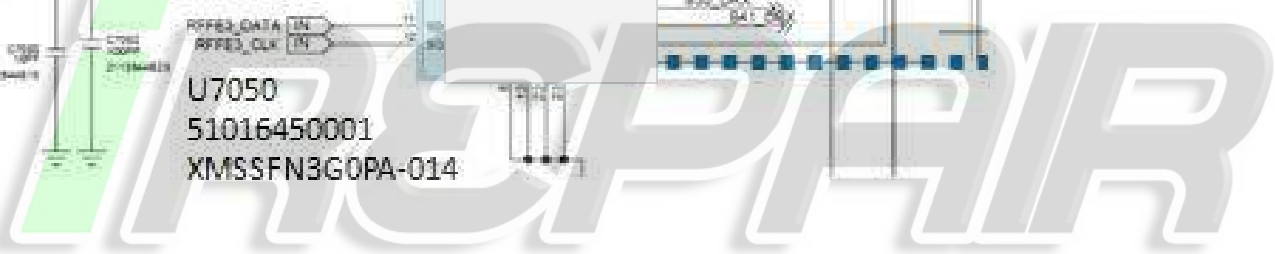
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DATE  
7/22/2015



## PAGE 47 OF 204 PAGES

## Bounce



DATE  
7/22/2015

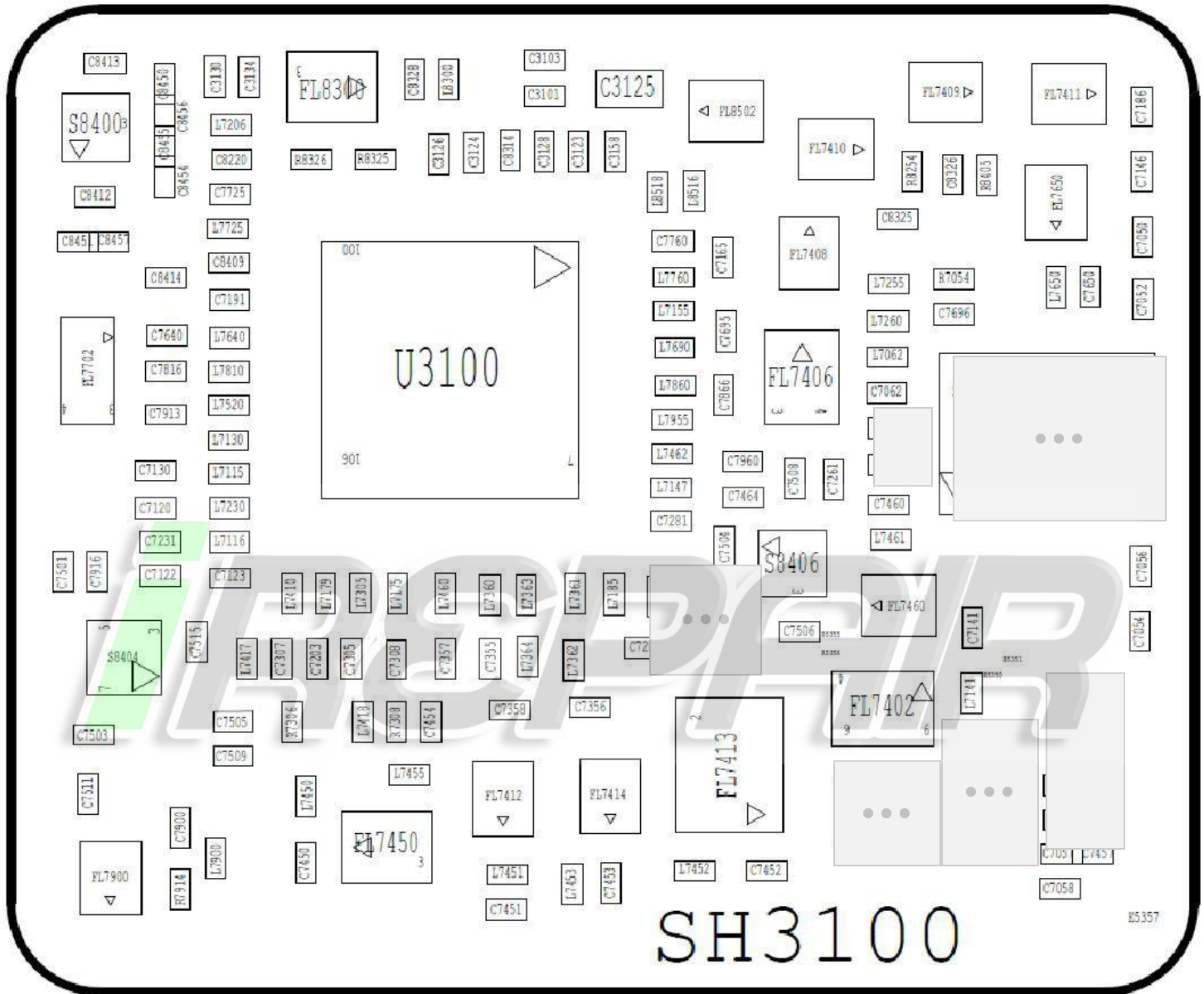
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MODEL

Bounce



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|                                                            |                                   |
|------------------------------------------------------------|-----------------------------------|
| <p>TITLE</p> <p><b>Bounce RF Troubleshooting Guide</b></p> | <p>MODEL</p> <p><b>Bounce</b></p> |
|------------------------------------------------------------|-----------------------------------|

## B4 TX - LTE & WCDMA



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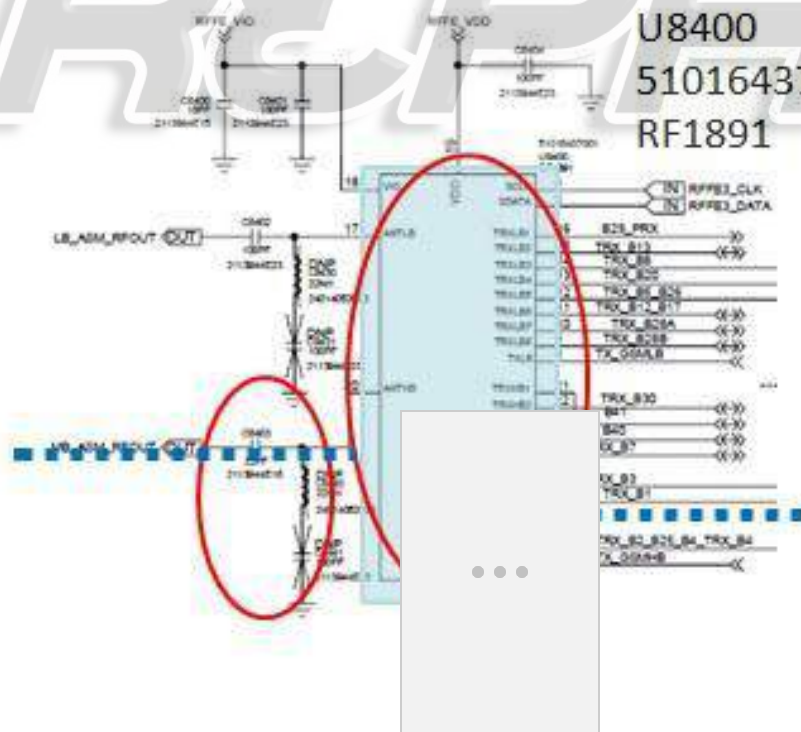
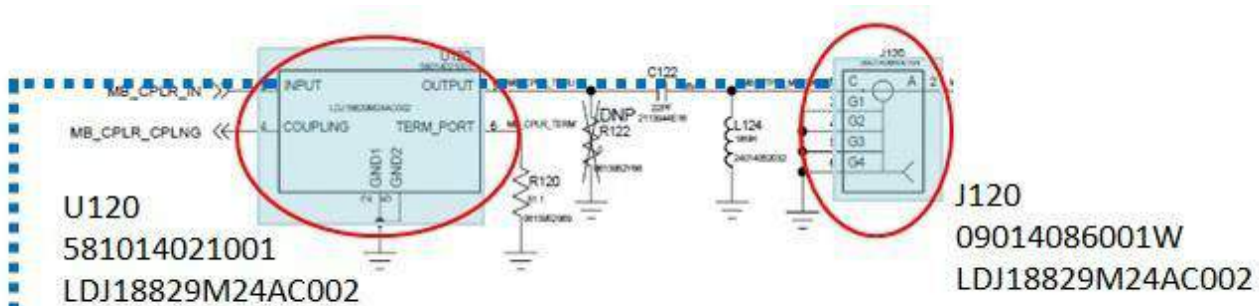
PAGE 50 OF 204 PAGES

TITLE

## Bounce RF Troubleshooting Guide

## MODEL

## Bounce



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## PAGE 51 OF 204 PAGES

## Bounce

TITLE

Bounce RF Troubleshooting Guide

MODEL

Bounce



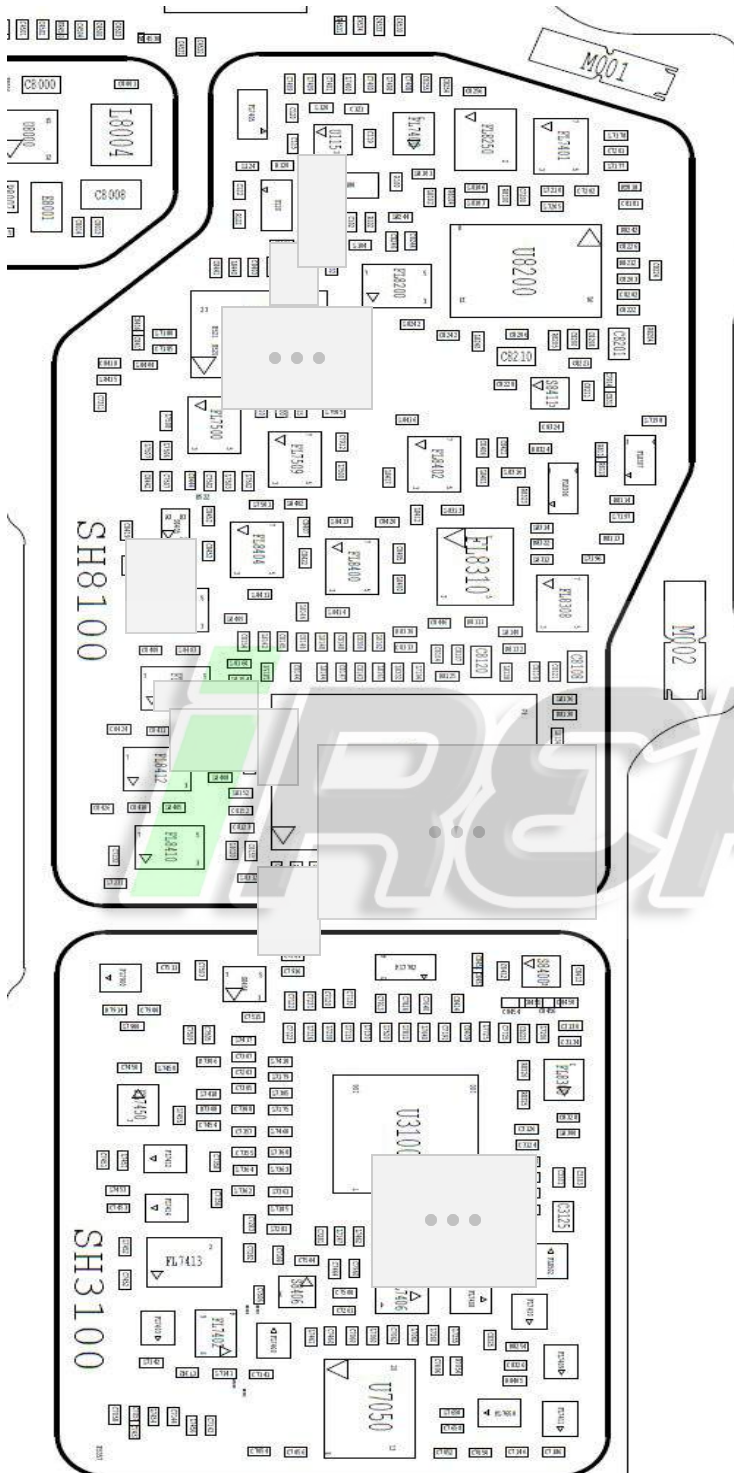
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MODEL

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## B4 PRX LTE/WCDMA



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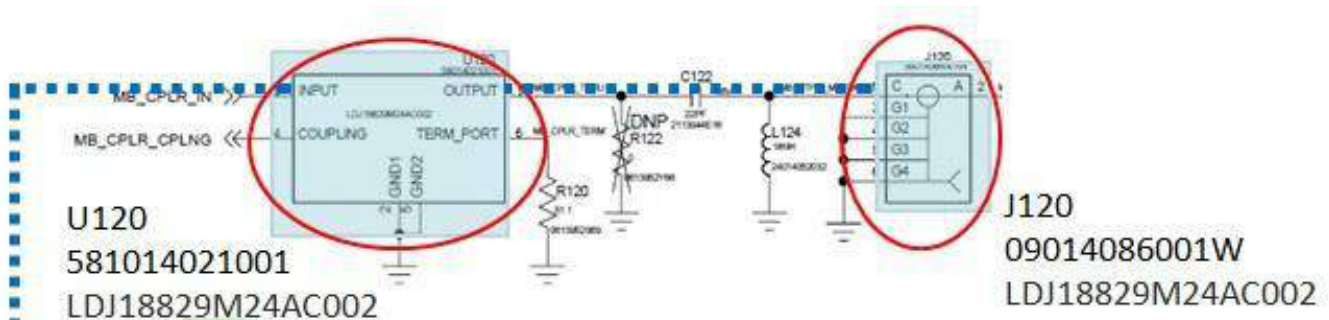
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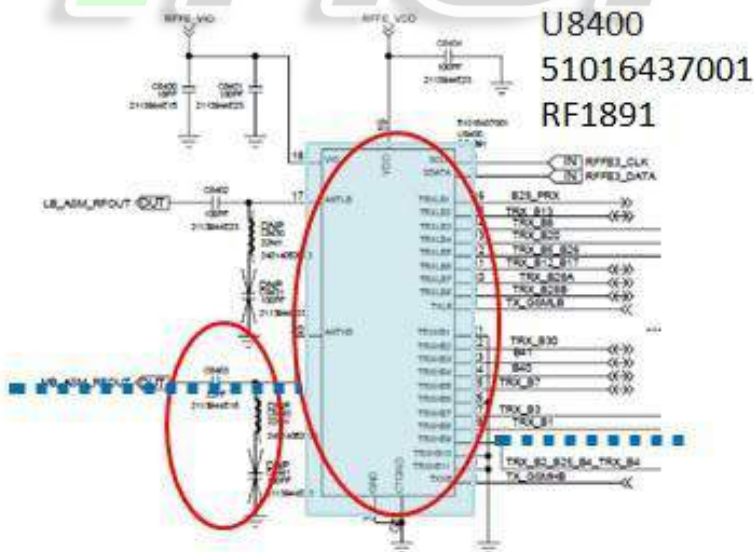
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Antenna Connector Check: J120

Coupler Checks: U120



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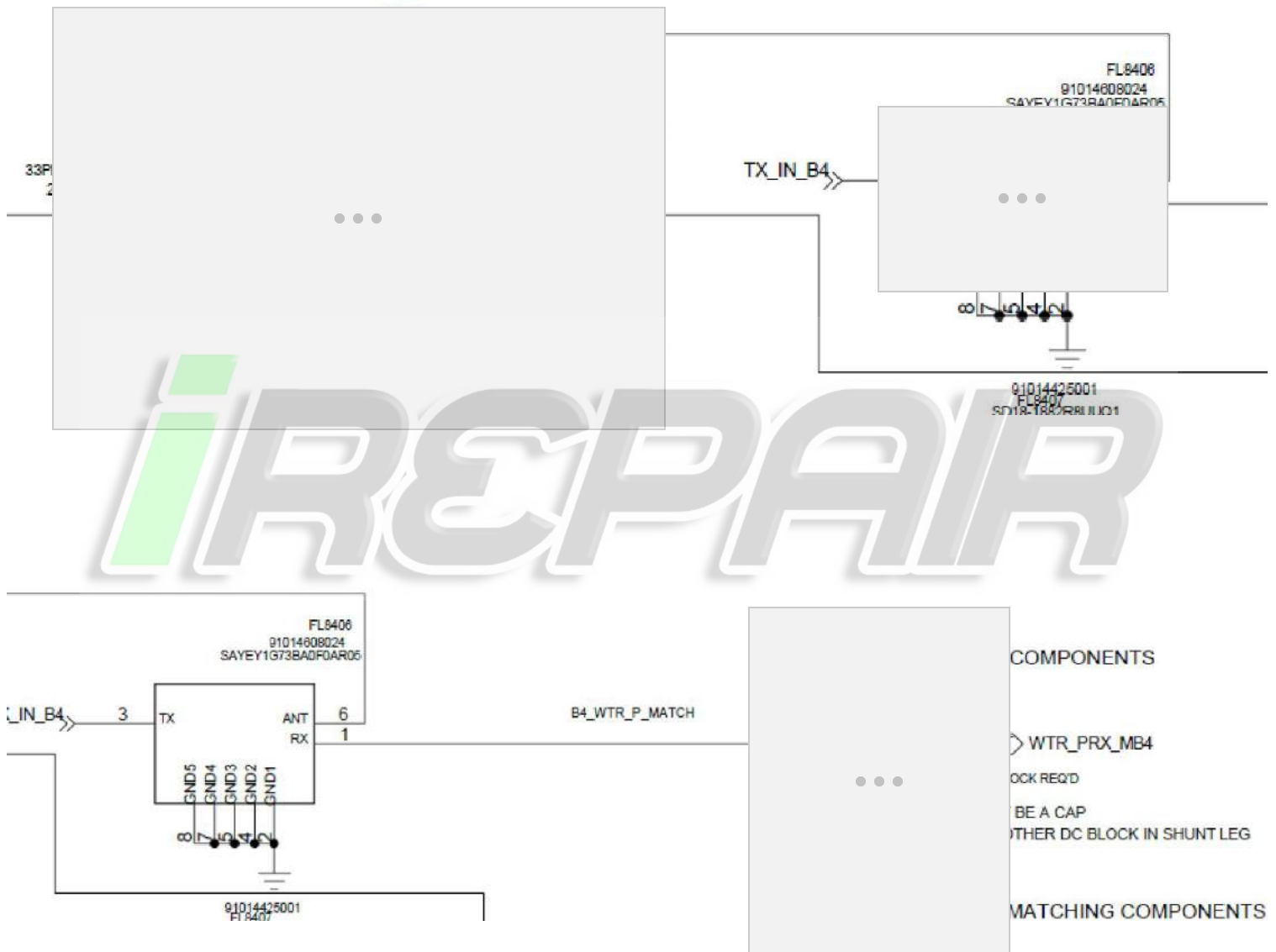


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|       |                                 |       |        |
|-------|---------------------------------|-------|--------|
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### B4 Duplexer Check:



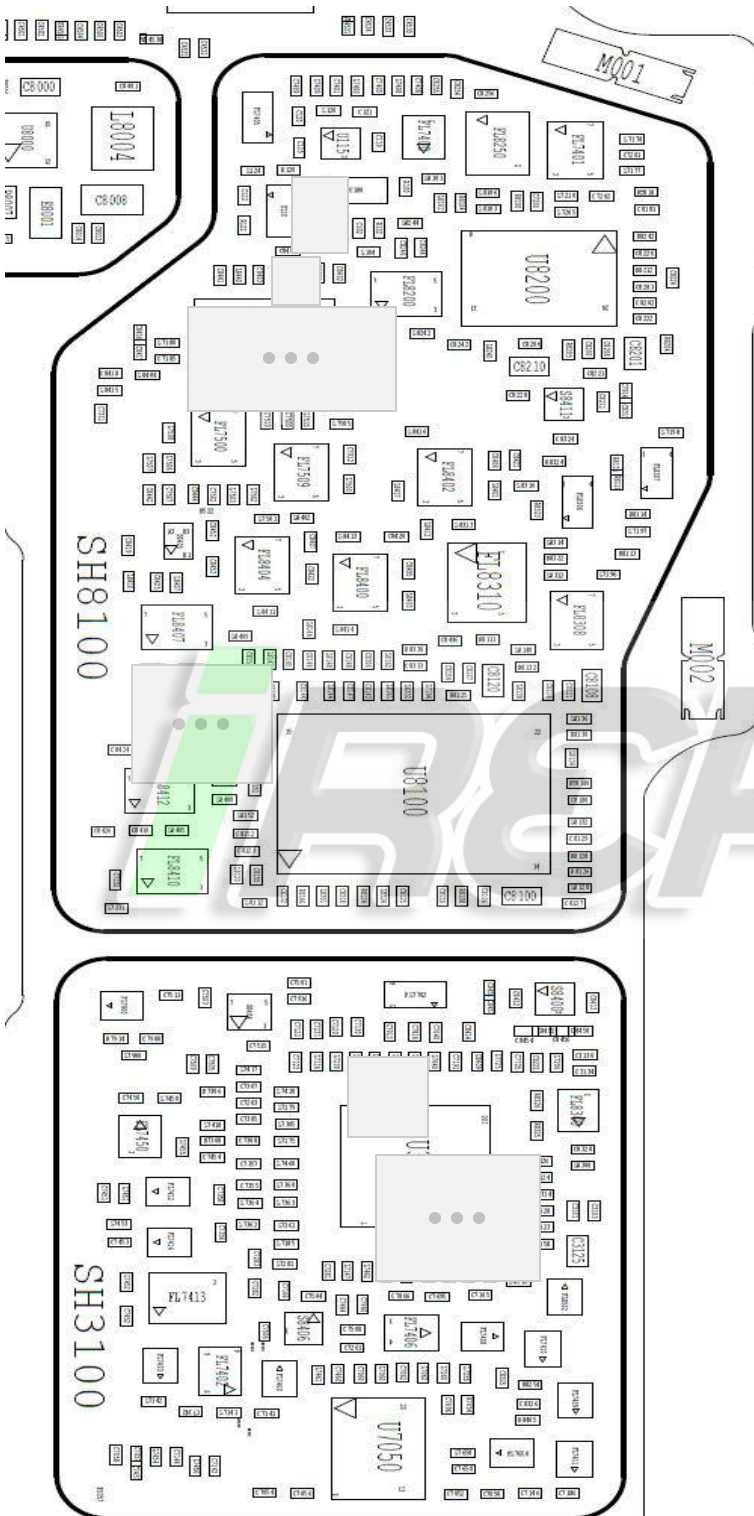
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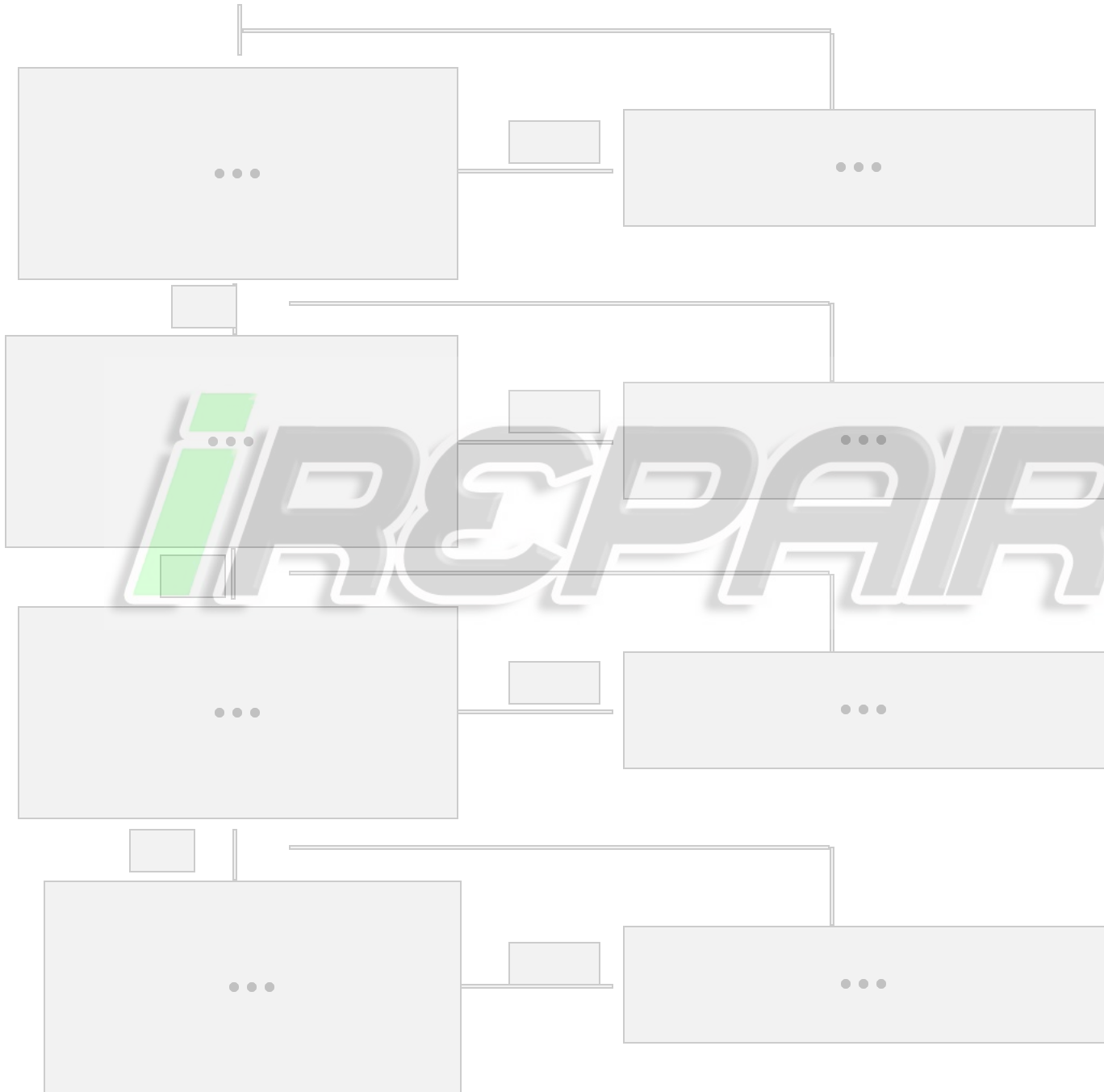
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MODEL

Bounce

## B4 DRX LTE/WCDMA



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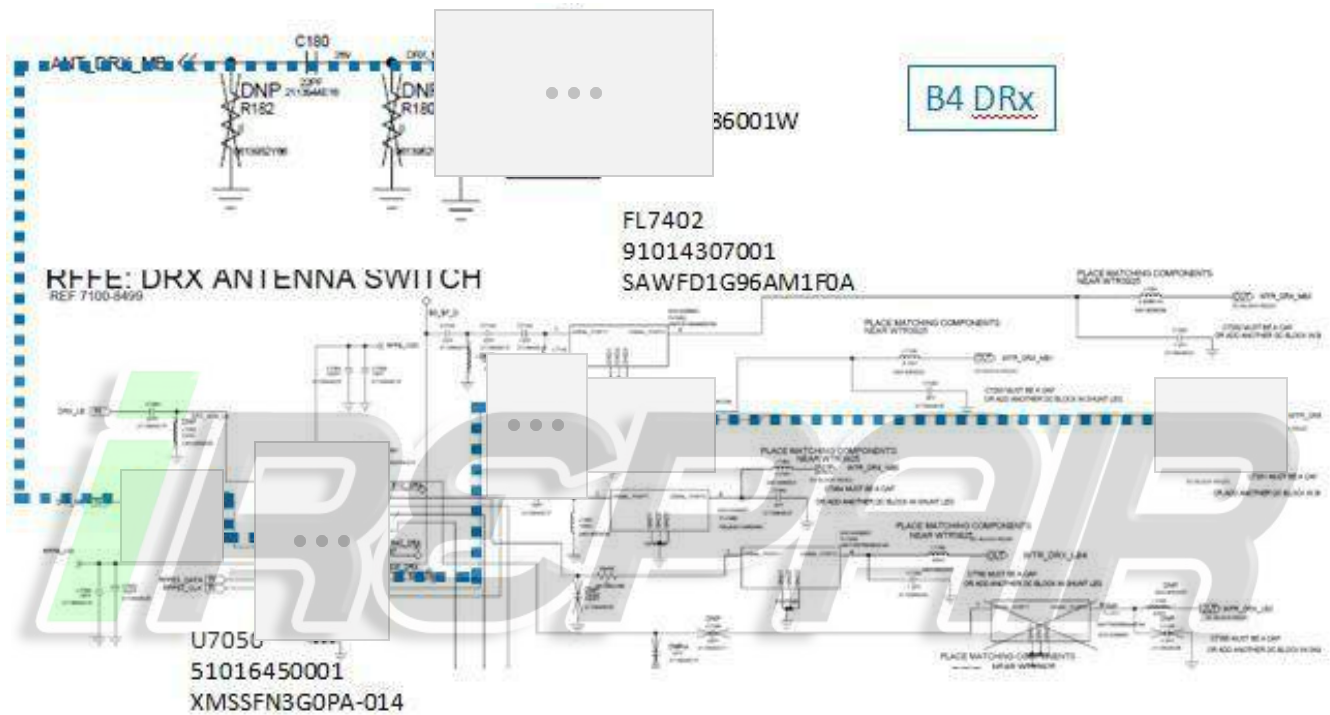
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7/22/2015



## PAGE 59 OF 204 PAGES

## Bounce RF Troubleshooting Guide

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**Bounce**The image shows the 'iREPAIR' logo in a stylized, italicized font. The 'i' is green, and the rest of the letters are gray with a white outline. A large, semi-transparent gray crosshair is centered over the logo.**WRITER**  
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## PAGE 61 OF 204 PAGES

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| <div>TITLE</div> <div>Bounce RF Troubleshooting Guide</div> | <div>MODEL</div> <div>Bounce</div> |
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LTE B5/26, CDMA BC0/10, WCDMA B5/19, TX



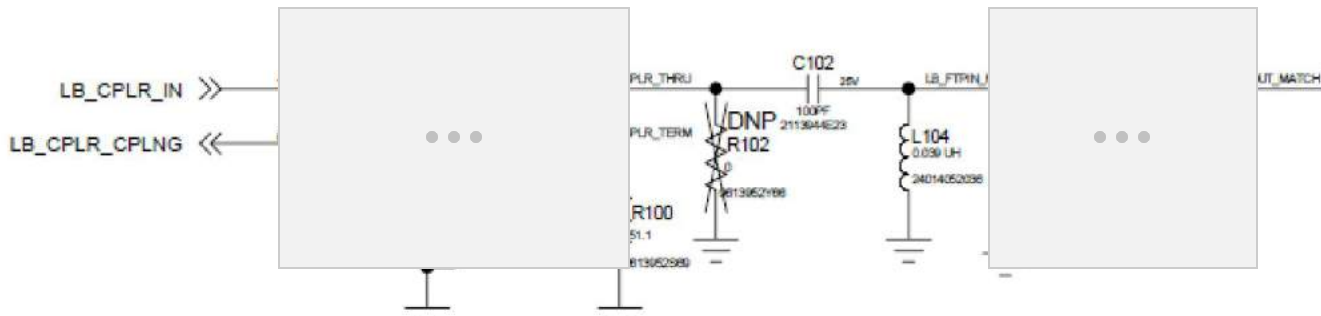
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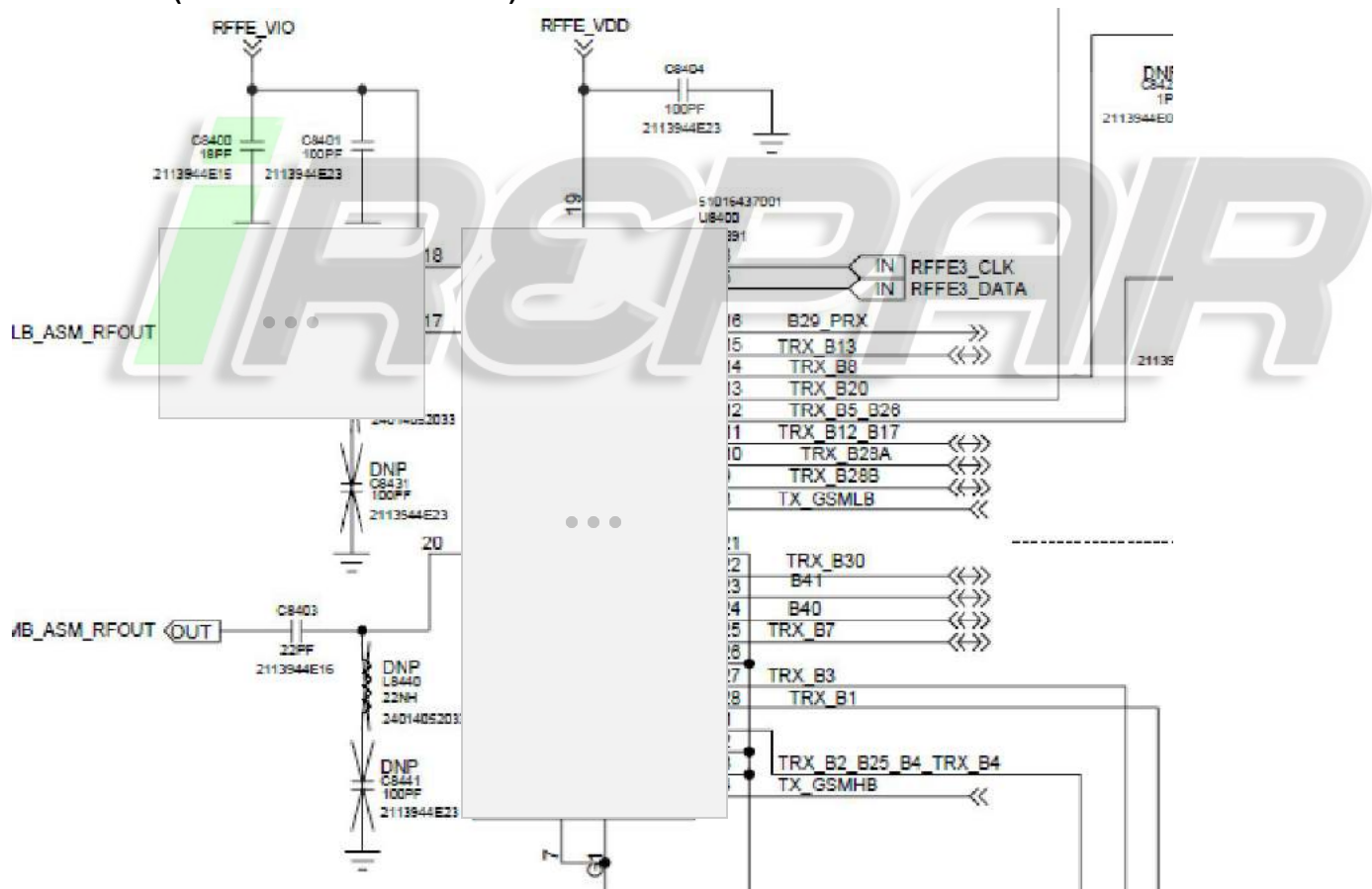
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RF coupler and Connector:



TRX switch (Antenna Switch Module) ASM:



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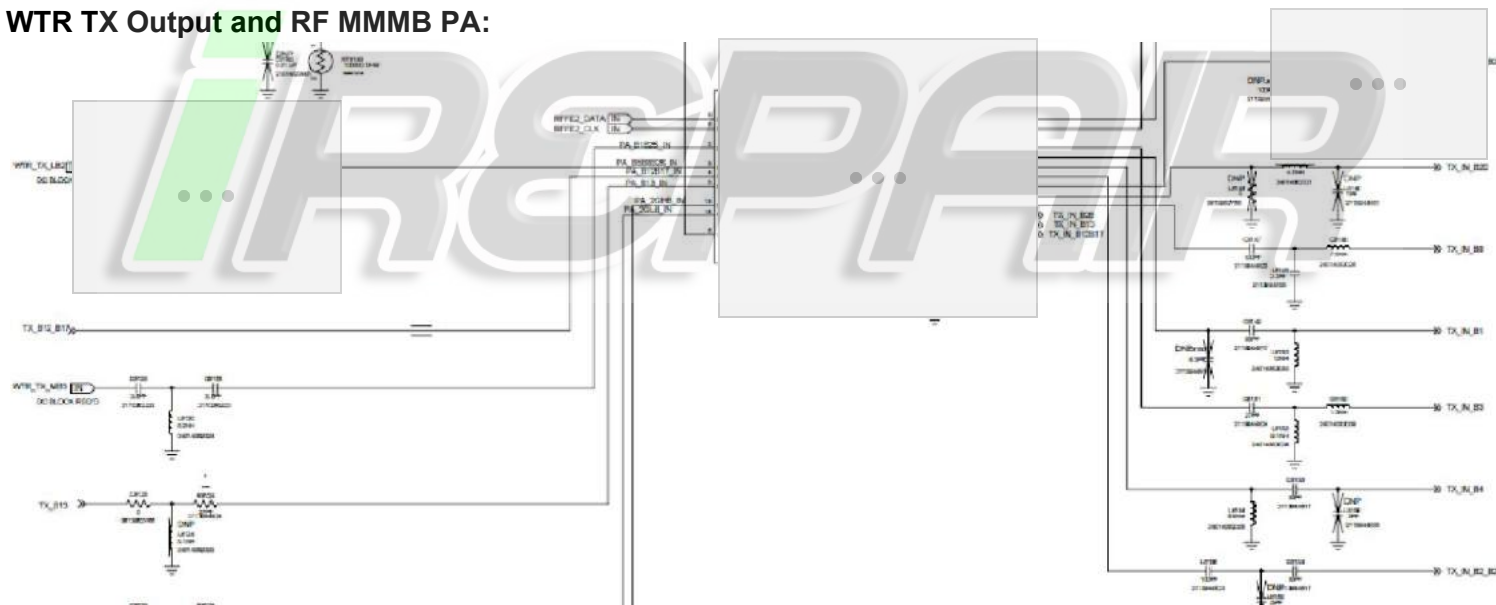
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## MMMB PA output and Duplexer:



## WTR TX Output and RF MMMB PA:



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# ENGINEERING LOG SHEET

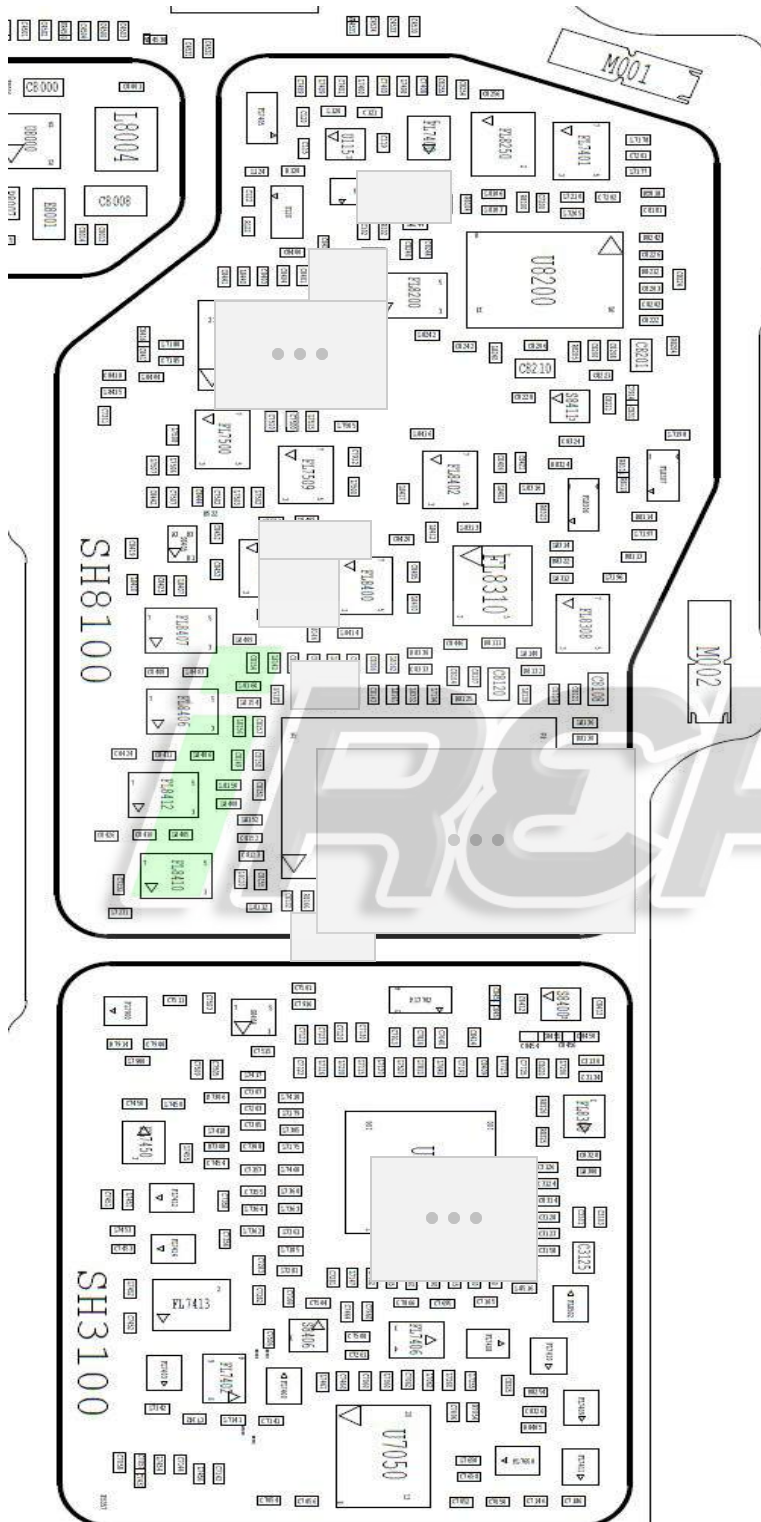
PAGE 65 OF 204 PAGES

TITLE

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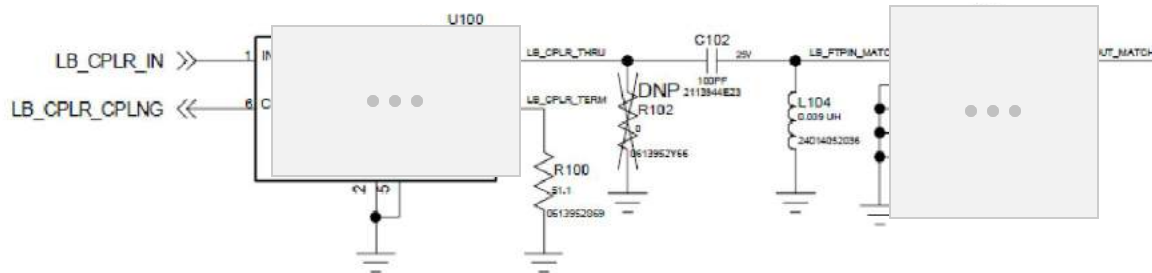
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LTE B5/26, CDMA BC0/10, WCDMA B5/19, PRX

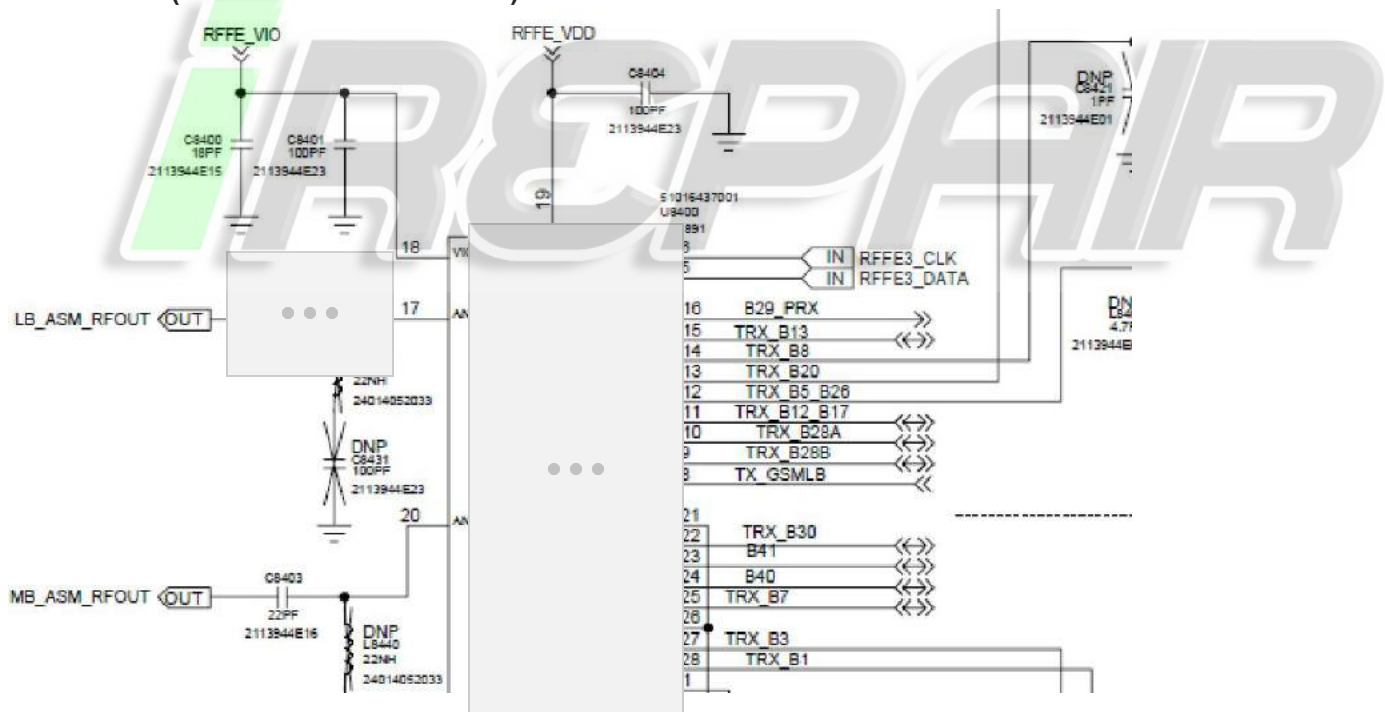


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### RF coupler and Connector:



## TRX switch (Antenna Switch Module)



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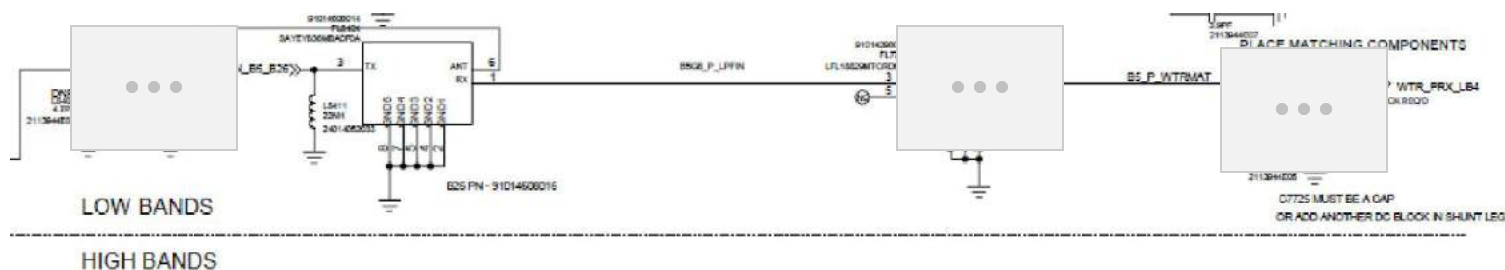
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## Duplexer/ RX filter/ WTR match



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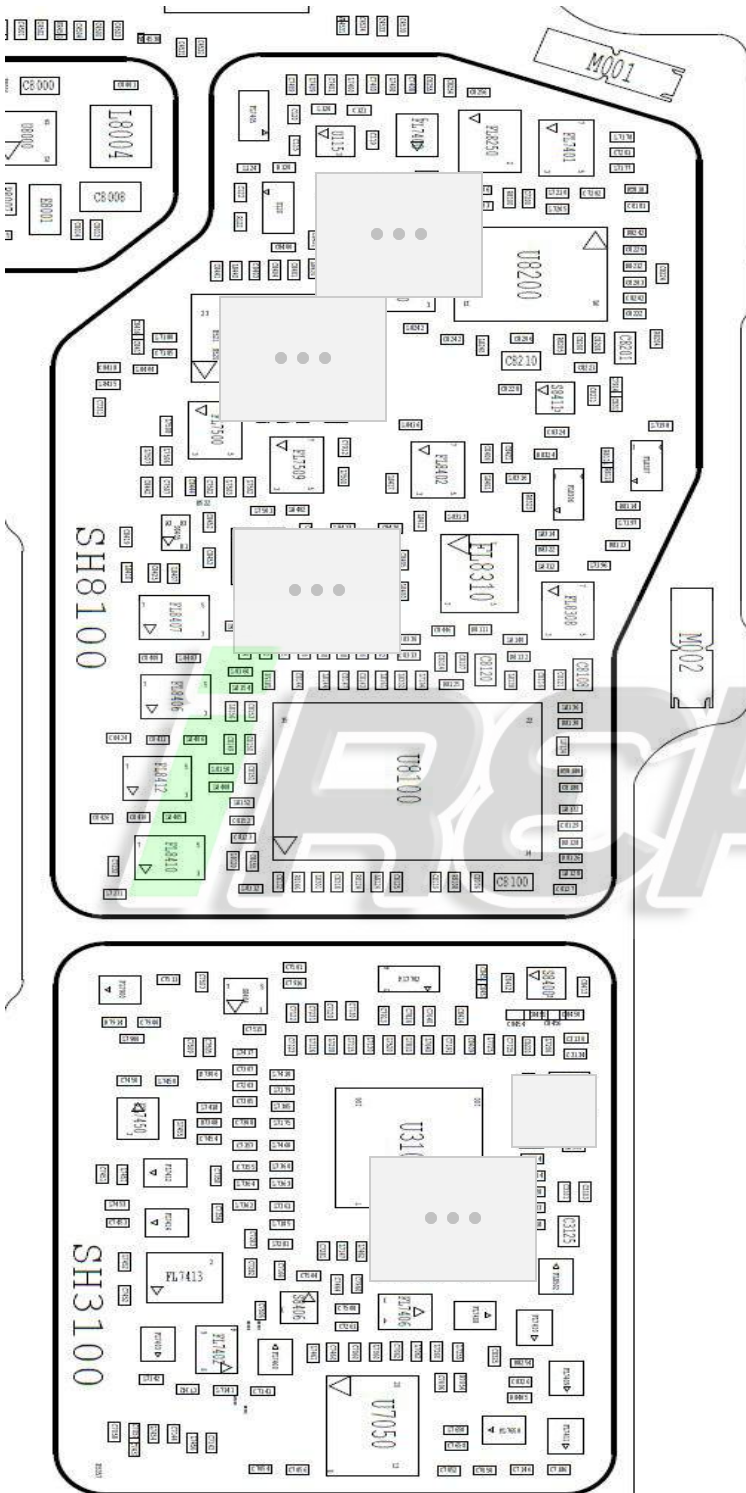
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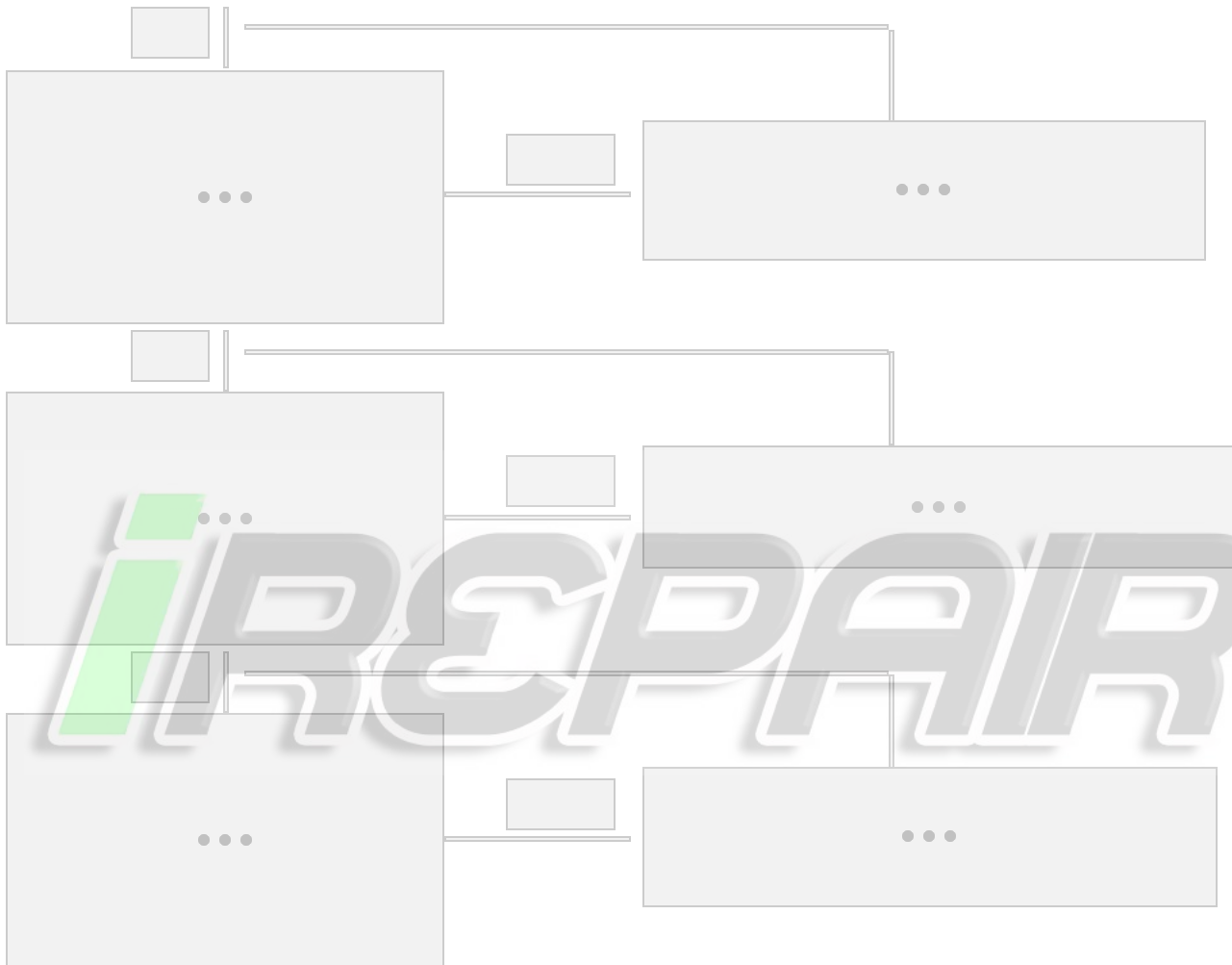
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MODEL

Bounce

LTE B5/26, CDMA BC0/10, WCDMA B5/19, DRX



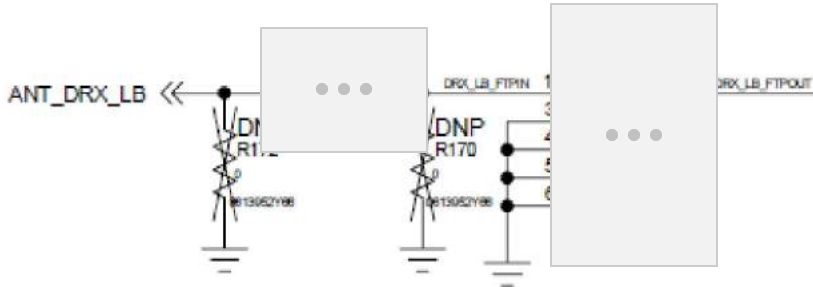
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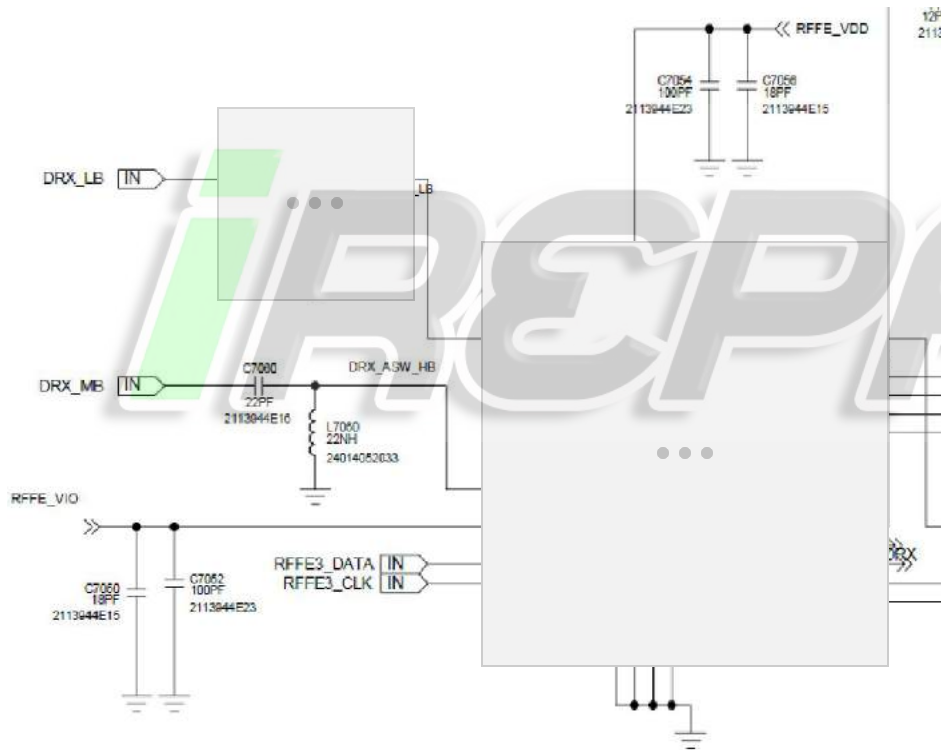


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|-------------------------------------------------|------------------------|

## DRX LB RF Connector:



## DRX ASM:



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MODEL

Bounce

DRX Filter and WTR match:



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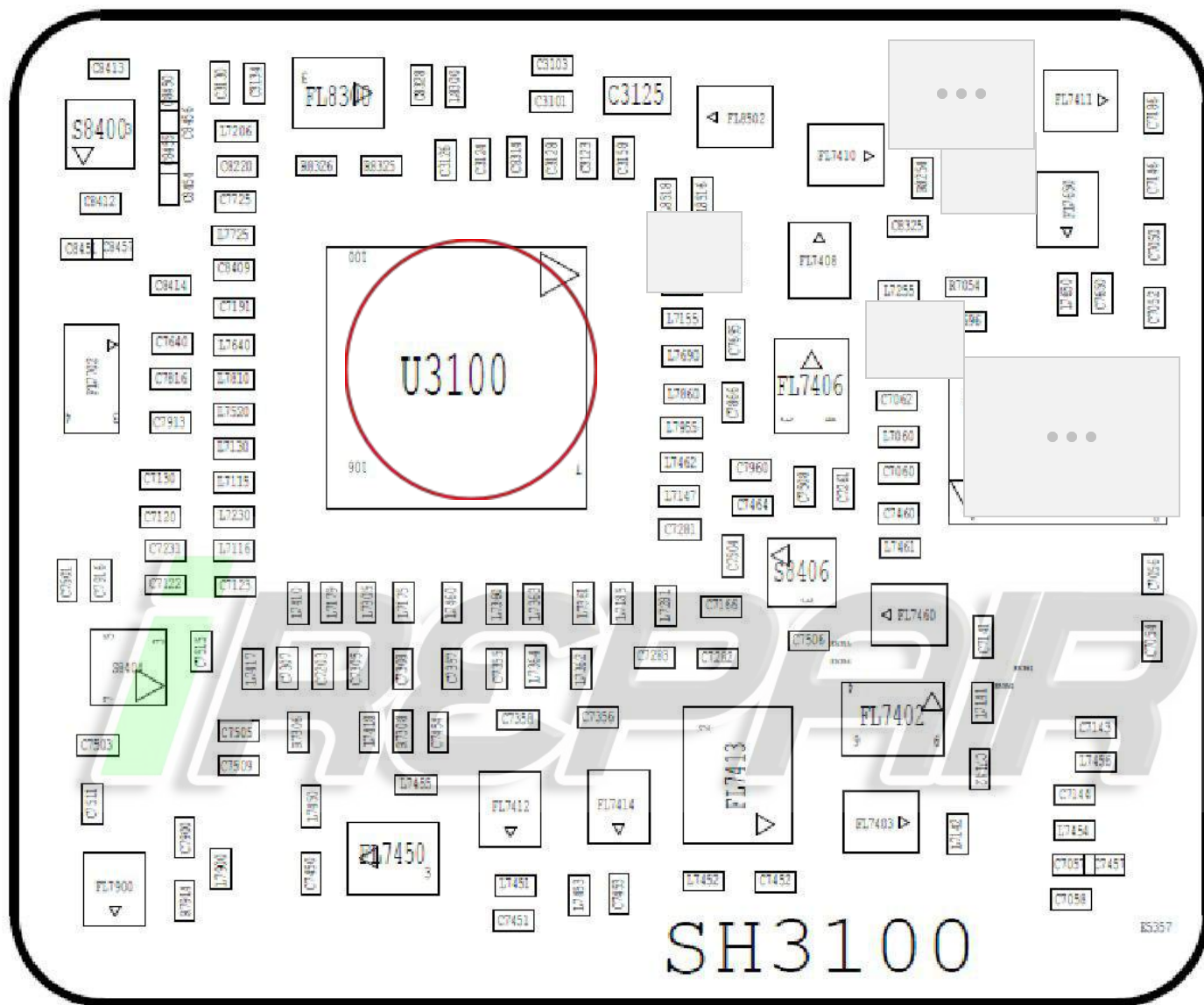
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## PAGE 73 OF 204 PAGES

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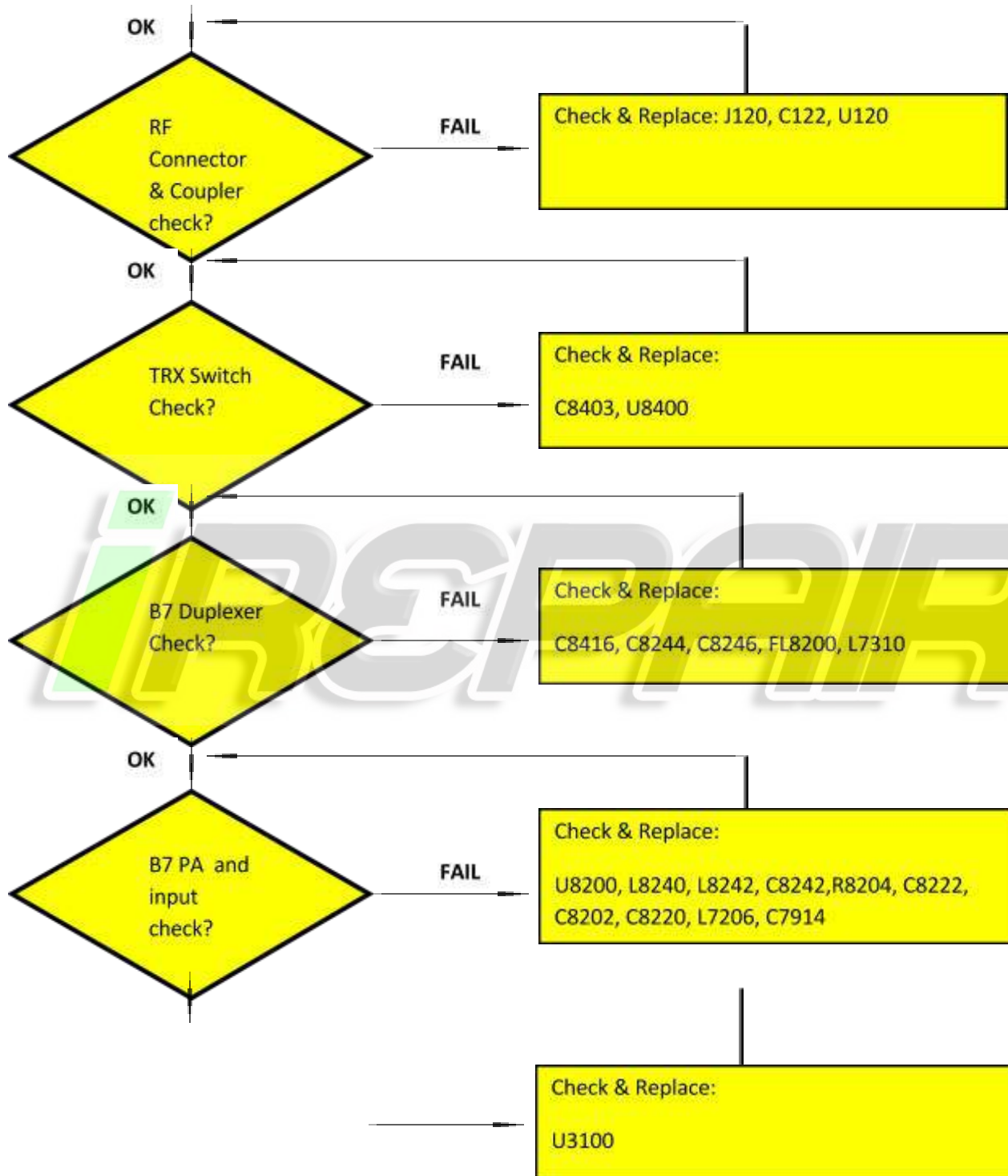
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## LTE B7 TX



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B7 TX Schematic

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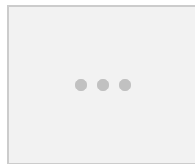
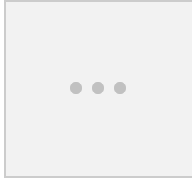
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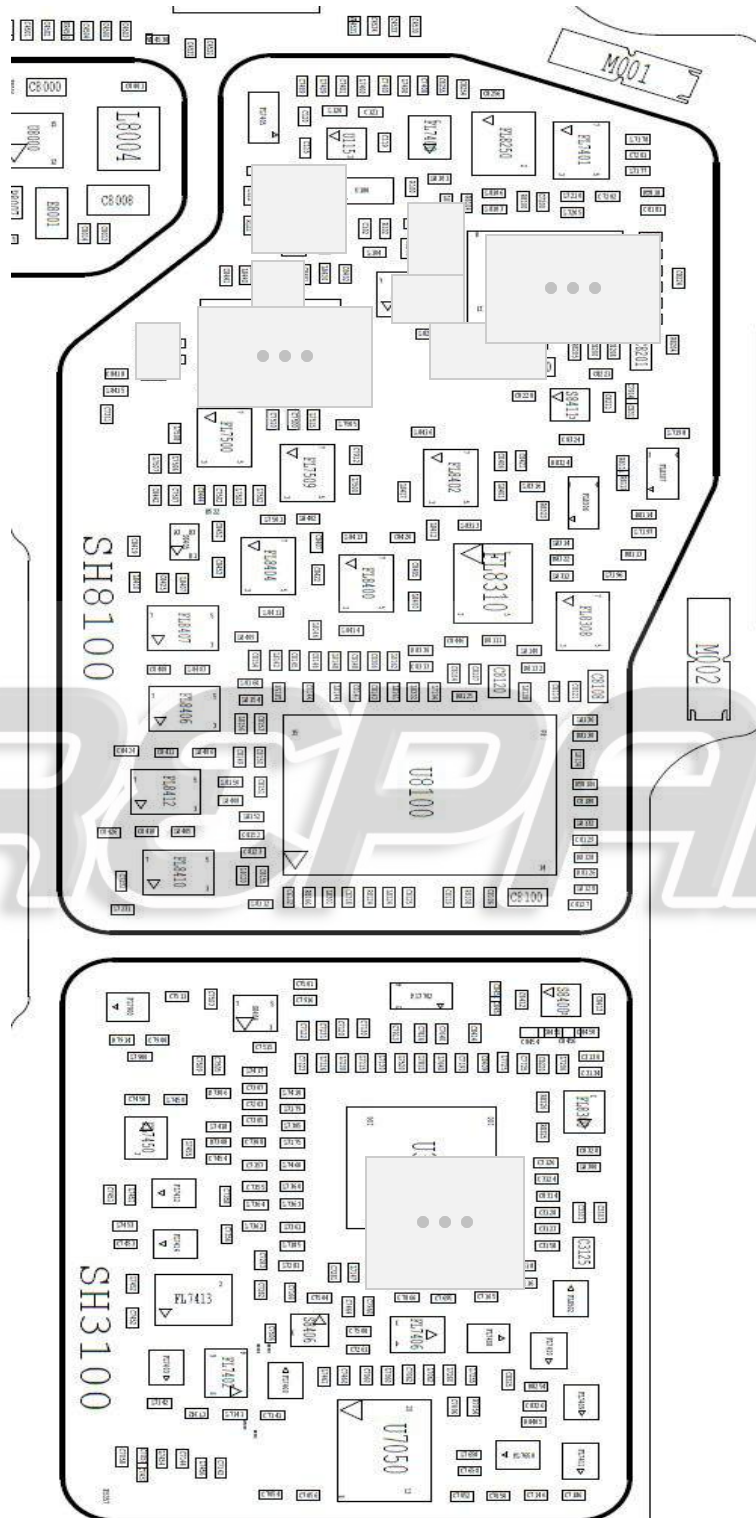


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## LTE B7 PRX



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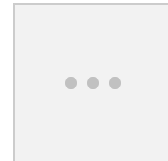
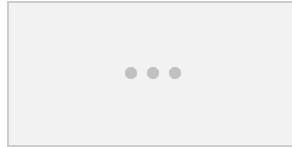
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Bounce

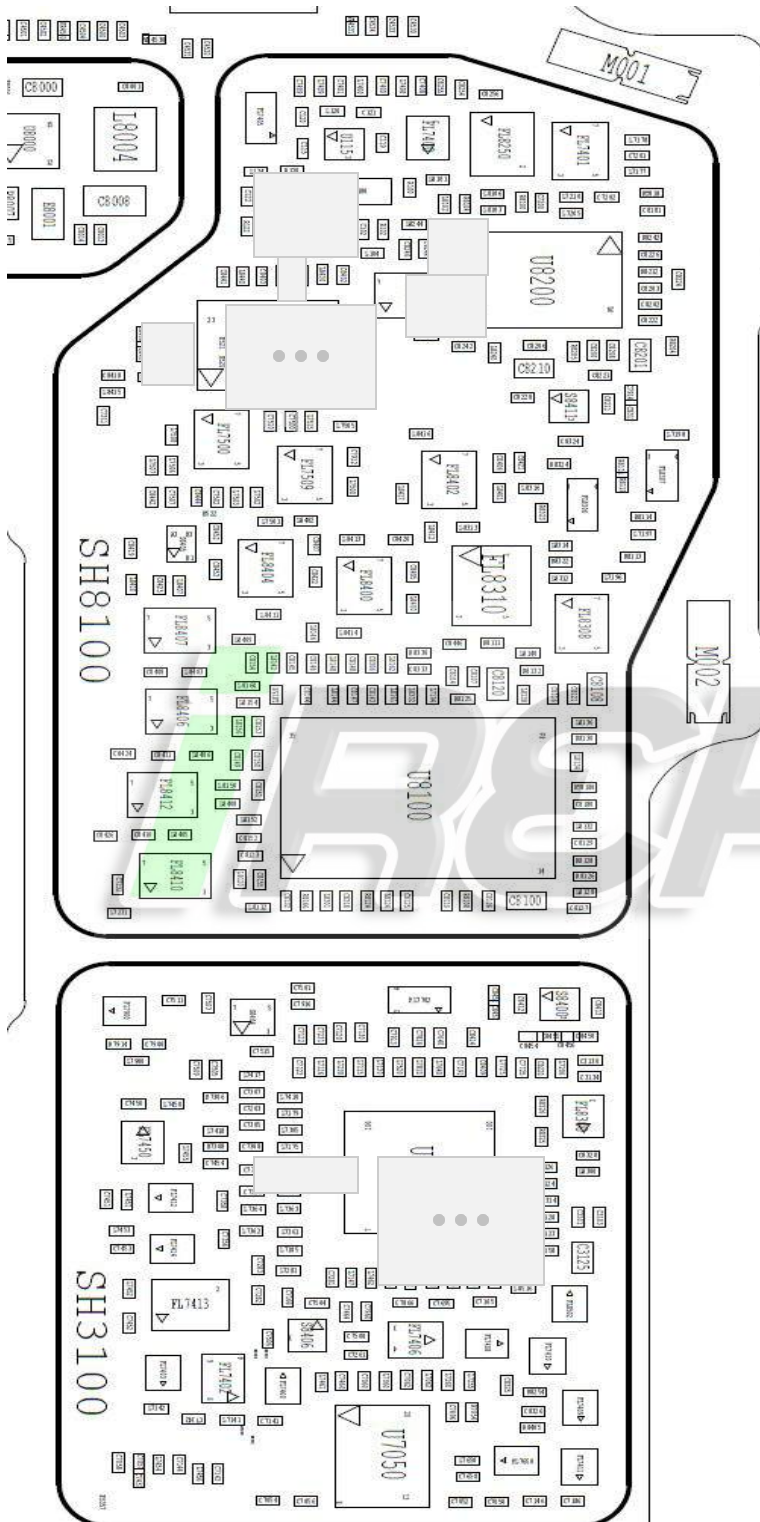


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## LTE B7 DRX

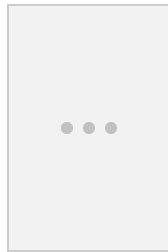


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B7 DRX Schematic



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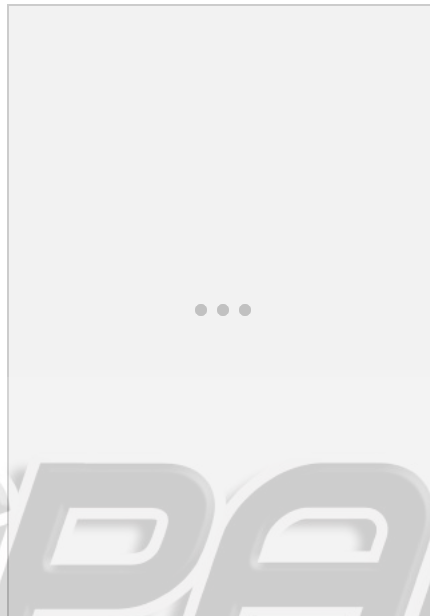
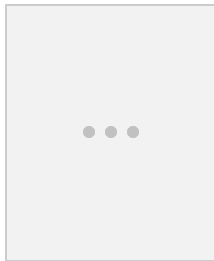
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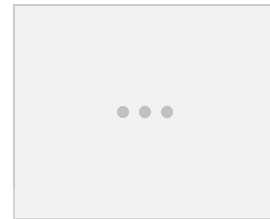
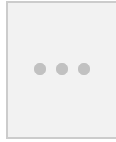
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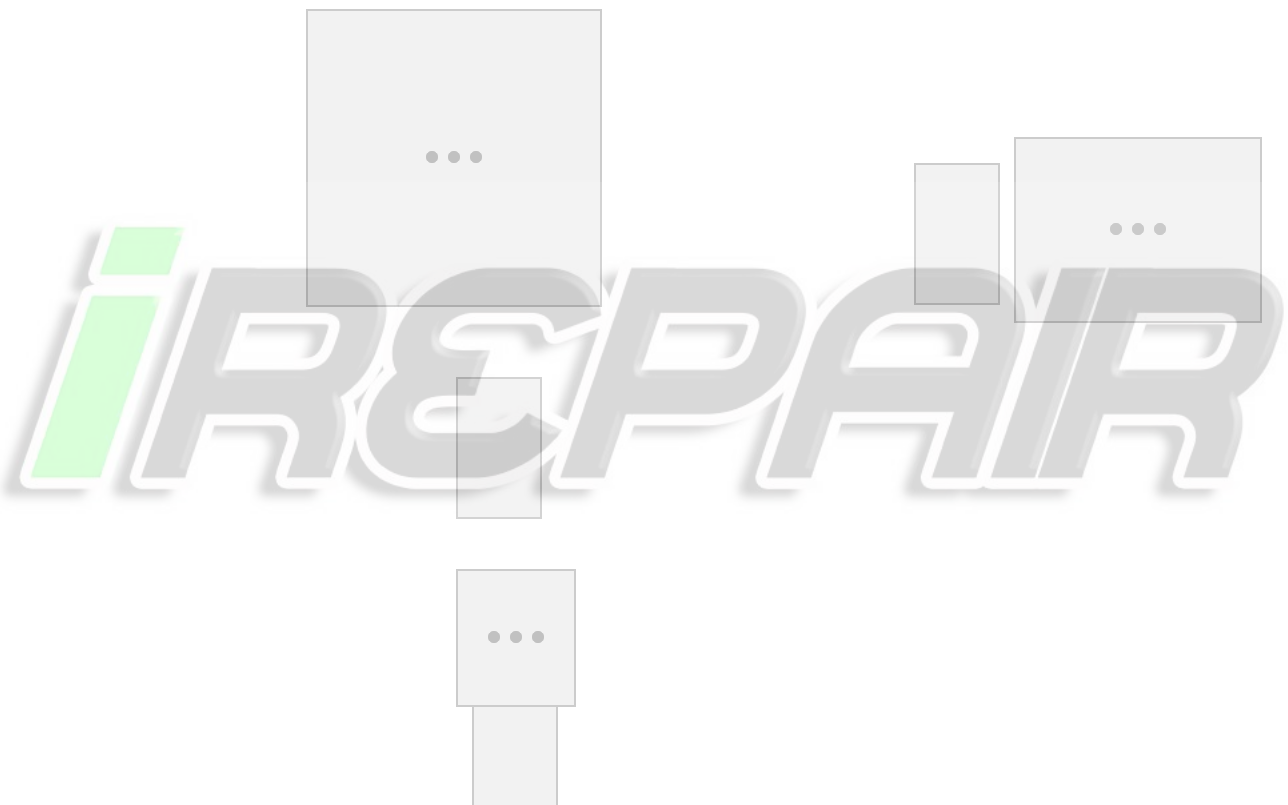
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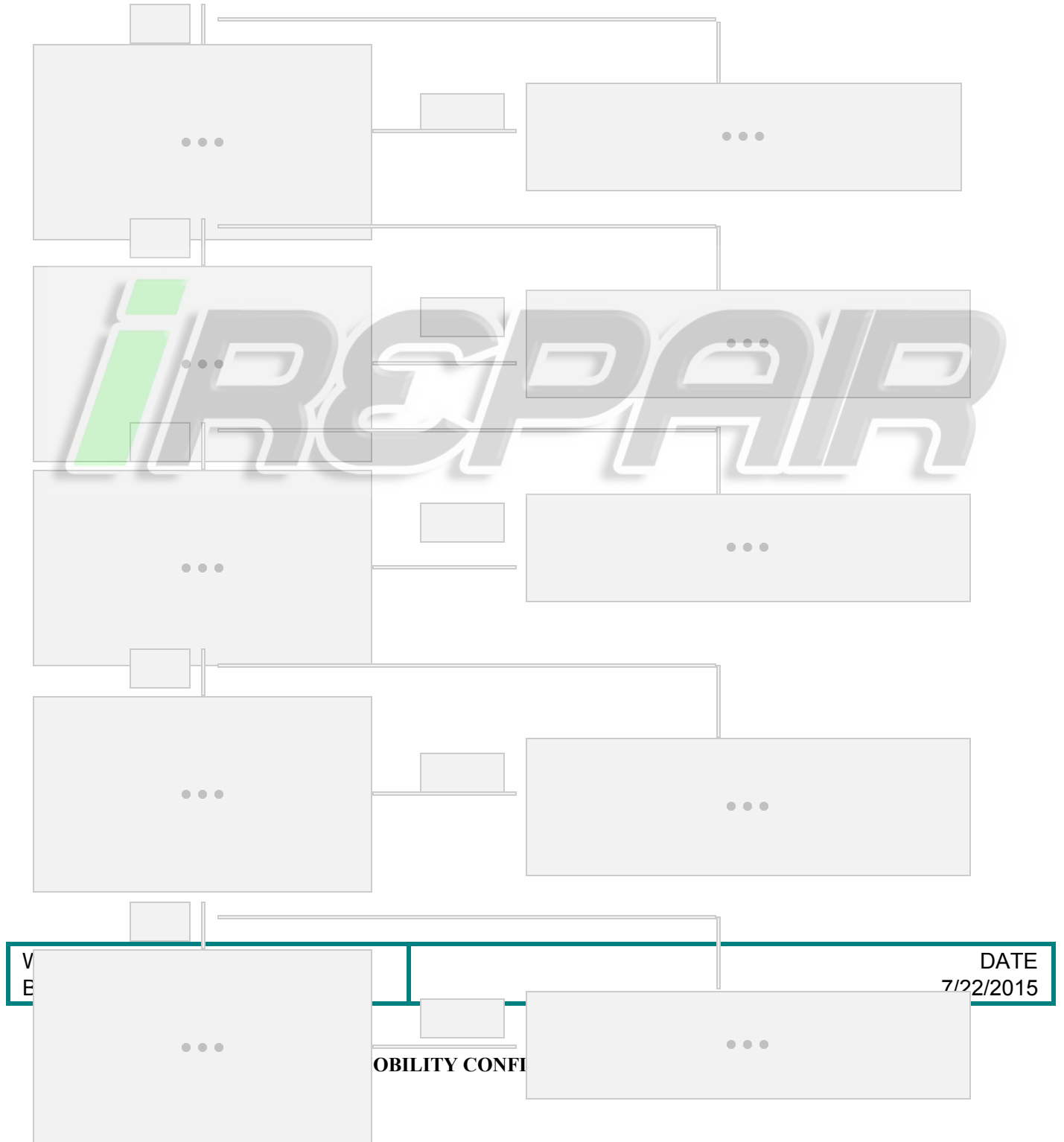


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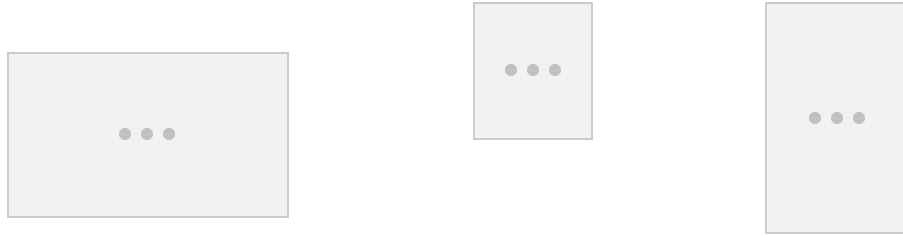
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| TITLE                           | MODEL  |
| Bounce RF Troubleshooting Guide | Bounce |

## B8 LTE & WCDMA TX



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|------------------------------------------------------------|-----------------------------------|
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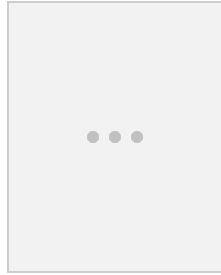
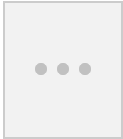
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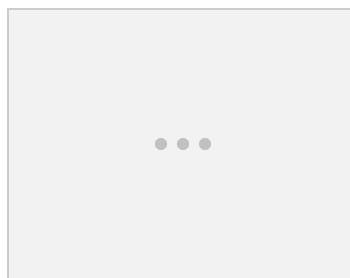
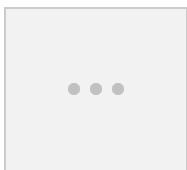
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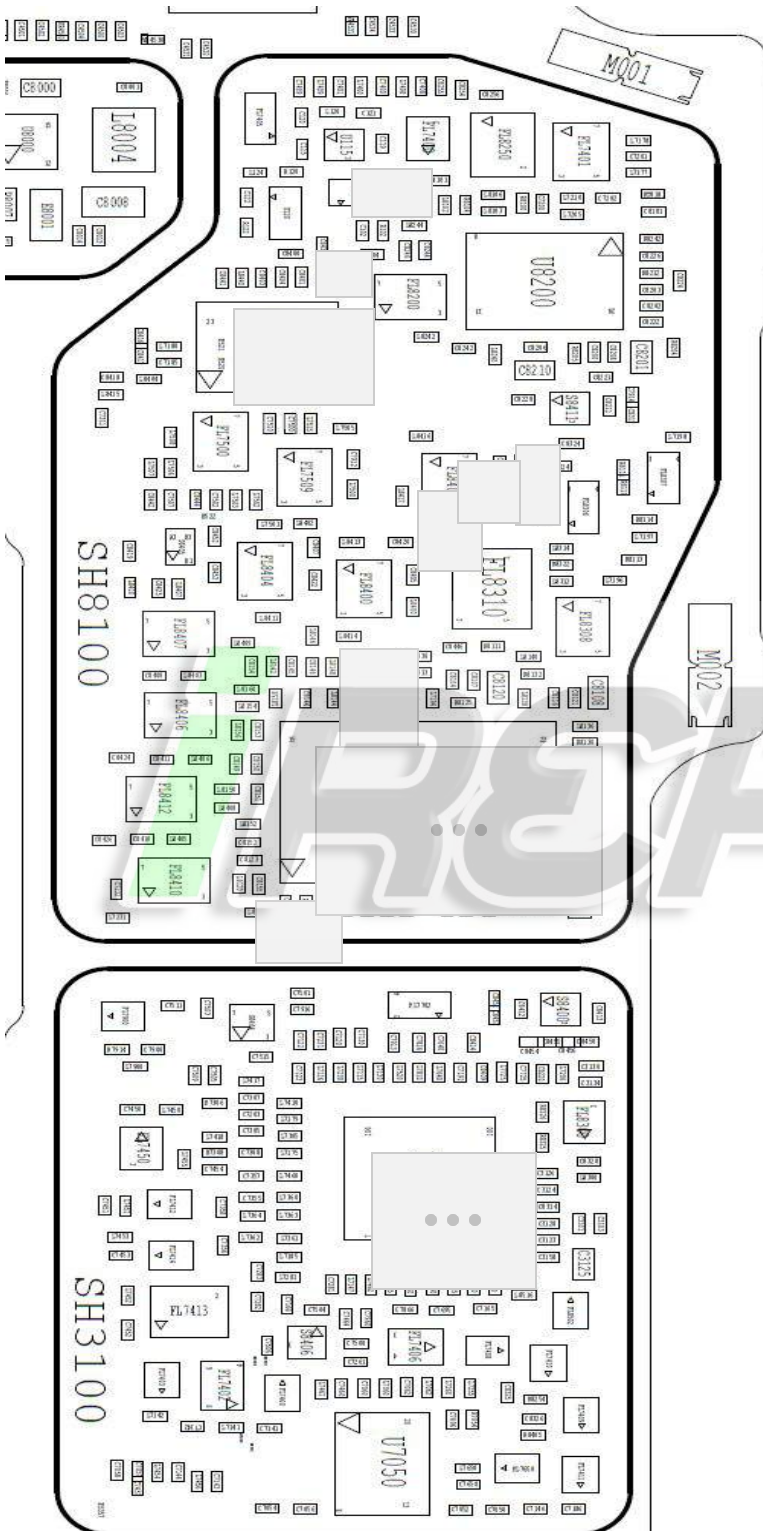
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MODEL

Bounce



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## LTE &amp; WCDMA B8 PRX



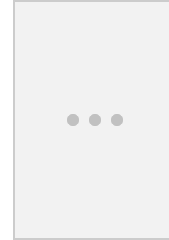
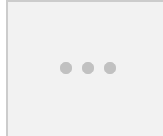
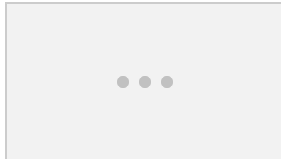
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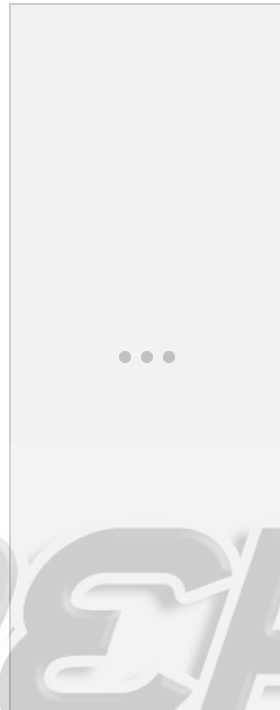
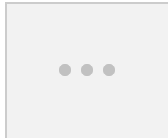
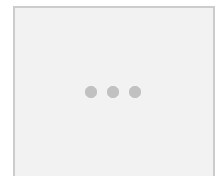
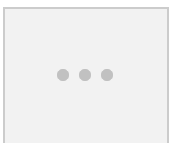
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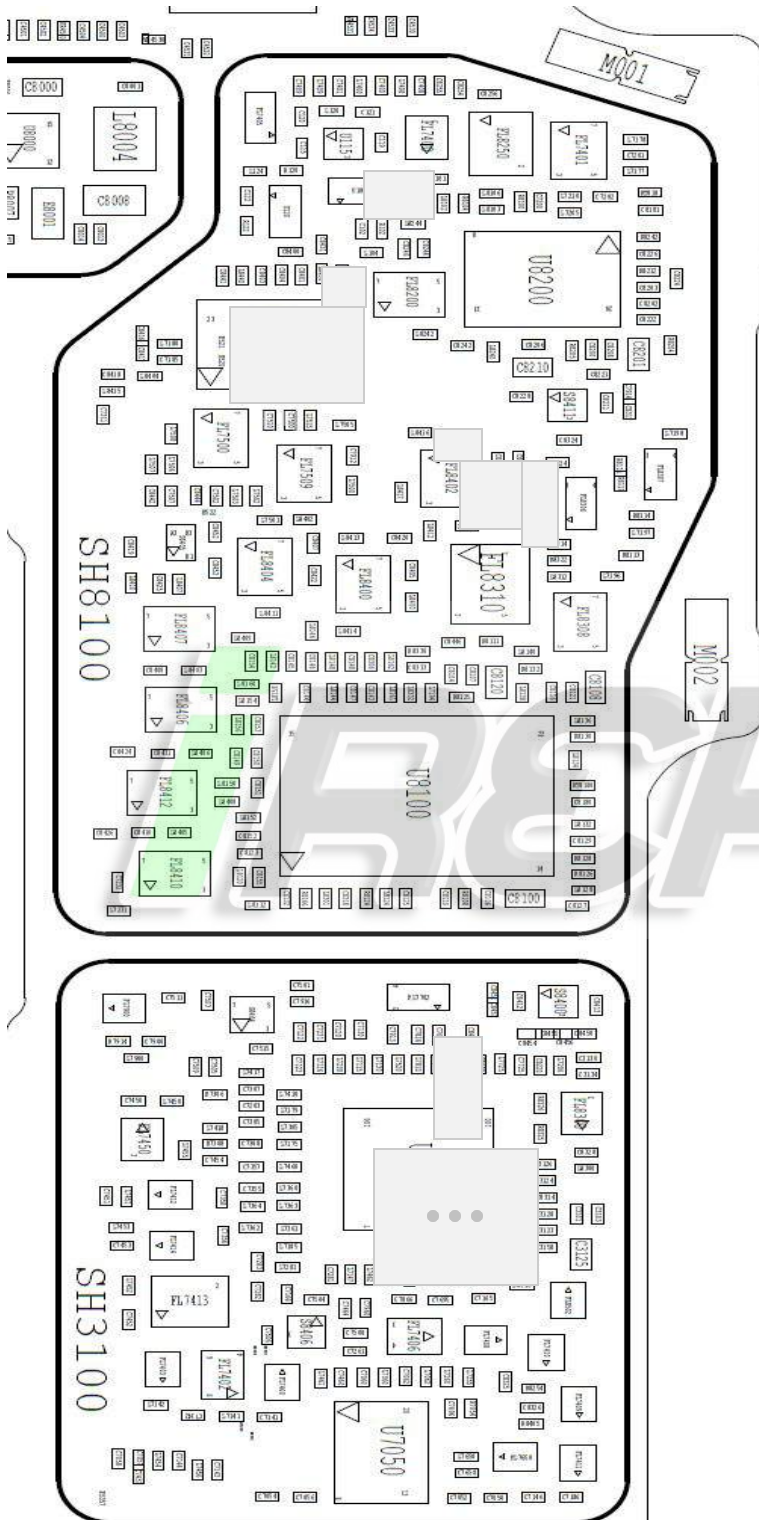
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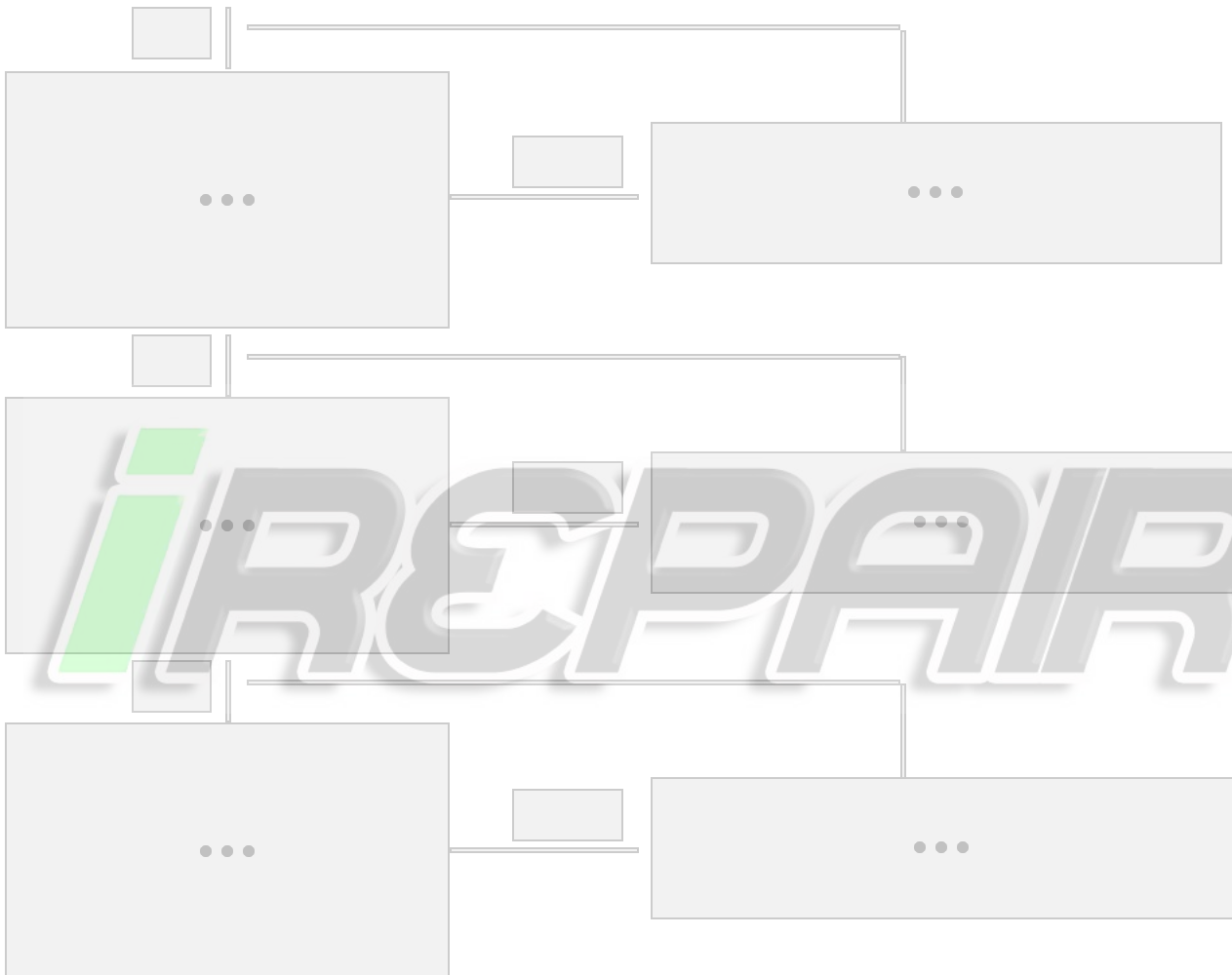
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## B8 LTE &amp; WCDMA DRX



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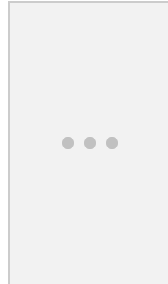
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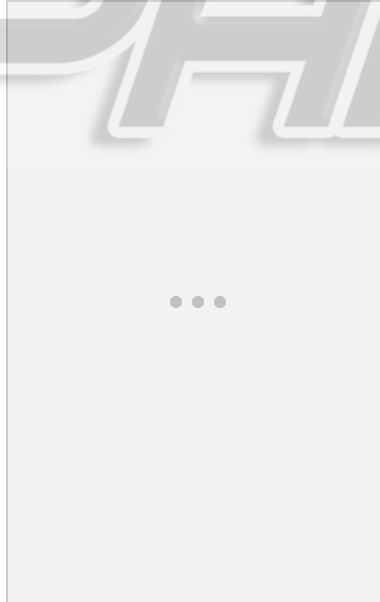
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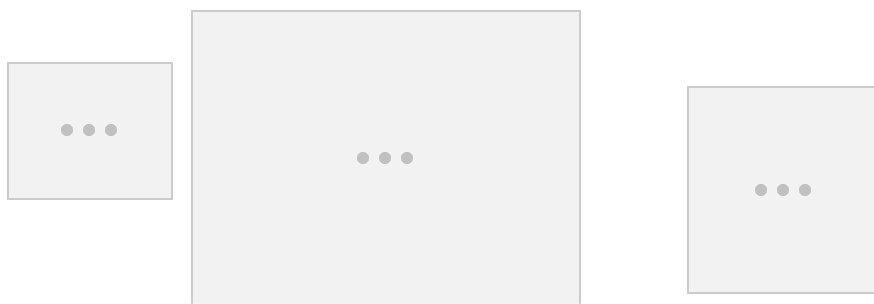
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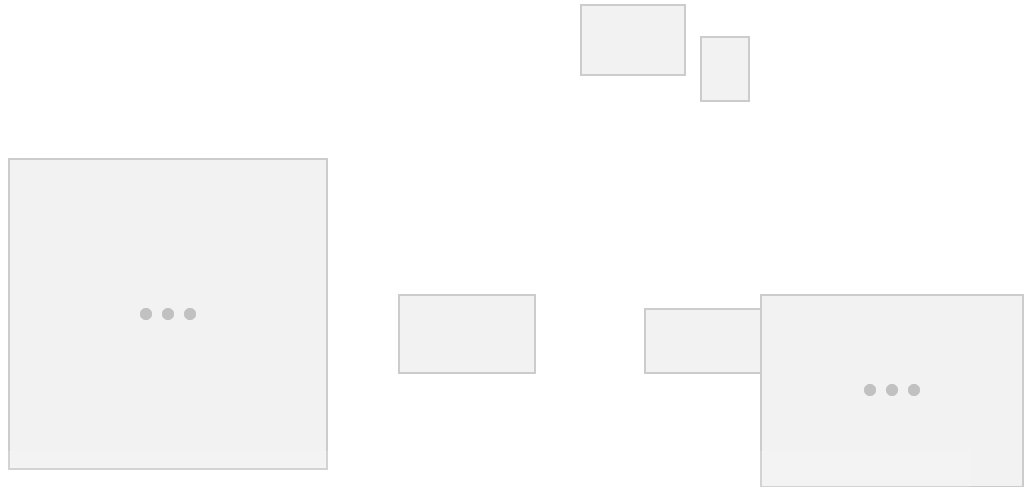


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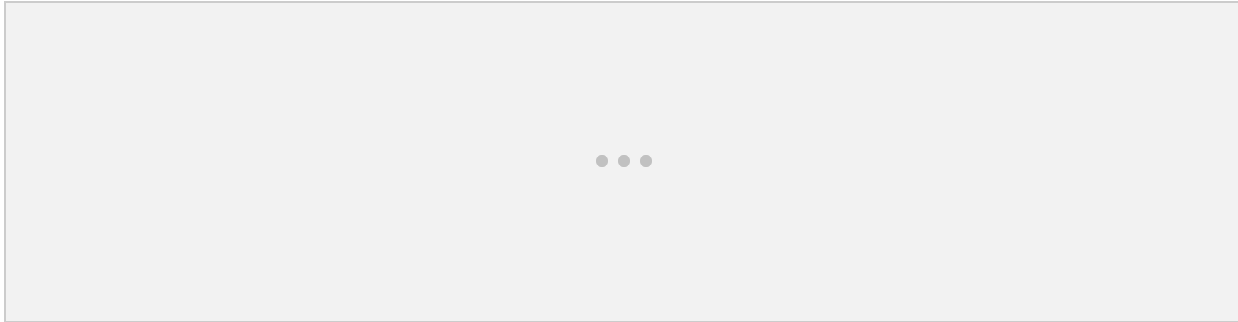
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LTE B12 & B17 TX

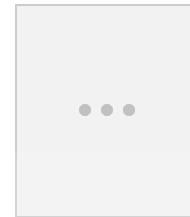
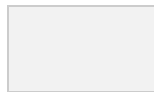
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**B12/17 TX Schematic- Connector and Coupler Check:**

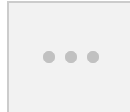
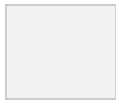


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B12/B17 ASM Check:

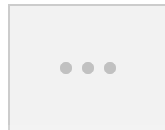
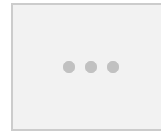


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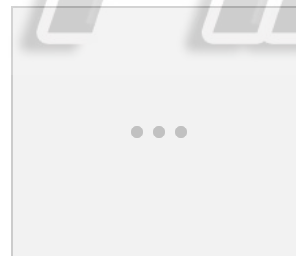
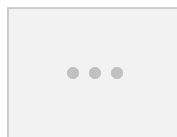
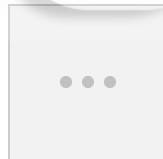
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B12/B17 Filter and Duplexer check:



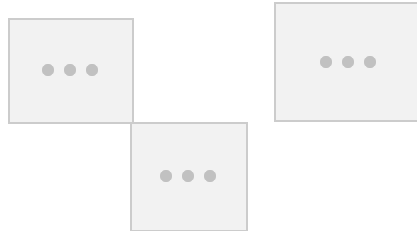
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WTR TX input:



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**B12/B17 TX Layout :**

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LTE B12 & B17 PRX (ROW SKUs)



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LTE B12 & B17 PRX Schematics (**ROW SKU**)



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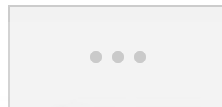
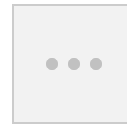
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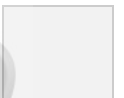
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LTE B12 & B17 PRX Schematics (**ROW SKU**)– cont'd

Filter and Duplexer Check



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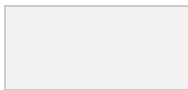
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B12/B17- B13 Switch Check:

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B12/B17 PRX WTR input:



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**B12/B17 RX Layout (ROW SKU):**

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## LTE B12/B17 DRX



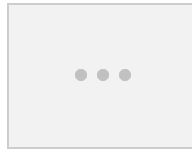
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RF Path from WTR to Duplexer to ASM to RF Jack.

Schematics:

B12/B17 DRX Antenna connector:

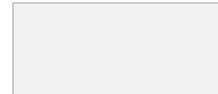


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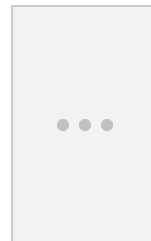
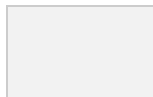
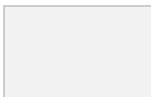
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DRX ASM LB



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B12 DRX filter:



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**B12/B17 DRX Layout: RF Path from WTR to Duplexer to ASM to RF Jack.**

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B12/B17 WTR DRX Input:

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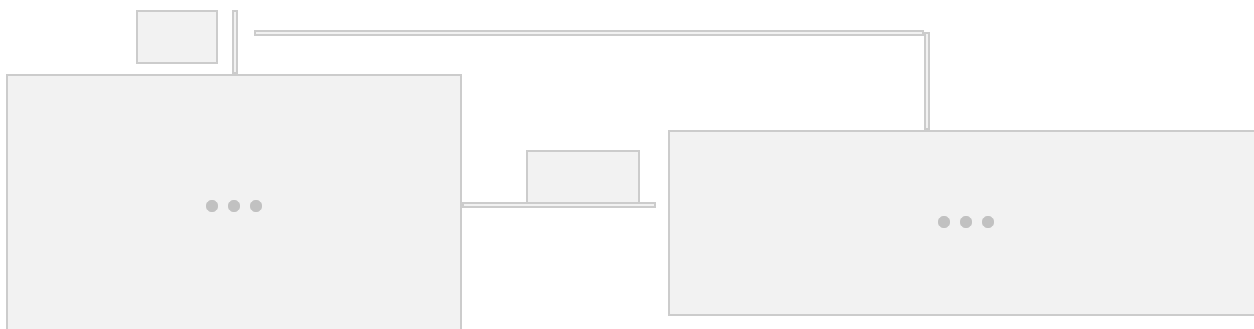
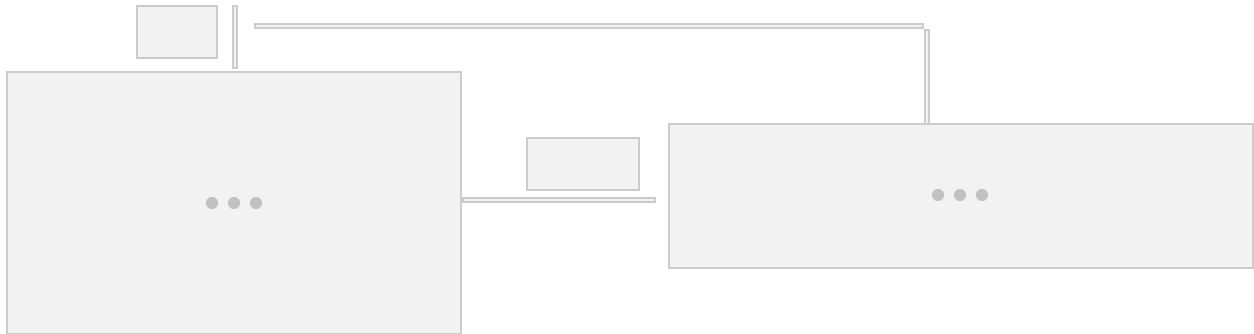
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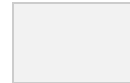
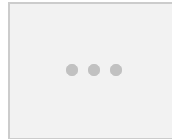
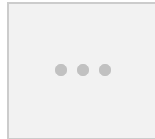
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**B13 TX Schematics:****Connector and Coupler check:**

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ASM Check:

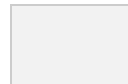


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PA Check:



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PA Input:



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WTR TX input to PA:



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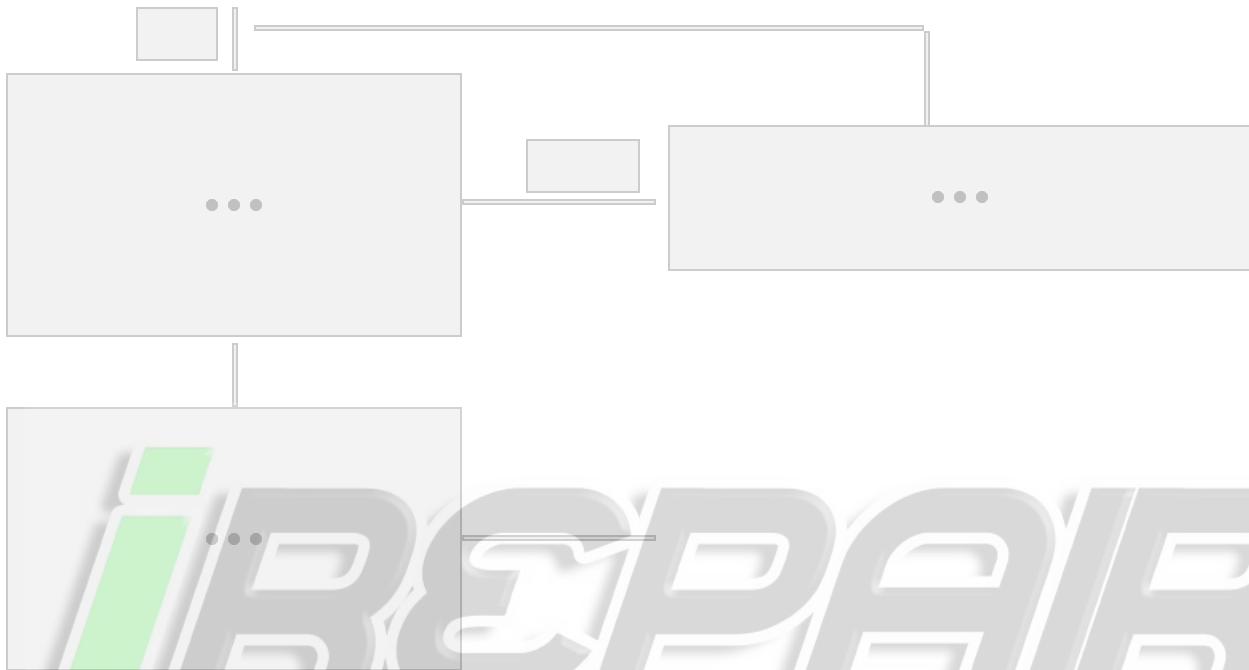
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LTE B13 PRX

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**B13 RX Schematics:**

Antenna connector check: J100, C102, U100, L104

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ASM Switch check: U8400, C8402, FL8306, R8324

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**Duplexer and Filter Check:** FL8310, R8322,FL8306

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WTR Input Check:

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PCB LAYOUT B13 PRX WTR input.

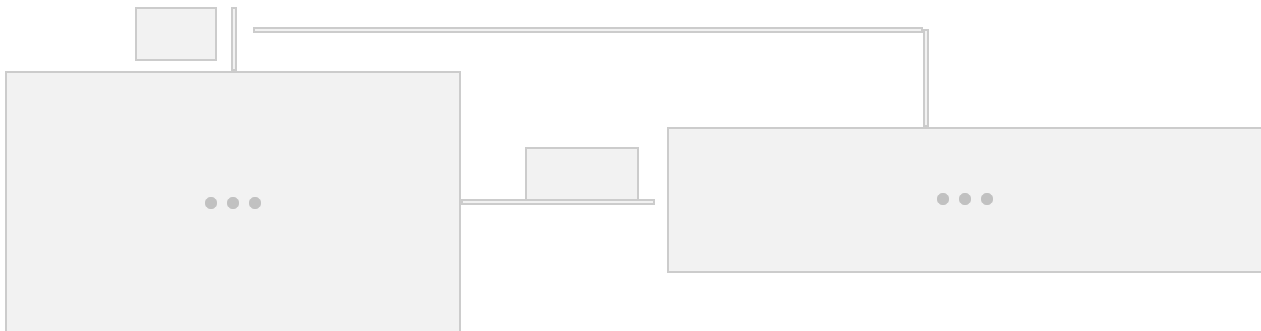
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LTE B13 DRX

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**B13 DRX Schematics:**

**B13 DRX RF Connector:**

**ASM Check:**

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B13 DRX WTR input:

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**B13 DRX Layout:**

RF Connector check:

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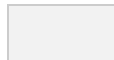
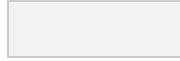
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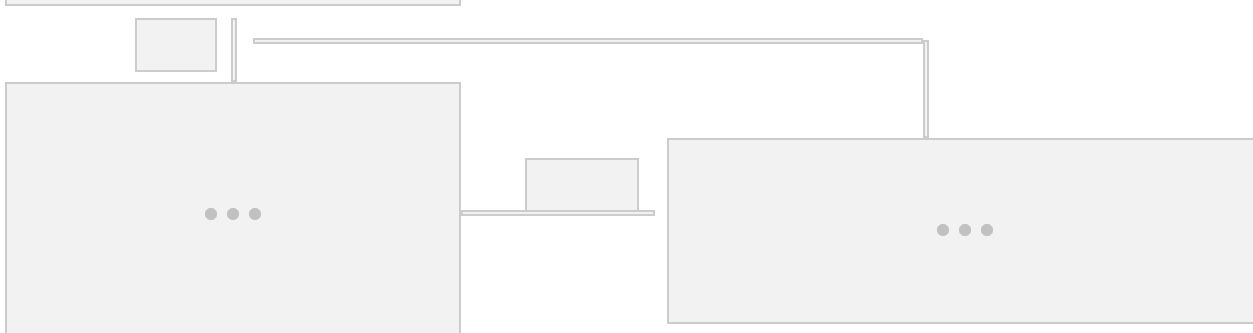
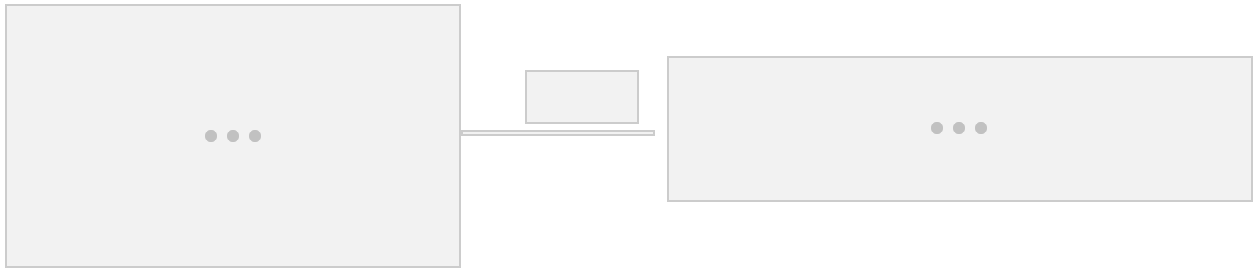
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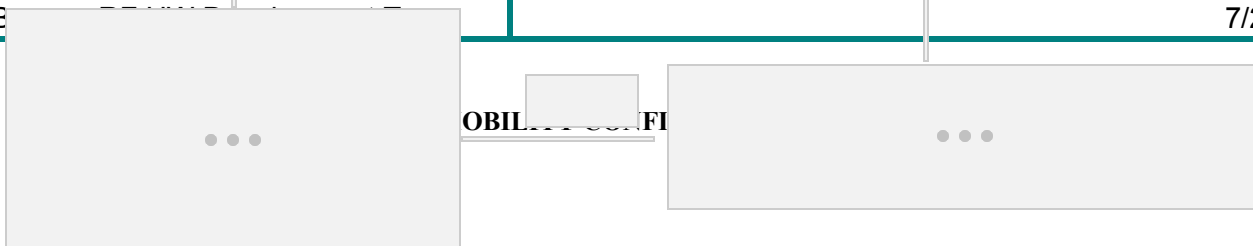


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## B20 TX:



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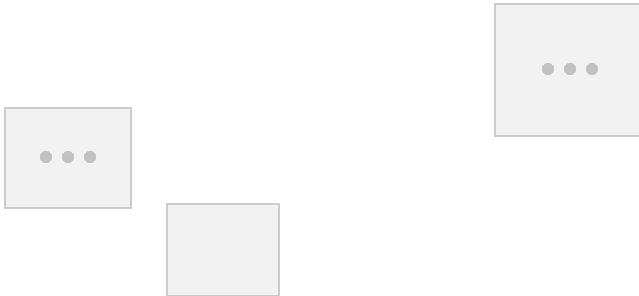
Antenna connector, Coupler check:

ASM Switch Check:

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**B20 Duplexer Check****B20 PA Check: WTR output to PA**

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PA Output:

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**B20 TX Layout**

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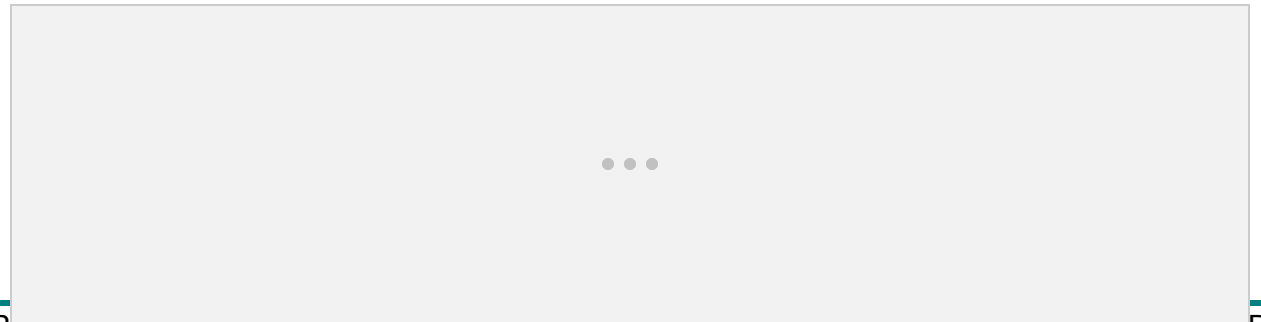
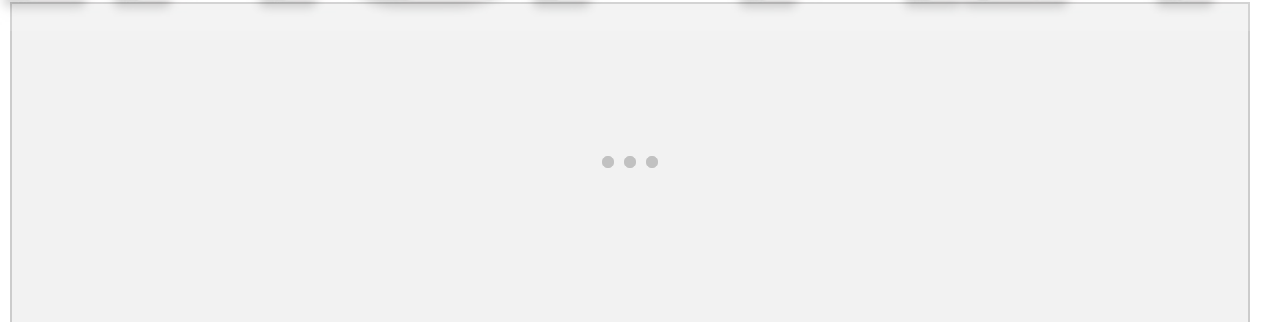
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LTE B20 RX Schematics:

Antenna connector, Coupler check:

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ASM Switch Check:

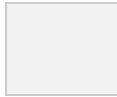
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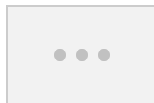
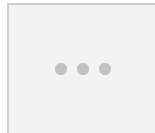
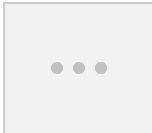


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**B20 Duplexer Check**

WTR B20 PRX Input check:

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**B20 DRX Schematic****B20 DRX RF Connector:**

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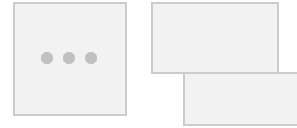
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B20 DRX Filter and WTR input:



B20 DRX WTR input

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**B20 DRX Layout:**

**RF Connector Check:**

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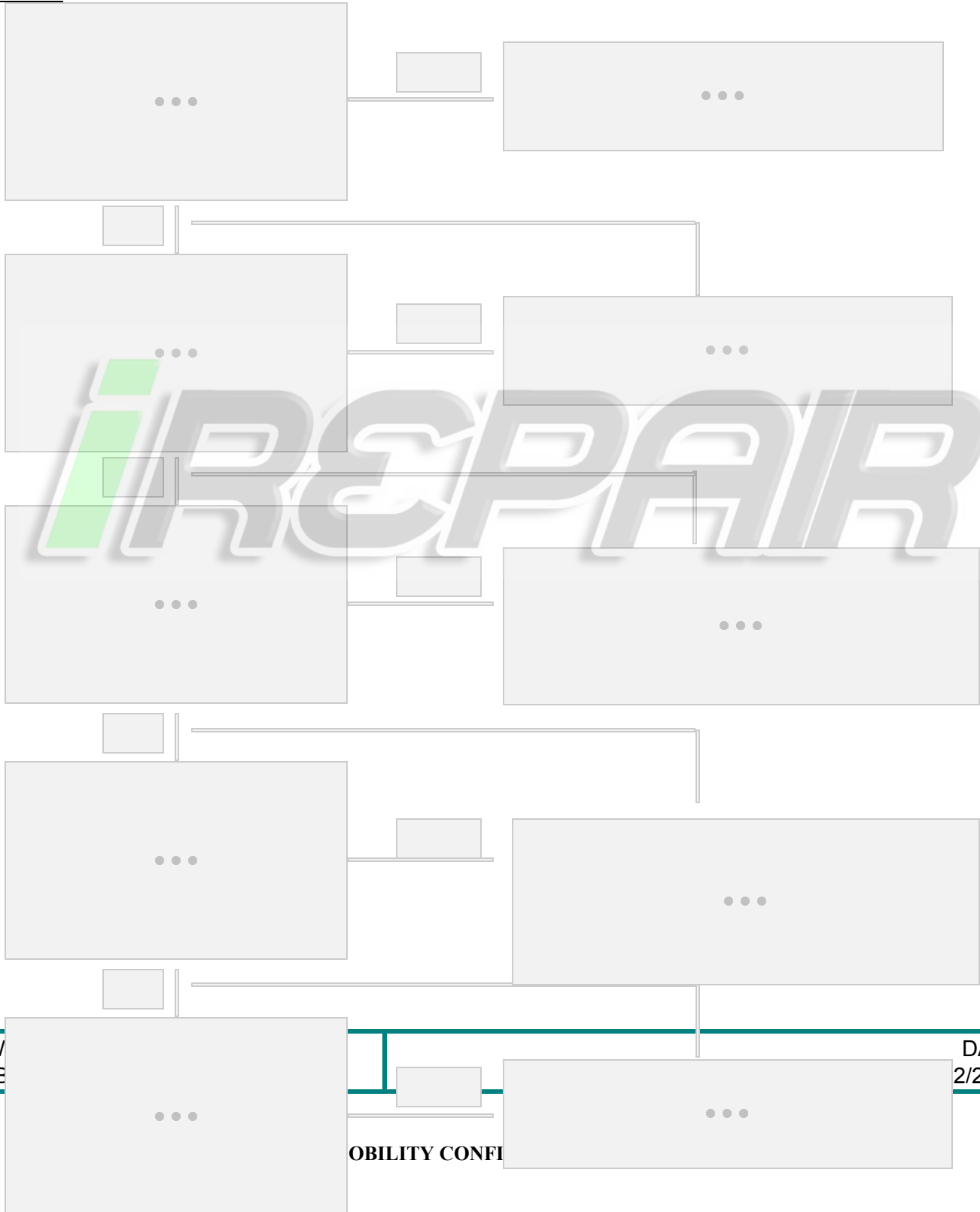
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## B28A/B TX



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**Antenna Connector and Coupler Check:****LTE B28 A/B TX Schematics:** J100, C102, U100, L104

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LTE B28A/B ASM check.

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LTE B28A Duplexer

B28B Duplexer:

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B28A/B TX: WTR input to PA.

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B28A/B PA Output check.

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WTR output:

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B28A and B28B TX Switch Schematic:

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B28B TX PCB Layout:

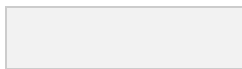
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B28A-B28B TX Switch Check:

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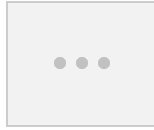
## B28A/B RX



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LTE B28 A/B RX Schematics: J100, C102, U100, L104



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LTE B28A/B ASM check.

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LTE B28A Duplexer

B28B Duplexer:

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B28A and B28B RX Switch:

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LTE B28 DRX:



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B28 DRX Antenna Connector Check: J170 and C170

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B28DRX ASM Check: U7050, C7062.

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**B28 DRX Filter and Switch Check: FL7406,L7260,C7261,S8406**

**B28 DRX WTR Input Check: C7508,L7955,C7960**

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**B28 DRX Layout:**

RF Path from WTR to Duplexer to ASM to RF Jack.

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B28 DRX Filter and switch layout:

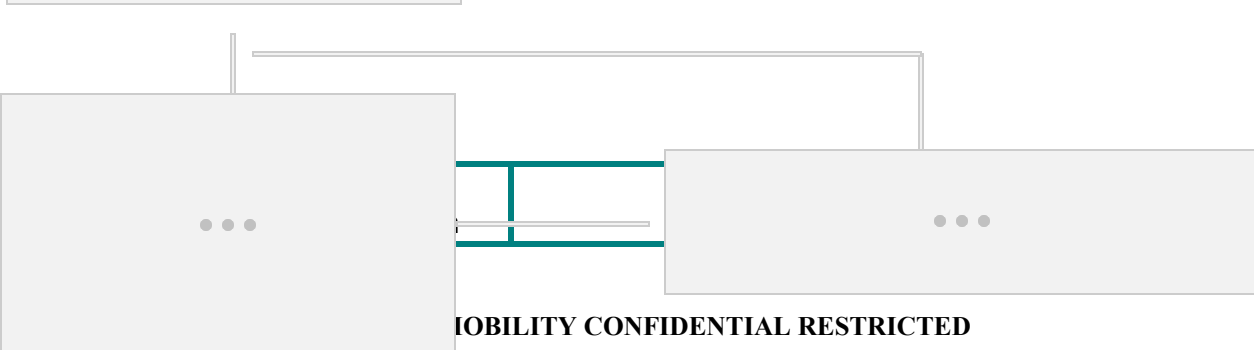
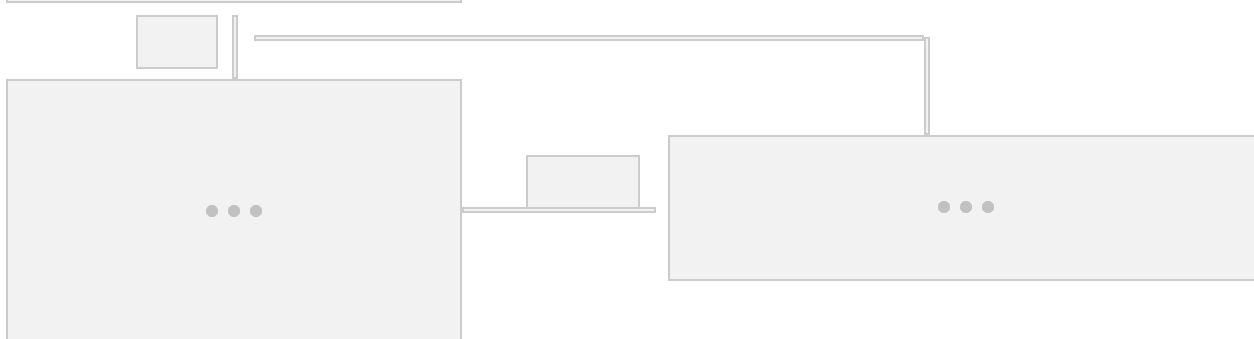
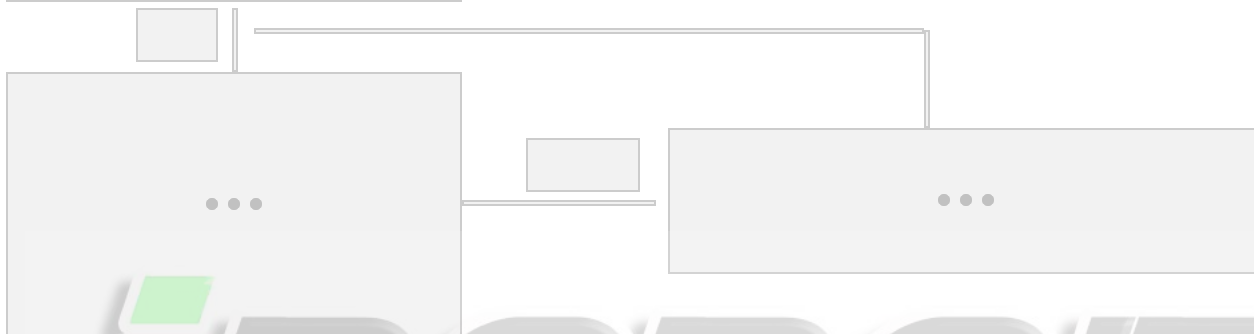
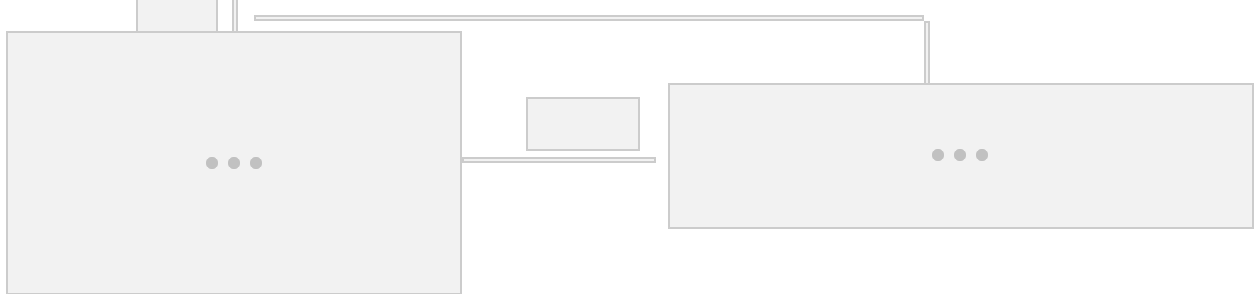
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|-------------------------------------------------|------------------------|

**LTE B40 TX**



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**B40 Connector and Coupler Check: U120, C122, J120**

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B40 TX Schematics: ASM Check:

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**B40 TRX Filter:**

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**B40 PA and Switch:**

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B40 WTR TX output:

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**B40 TX-RX Layout**

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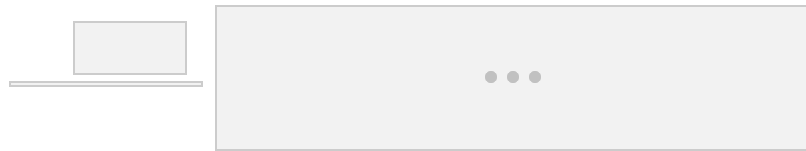
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**LTE B40 PRX**



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|---------------------------|--------------------------------------|

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**B40 PRX Schematics:**

**B40 Connector and Coupler Check: U120, C122, J120**

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B40 RX Schematics: ASM Check:

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**B40 TRX Filter:**

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**B40 PA and Switch:**

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B40 RX input to WTR:

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**B40 TX-RX Layout**

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**B40 RX Layout: B40 PRX input to WTR(U3100)**

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B40 DRX Schematics:

Antenna Connector: J180

B40 DRX ASM: Check U7050,C7060,L7060

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**B40 DRX Layout:** Connector J180

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**RF MIPI Devices Debug:**

Failure Mode: In the event of QC\_DO\_SELF\_CAL and consecutive steps fail during RF calibration, the MIPI RF Devices need to be checked.

Tools needed: QRCT and Bounce DUT should be in BP Tools mode:

RF Components that are MIPI devices are listed in the table below:

| <u>Channel</u> | <u>Ref Des</u> | <u>Part Description</u>        | <u>Prod ID</u> | <u>Slave ID</u> |
|----------------|----------------|--------------------------------|----------------|-----------------|
| 0              | U3100          | WTR3925                        | 0xC0           | 0x1             |
| 1              | U8000          | QFE2101 - APT                  | 0x31           | 0x4             |
| 1              | U8100          | SKY77647-MMMBPA                | 0x1D           | 0x2             |
| 1              | U8200          | SKY77807-MMMBPA_HB             | 0x82           | 0x7             |
| 2              | U8400          | RF1891- ASM Main               | 0x2B           | 0xB             |
| 2              | U7050          | XMSSFN3G0PA-ASM DRx            | 0x10           | 0xA             |
| 2              | U350           | STHVDAC-253MTG- ANT TUNER Main | 0x04           | 0x0C            |
| 2              | U380           | STHVDAC-253MTG- ANT TUNER DRX  | 0x44           | 0x0D            |

Table 1.0. MIPI RF Devices - Product and Slave ID

|                               |           |
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**Open QRCT:**

**Step 1:** a. Click and Select COM Port from the drop down list.

b. Mobile Mode Control -> FTM



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Step 2: Open FTM Command -> Common -> Main Controls:



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Step 3: Set the Channel and Slave ID of the RF component from the table 1.0.

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**Step 4: Read the Device Info:**



Once the Product ID is read, compare with the values in table 1.0.

If the value matches the table, it means that the RF component is detected and working correctly, if an invalid value is read back for Product ID, check the component for possible open pin (no solder)OR short to ground on the RFFEx\_CLK and RFFEx\_DATA pins.

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